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SCHOOLOF COMPUTING

Department of Computer Science and Engineering

ComputerNetworks

(212CSE3302)

StudentName : …….

RegisterNumber : ……

Slot / Section : ……..

Ex.No:	1a
Name of the Experi ment	Study of Network Tool - Cisco Packet tracer
Date	

AIM

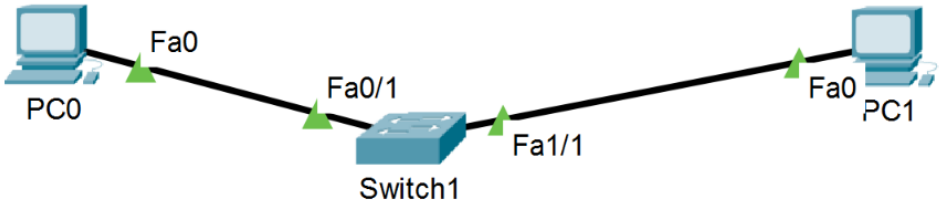
To study and familiarize with the Cisco Packet Tracer simulation software and execute basic networking commands (ipconfig, ping, hostname, nslookup, tracert, netstat, getmac, arp) for network troubleshooting and connectivity verification.

HARDWARE & SOFTWARE REQUIREMENTS

1. **Software:** Cisco Packet Tracer Simulation Tool
2. **Devices:**
    - End Devices: PC0, PC1
    - Networking Device: Switch (2960 Series)
    - Cables: Copper Straight-Through Wires

CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask	Default Gateway
PC0	Fa0	172.16.108.25	255.255.0.0	172.16.1.1
PC1	Fa0	172.16.108.26	255.255.0.0	172.16.1.1



PC1

Physical | Config | Desktop | Programming | Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

- DHCP
- Static

IPv4 Address	172.16.108.26
Subnet Mask	255.255.255.0
Default Gateway	172.16.1.1
DNS Server	0.0.0.0

IPv6 Configuration

- Automatic
- Static

IPv6 Address	/
Link Local Address	FE80::2D0:97FF:FEBB:CDE7
Default Gateway	
DNS Server	

802.1X

- Use 802.1X Security

Authentication	MD5
Username	
Password	

Top

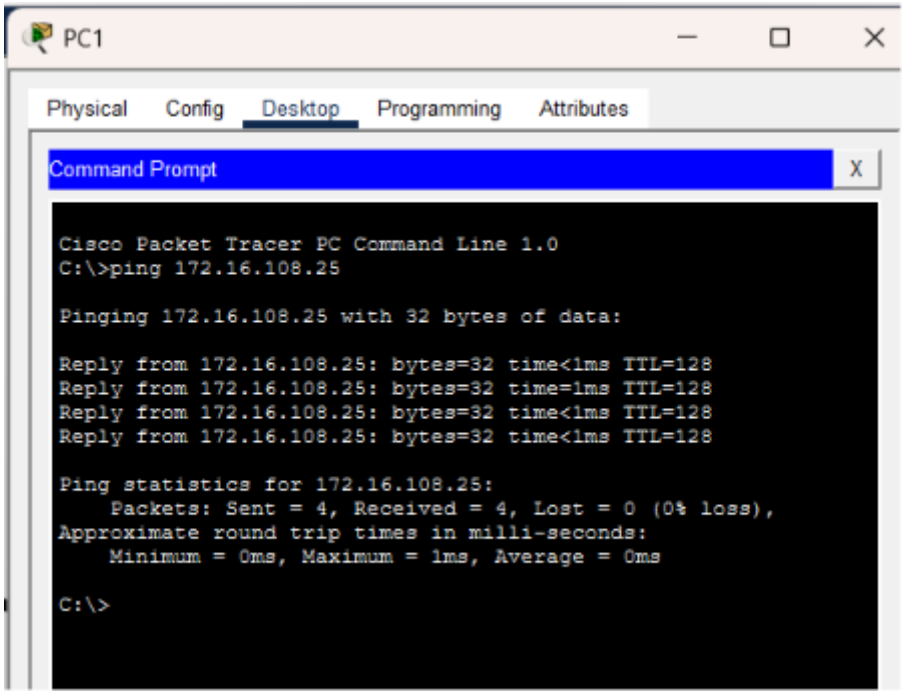
PROCEDURE

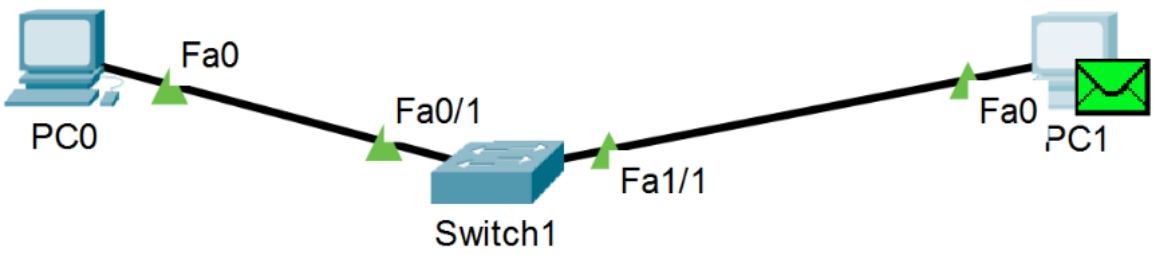
1. **Building the Topology:**
    - Launch **Cisco Packet Tracer**.
    - Drag and drop two Generic **PCs (PC0 and PC1)** and one **2960 Switch** onto the workspace.
    - Connect PC0 (FastEthernet0) to Switch0 (FastEthernet0/1) using a **Copper Straight-Through cable**.
    - Connect PC1 (FastEthernet0) to Switch0 (FastEthernet0/2) using a **Copper Straight-Through cable**.
2. **IP Configuration:**
    - Click on **PC0** -> Navigate to **Desktop** -> Open **IP Configuration**.
    - Select **Static** and enter IPv4 Address: 172.16.108.25 and Subnet Mask: 255.255.0.0.
    - Click on **PC1** -> Navigate to **Desktop** -> Open **IP Configuration**.
    - Select **Static** and enter IPv4 Address: 172.16.108.26 and Subnet Mask: 255.255.0.0.
3. **Executing Commands & Connectivity Verification:**
    - Open the **Command Prompt** on PC0.
    - Run ipconfig to verify local IP settings.
    - Run ping 172.16.108.26 to verify ICMP reachability to PC1.
    - Execute additional system commands (hostname, netstat, arp -a, getmac) to inspect system and network states.

COMMAND EXECUTION DETAILS

- **ipconfig**: Displays the IP address, Subnet Mask, and Default Gateway of the active interface.
- **ping <IP>**: Sends ICMP Echo Request packets to check network-layer connectivity with another host.
- **hostname**: Displays the unique computer host name assigned to the local machine.

- **arp -a**: Displays the current Address Resolution Protocol (ARP) cache mapping IP addresses to MAC addresses.
- **tracert <IP>**: Traces the hop-by-hop route packets take to reach a destination address.
- **getmac**: Retrieves the physical (MAC) address of the network adapters installed on the machine.





Rubrics for Experiment Assessment:

Rubrics	**Good**	**Normal**	**Poor**	**Marks**
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

RESULT:

Thus the Study of Network Tool-Packet Tracer has been done successfully.

Ex.No:	1b
Name of the Experi ment	Study and Design a simple LAN Network
Date	

AIM

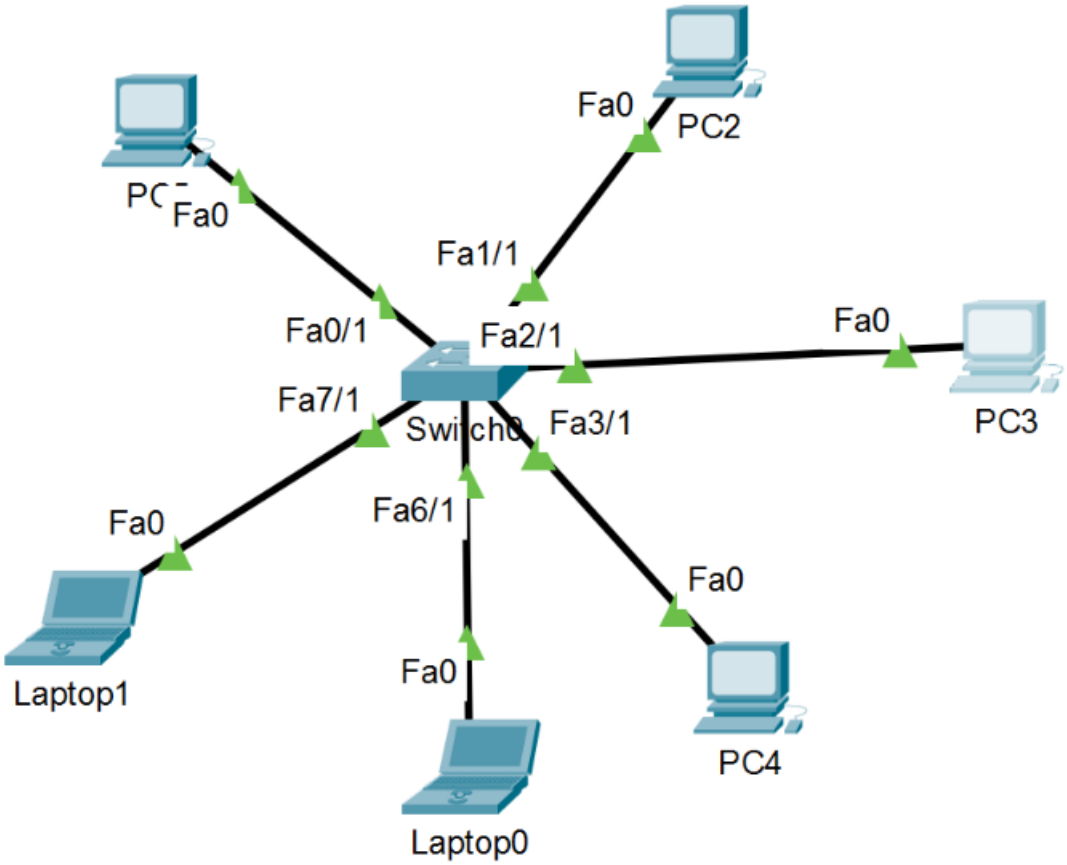
To design and configure a simple Local Area Network (LAN) using a switch, multiple PCs, and laptops, and to verify network connectivity using basic network commands in Cisco Packet Tracer.

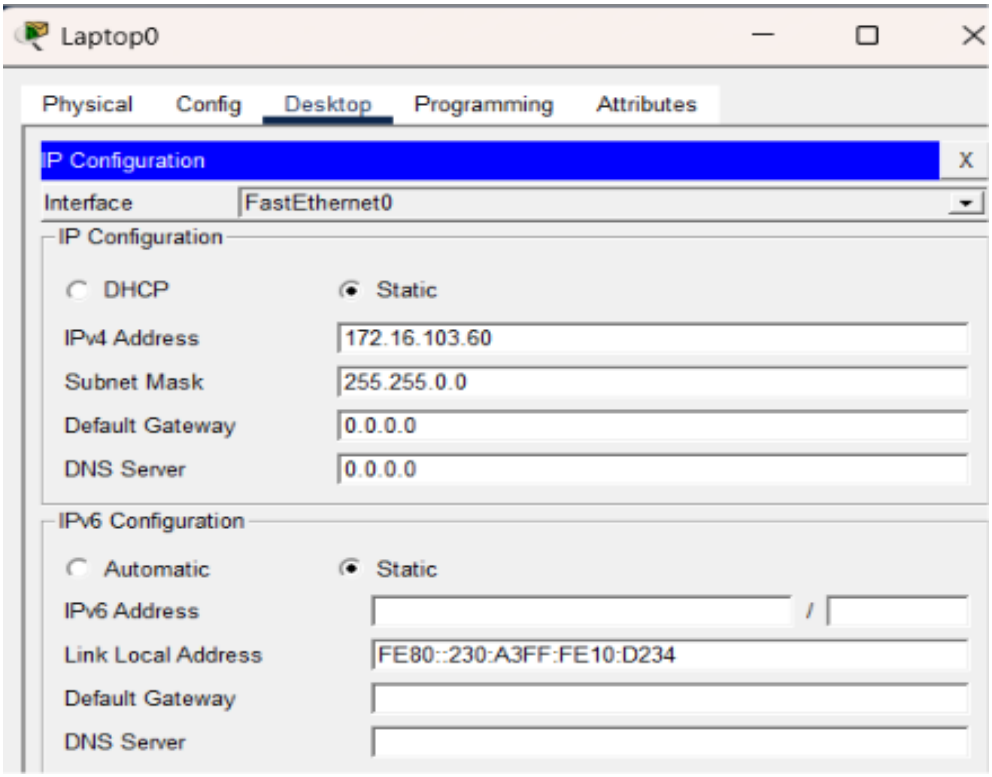
HARDWARE & SOFTWARE REQUIREMENTS

1. **Software**: Cisco Packet Tracer
2. **Hardware / Devices:** 1 x Switch (e.g., 2960-24TT), End Devices: PC0, PC1, PC2, PC3, Laptop0, and Laptop1 (Total 6 end devices), Copper Straight-Through Cables (Wires).

CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet mask
PC1	Fa0	172.16.103.55	255.255.0.0
PC2	Fa0/5	172.16.103.56	255.255.0.0
PC3	Fa0/6	172.16.103.57	255.255.0.0
PC4	Fa0/3	172.16.103.58	255.255.0.0
Laptop0	Fa0/2	172.16.103.60	255.255.0.0
Laptopl	Fa0/4	172.16.103.61	255.255.0.0


Fa0
PC2
Fa0
Fa1/1
Fa0/1
Fa2/1
Fa0
PC3
Fa7/1
Fa3/1
Fa6/1
Fa0
Fa0
Fa0
PC4
Laptop1
Laptop0


Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.103.60
Subnet Mask
255.255.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::230:A3FF:FE10:D234
Default Gateway
DNS Server

PROCEDURE

Topology Creation:

- Open Cisco Packet Tracer.
- Place a Switch in the workspace.
- Add the required end devices: PC0, PC1, PC2, PC3, Laptop0, and Laptop1.
- Connect each end device to the switch using Copper Straight-Through cables via their respective FastEthernet interfaces (Fa0, Fa0/2 to Fa0/6).

IP Configuration:

- Click on each end device, navigate to the Desktop > IP Configuration tab, and assign the corresponding static IP address and subnet mask as specified in the configuration table below.

Connectivity Verification:

- Access the Command Prompt (cmd) on any PC or Laptop.
- Use the ipconfig command to verify the assigned IP parameters.
- Use the ping command to test data transmission and network reachability between different nodes on the LAN.

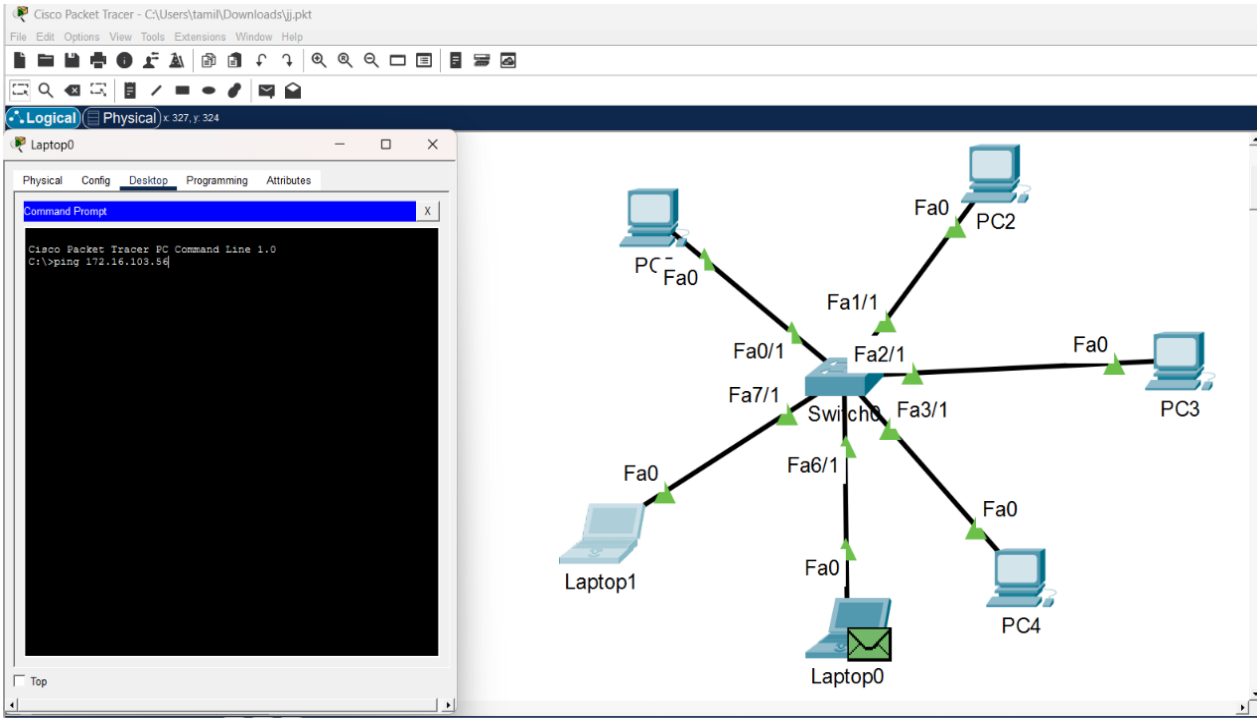
COMMAND EXECUTION DETAILS

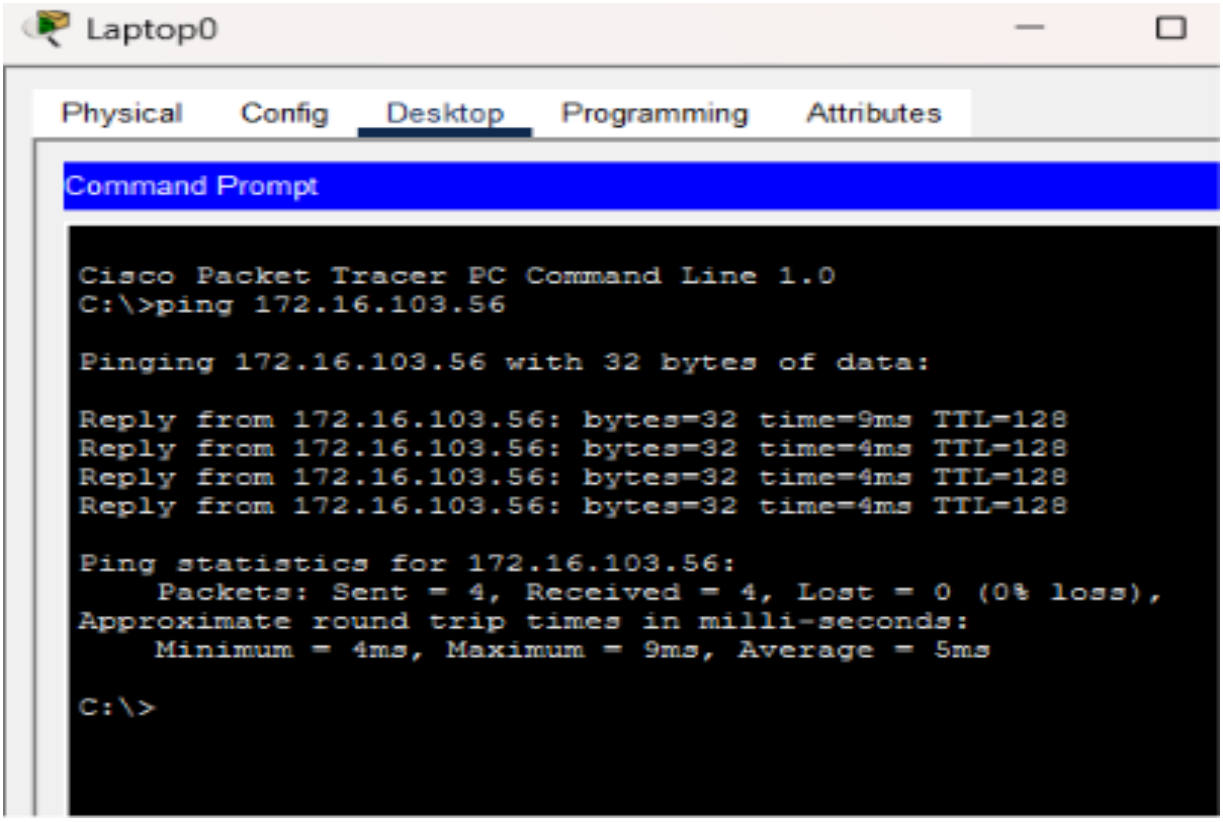
- **ipconfig:** Displays the IPv4 address, subnet mask, and default gateway configured on the network adapter.

  Syntax: **C:\> ipconfig**

- **ping:** Sends ICMP echo request packets to test network connectivity between host devices on the same subnet.

  Syntax: **C:\> ping <IP_Address> (e.g., ping 172.16.103.56)**


Cisco Packet Tracer - C:\Users\tamil\Downloads\jj.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.103.56
Top
Fa0
PC2
Fa0
Fa1/1
Fa0/1
Fa2/1
Fa0
PC3
Fa7/1
Switch0
Fa3/1
Fa0
Fa6/1
Fa0
Laptop1
Fa0
PC4
Laptop0


Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.103.56
Pinging 172.16.103.56 with 32 bytes of data:
Reply from 172.16.103.56: bytes=32 time=9ms TTL=128
Reply from 172.16.103.56: bytes=32 time=4ms TTL=128
Reply from 172.16.103.56: bytes=32 time=4ms TTL=128
Reply from 172.16.103.56: bytes=32 time=4ms TTL=128
Ping statistics for 172.16.103.56:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 9ms, Average = 5ms
C:\>

Rubrics for Experiment Assessment:

Rubrics	**Good**	**Normal**	**Poor**	**Marks**
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

RESULT:

Thus, a simple Local Area Network (LAN) was successfully designed, configured, and tested for end-to-end connectivity using Cisco Packet Tracer.

Ex.No:	2a
Name of the Experi ment	Study of Network Devices - Building a peer to peer network
Date	

AIM

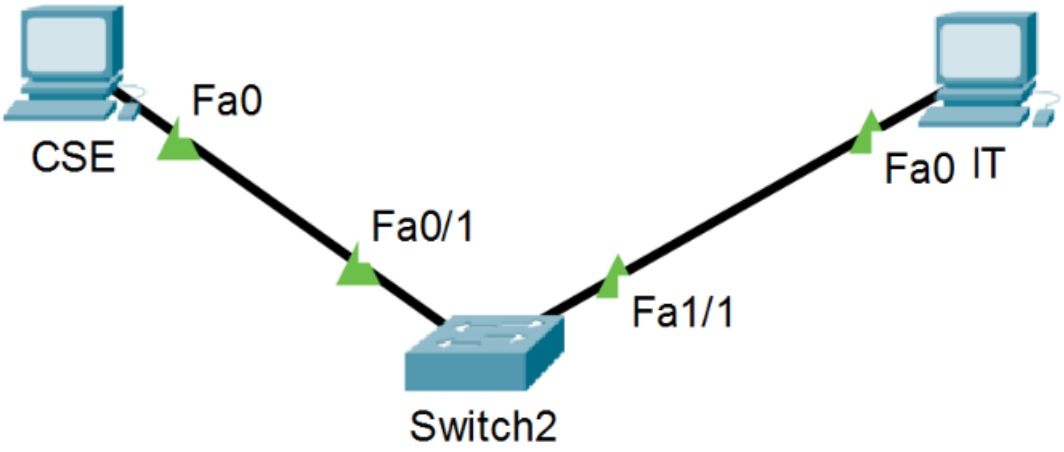
To study various network devices and design a basic Local Area Network (LAN) topology connecting multiple systems through a switch using Cisco Packet Tracer.

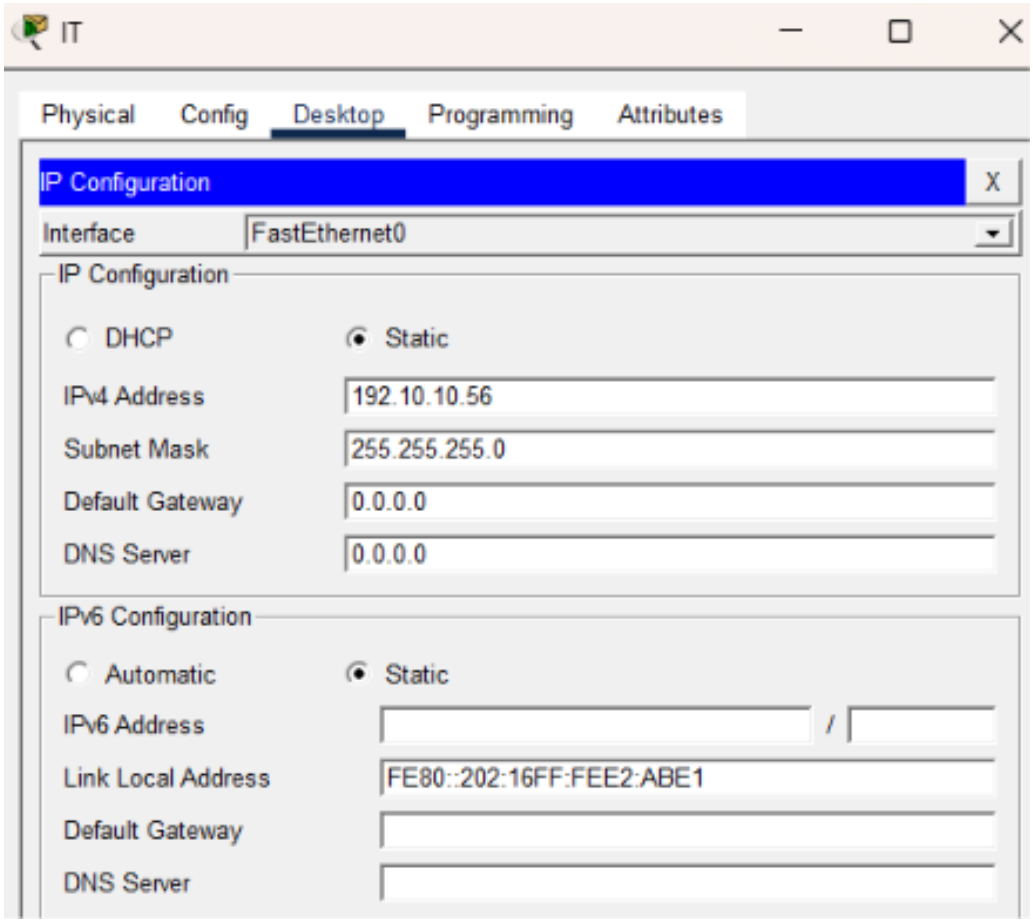
HARDWARE & SOFTWARE REQUIREMENTS

1. **Software**: Cisco Packet Tracer
2. **Hardware / Devices:** 1 x Switch, End Devices: PC0 (CSE), PC1 (IT), Copper Straight-Through Wires

CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask
CSE	Fa0	192.10.10.55	255.255.255.0
IT	Fa0	192.10.10.56	255.255.255.0


CSE
Fa0
Fa0/1
Switch2
Fa1/1
Fa0 IT


IT
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.10.10.56
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::202:16FF:FEE2:ABE1
Default Gateway
DNS Server

PROCEDURE

Topology Creation:

- Open Cisco Packet Tracer and place a **Switch** in the workspace.
- Add the required end devices (**PC0** labeled as CSE and **PC1** labeled as IT).
- Connect each device to the switch using **Copper Straight-Through** cables via their FastEthernet interfaces (Fa0).

IP Configuration:

- Access each device, go to Desktop > IP Configuration, and assign the appropriate static IP address and subnet mask (192.10.10.x with 255.255.255.0).

Connectivity Verification:

- Open the Command Prompt (cmd) on the devices.
- Run the ipconfig command to check interface IP parameters.
- Use the ping command to test network reachability between the CSE and IT nodes.

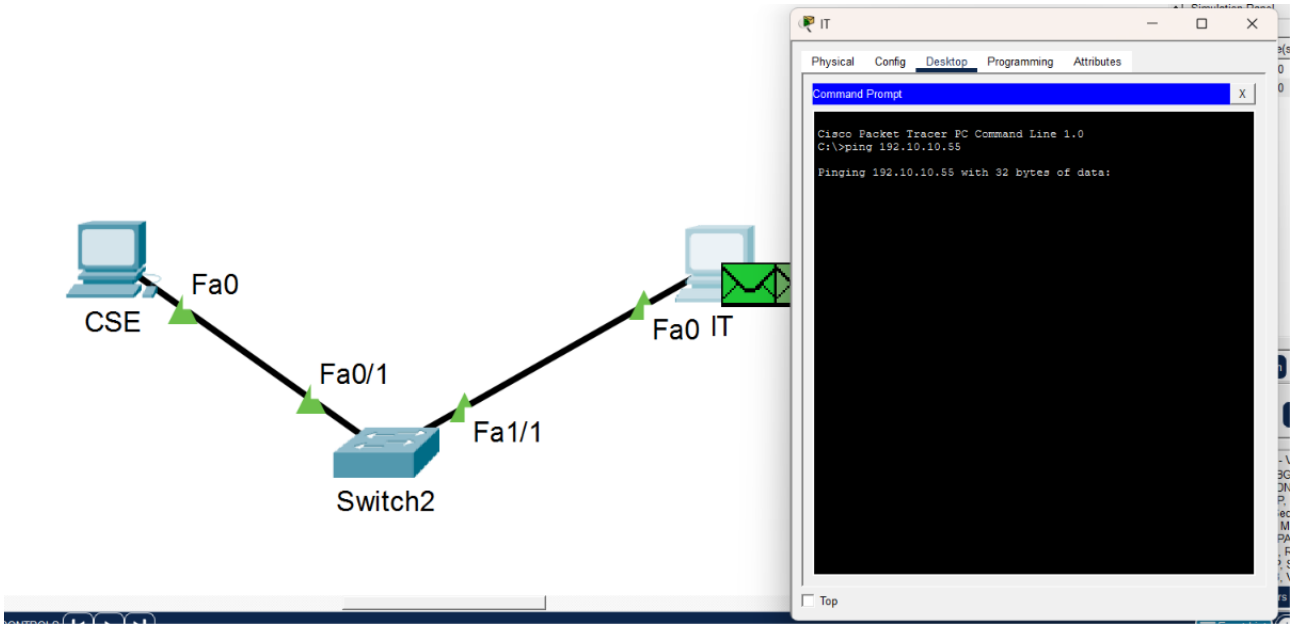
COMMAND EXECUTION DETAILS

- **ipconfig:** Displays the IPv4 address, subnet mask, and default gateway configured on the network adapter.

  Syntax: **C:\> ipconfig**

- **ping:** Sends ICMP echo request packets to test network connectivity between host devices on the same subnet.

  Syntax: **C:\> ping <IP_Address> (e.g., ping 172.16.103.56)**



IT

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.10.10.55

Pinging 192.10.10.55 with 32 bytes of data:

Reply from 192.10.10.55: bytes=32 time=8ms TTL=128
Reply from 192.10.10.55: bytes=32 time<1ms TTL=128
Reply from 192.10.10.55: bytes=32 time<1ms TTL=128
Reply from 192.10.10.55: bytes=32 time=1ms TTL=128

Ping statistics for 192.10.10.55:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>
```

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

RESULT:

Thus, the study of network devices and the configuration of a basic network topology were done successfully.

Ex.No:	2b
Name of the Experi ment	Study and Design of all Network Topologies
Date	

OBJECTIVE(S):

To design and implement network topologies using Cisco Packet Tracer

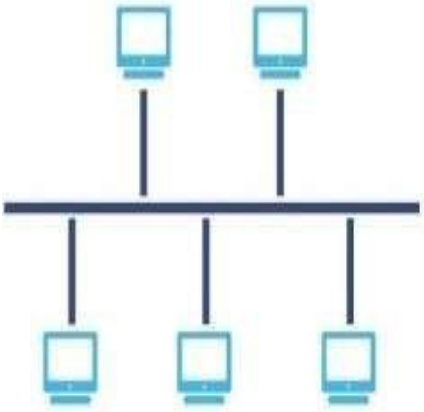
INTRODUCTION:

Network topology is the geometric representation of relationship of all the links connecting the devices or nodes. Network topology represent in two ways one is physical topology that define the way in which a network is physically laid out and other one is logical topology that defines how data actually flow through the network. In this lab, we will discuss how to design bus, star and mesh topology network and provide interfacing and simulation between end points using packet tracer software.

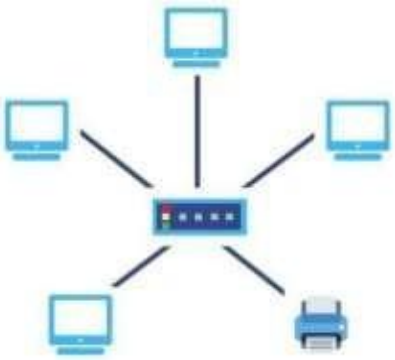
THEORETICAL BACKGROUND:

Bus Topology

In local area network, it is a single network cable runs in the building or campus and all nodes are connected along with this communication line with two endpoints called the bus or backbone. In other words, it is a multipoint data communication circuit that is easily control data flow between the computers because this configuration allows all stations to receive every transmission over the network. For bus topology we build network using three generic pc which are serially connected with three switches using copper straight through cable and switches are interconnected using copper cross over cable.

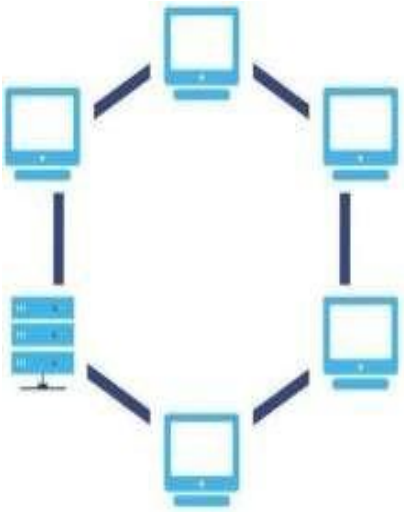


Star Topology



In star topology, all the cables run from the computers to a central location where they are all connected by a device called a hub. It is a concentrated network, where the end points are directly reachable from a central location when network is expanded. Ethernet 10 base T is a popular network based on the star topology. For star topology we build network using five generic pc which are centrally connected to single switch 2950-24 using copper straight through cable.

RING TOPOLOGY



As we mentioned earlier, the ring topology is similar to a daisy chain topology but with the loop closed so that the nodes are arranged in a ring or circle. Each node has exactly two peers and the data travels in one direction passing through each intermediate node on the ring until it reaches the destination node. Data can be made to pass in both directions by adding a second connection between the network nodes, creating a dual ring topology.

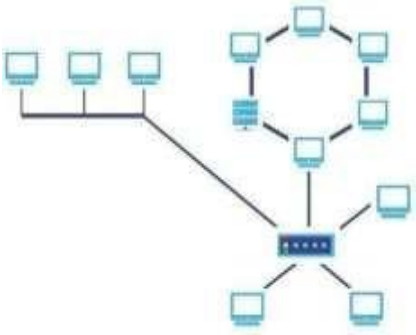
In a ring topology, an electrical “token” circulates around the network. Any node that wants to transmit data has to wait until it has possession of the token.

MESH TOPOLOGY



In mesh topology every device has a dedicated point to point link to every other device. The term dedicated stand for link carries traffic only between four devices it connects. It is a well-connected topology; in this, every node has a connection to every other node in the network. The cable requirements are high and it can include multiple topologies. Failure in one of the computers does not cause the network to break down, as they have alternative paths to other computers star topology, all the cables run from the computers to a central location.

Hybrid Topology



Hybrid topology combines two or more topologies. You can see in the above architecture in such a manner that the resulting network does not exhibit one of the standard topologies.

For example, as you can see in the above image that in an office in one department, Star and P2P topology is used. A hybrid topology is always produced when two different basic network topologies are connected.

HARDWARE & SOFTWARE REQUIREMENTS

1. **Software**: Cisco Packet Tracer
2. **Hardware / Devices:** Required - Switches, Hubs, End Devices PC, PC1,Copper Straight-Through Wires, Cross-Over cables.

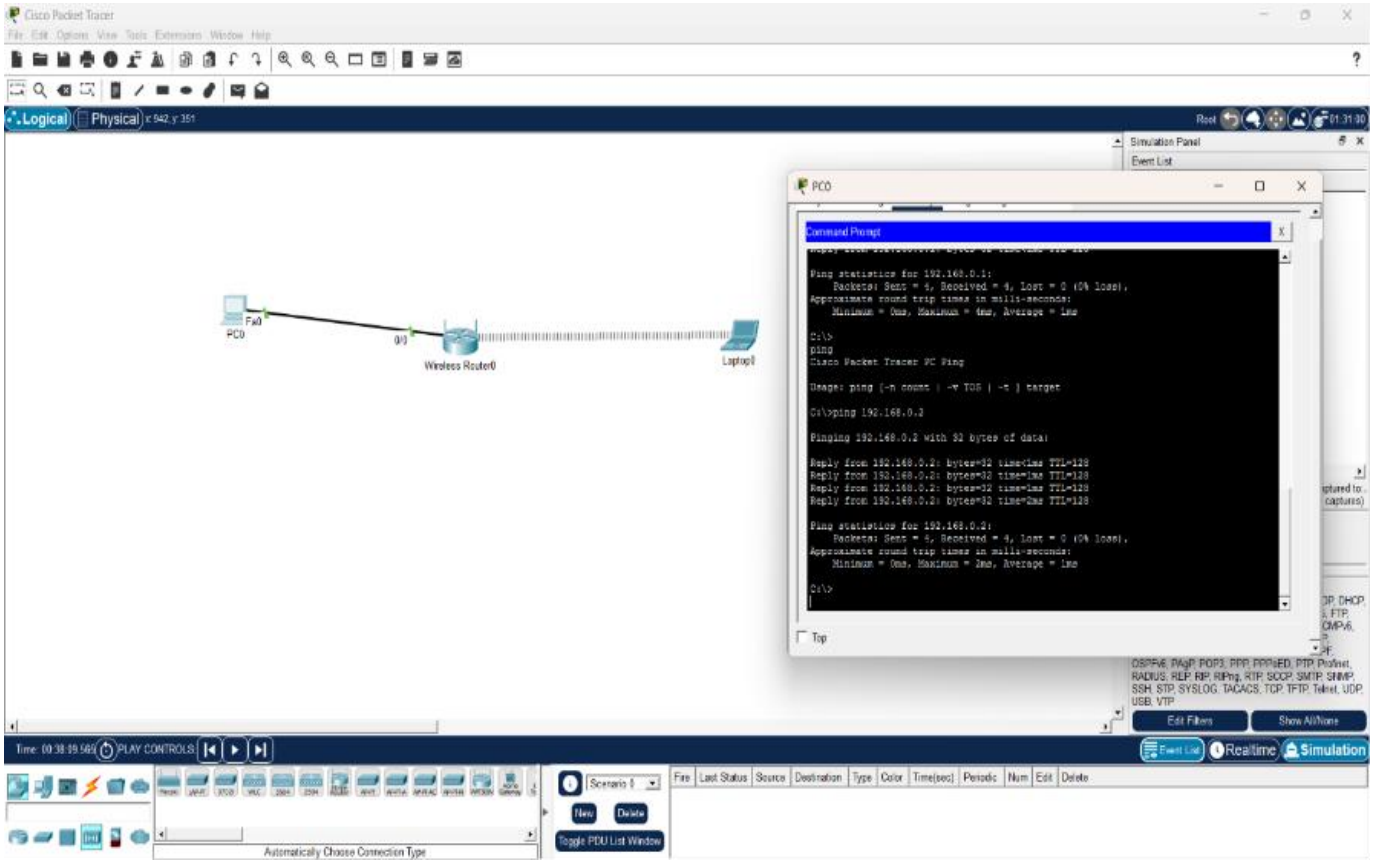
COMMAND EXECUTION DETAILS

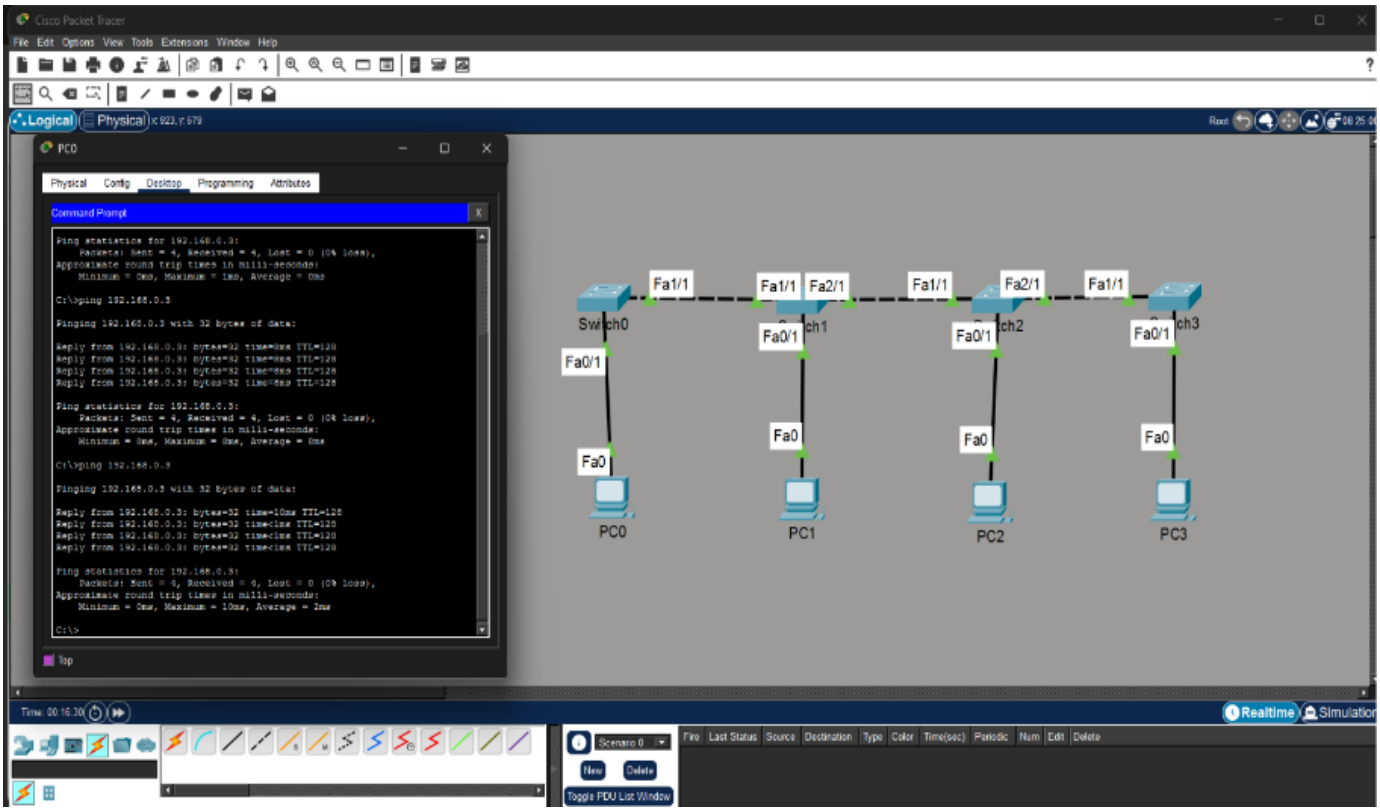
- **ipconfig:** Displays the IPv4 address, subnet mask, and default gateway configured on the network adapter.

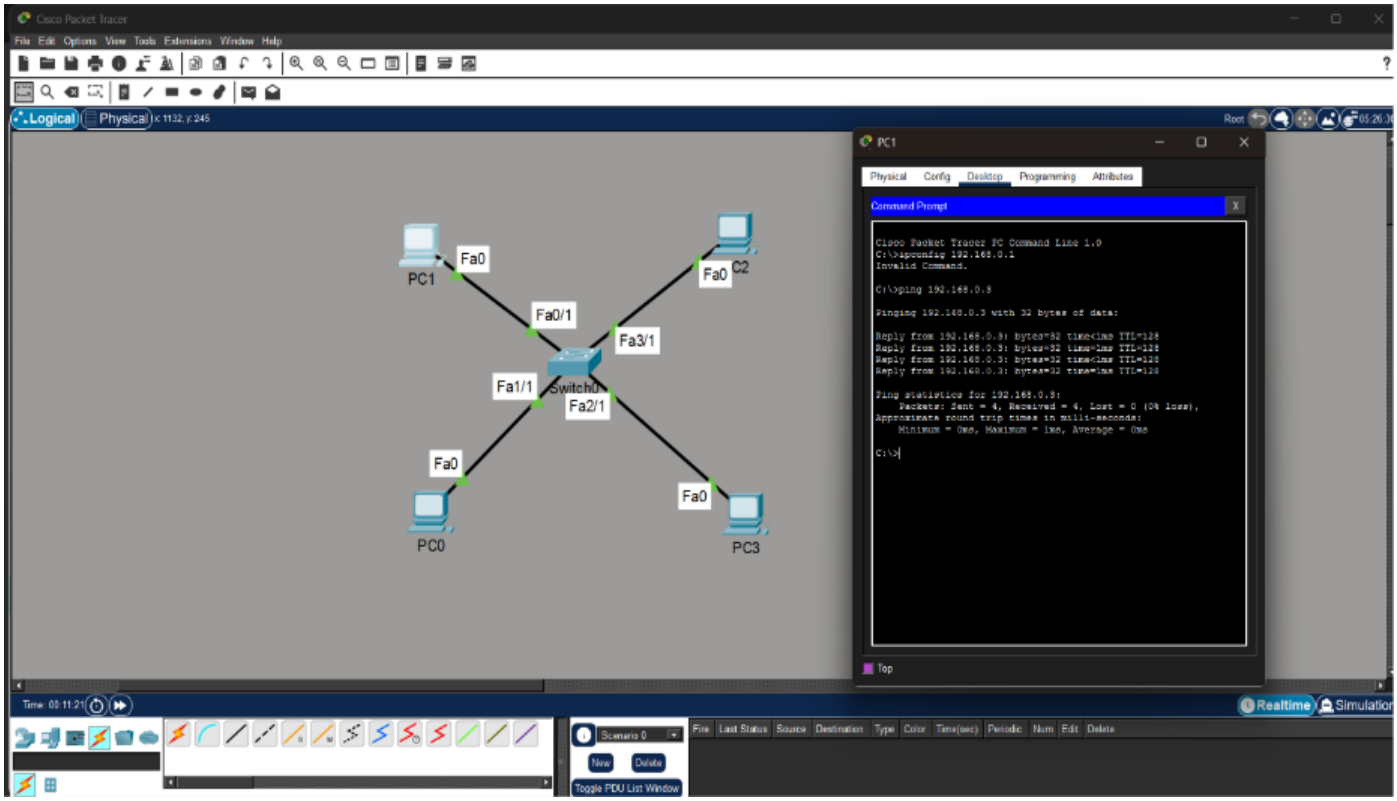
  Syntax: **C:\> ipconfig**

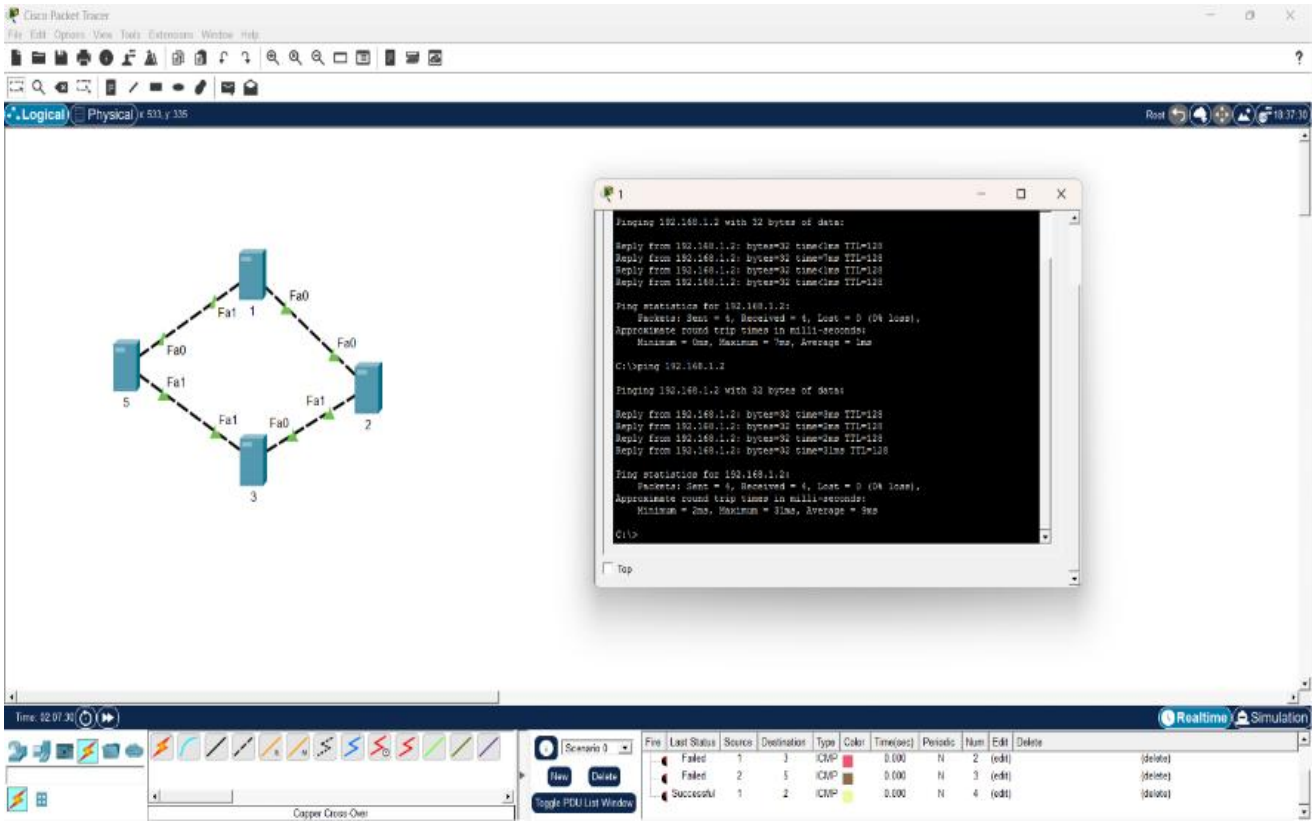
- **ping:** Sends ICMP echo request packets to test network connectivity between host devices on the same subnet.

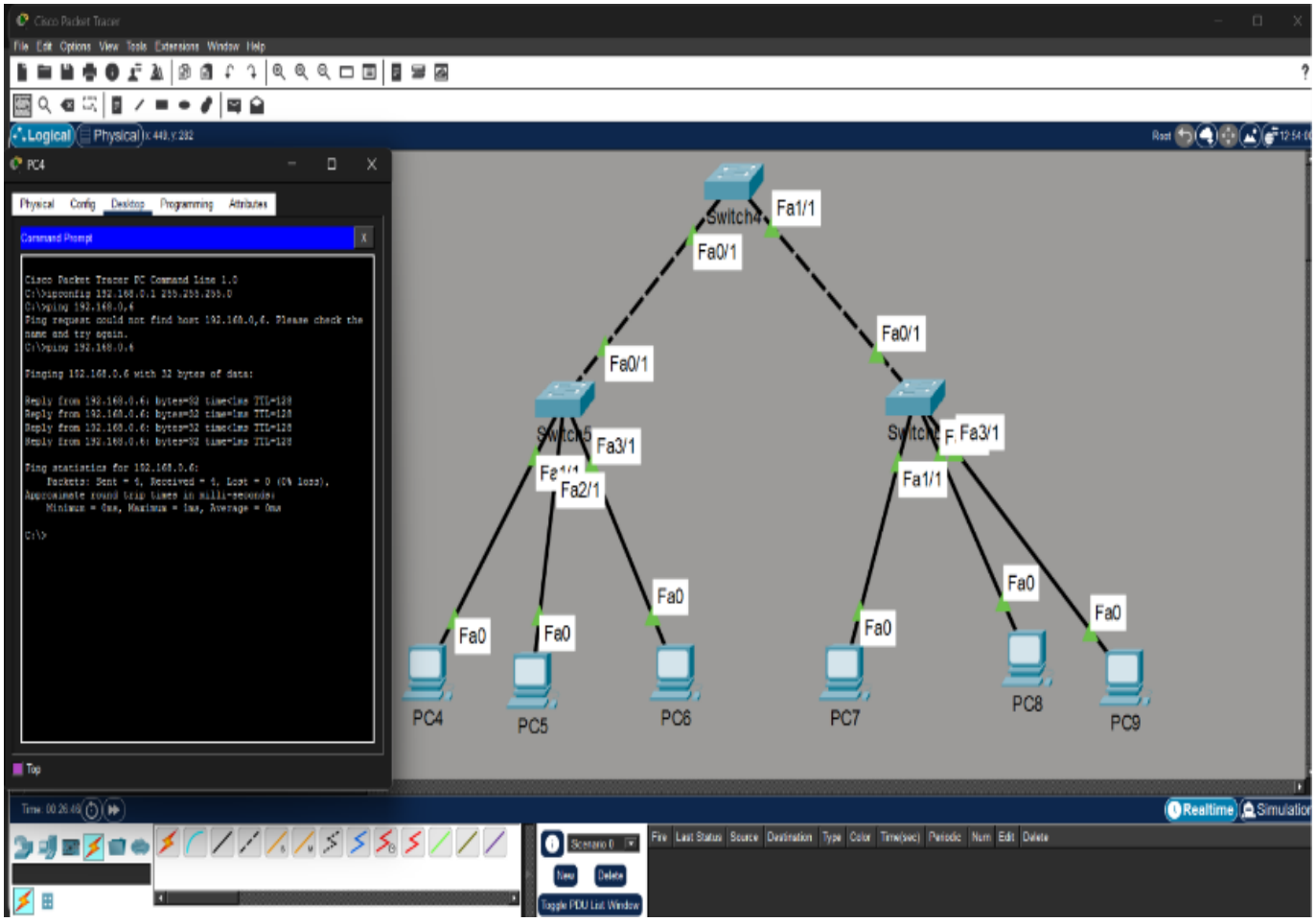
  Syntax: **C:\> ping <IP_Address> (e.g., ping 172.16.103.56)**

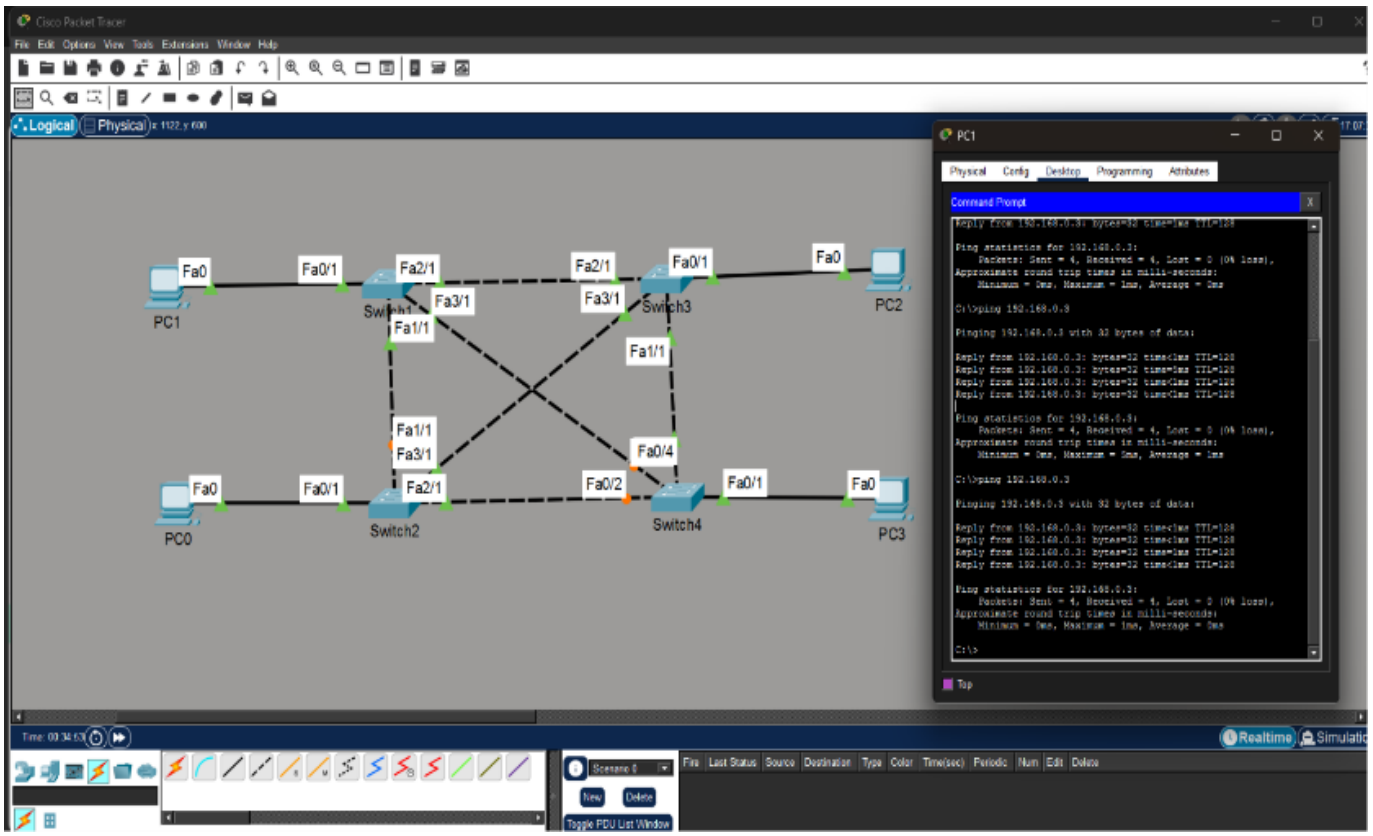

Cisco Packet Tracer
Logical
Physical
PC0
Command Prompt
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 1ms
C:\>
ping
Cisco Packet Tracer PC Ping
Usage: ping [-n count] [-w TOS] [-t] target
C:\>ping 192.168.0.2
Pinging 192.168.0.2 with 32 bytes of data:
Reply from 192.168.0.2: bytes=32 time<1ms TTL=128
Reply from 192.168.0.2: bytes=32 time=1ms TTL=128
Reply from 192.168.0.2: bytes=32 time=1ms TTL=128
Reply from 192.168.0.2: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 1ms
C:\>
Top
Fa0
0/0
PC0
Wireless Router0
Laptop0
Simulation Panel
Event List
Edit Filters
Show All/None
Realtime
Simulation
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete

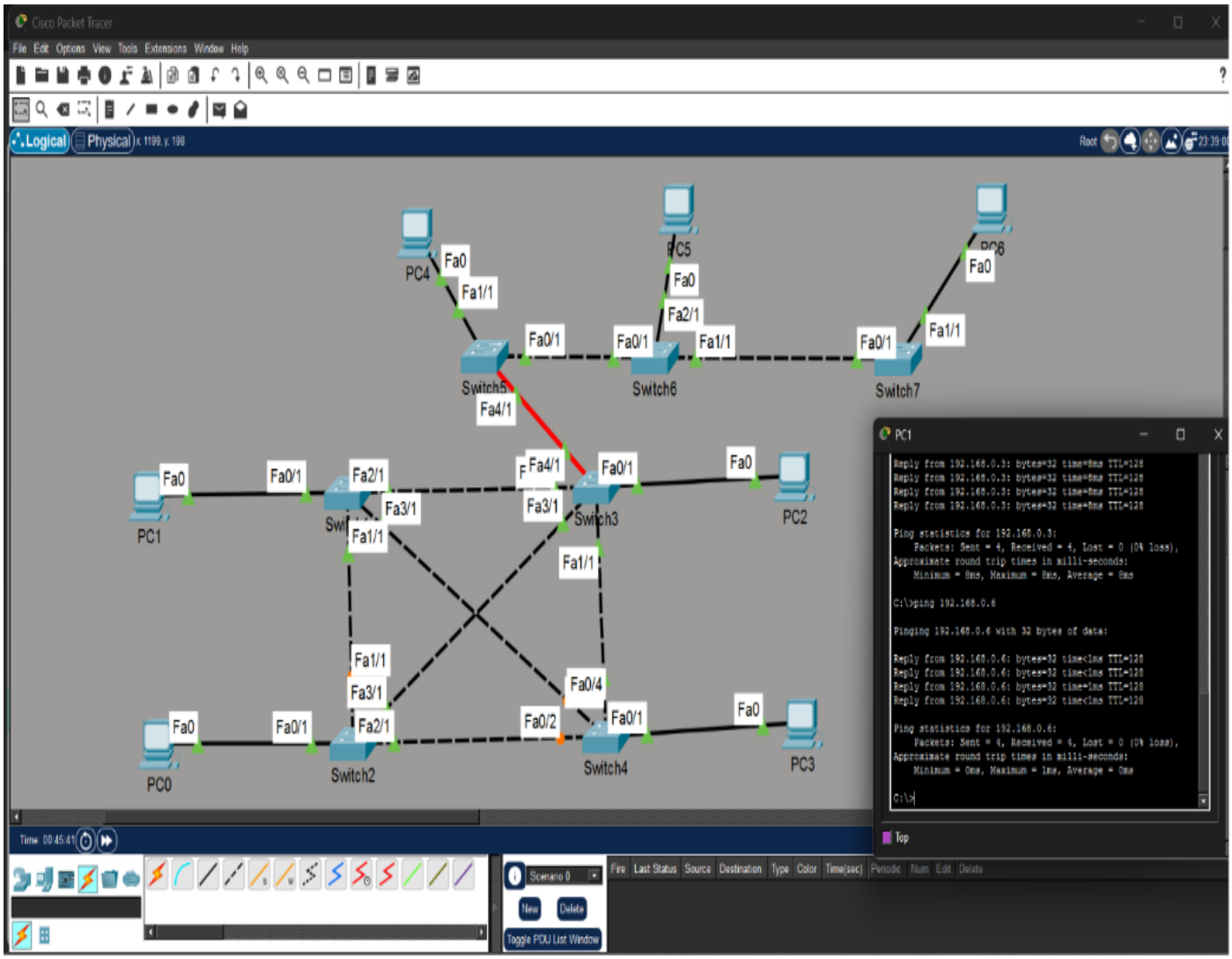

Cisco Packet Tracer
Logical
Physical
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.0.3
Pinging 192.168.0.3 with 32 bytes of data:
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Reply from 192.168.0.3: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.0.3
Pinging 192.168.0.3 with 32 bytes of data:
Reply from 192.168.0.3: bytes=32 time=10ms TTL=128
Reply from 192.168.0.3: bytes=32 time<1ms TTL=128
Reply from 192.168.0.3: bytes=32 time<1ms TTL=128
Reply from 192.168.0.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 2ms
C:\>
Top
Switch0
Fa1/1
Fa2/1
Fa0/1
Fa0
PC0
PC1
PC2
PC3
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Cisco Packet Tracer
Logical
Physical
PC1
Fa0
Fa0/1
Fa3/1
Fa1/1
Switch0
Fa2/1
Fa0
C2
Fa0
PC0
Fa0
PC3
PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig 192.168.0.1
Invalid Command.
C:\>ping 192.168.0.3
Pinging 192.168.0.3 with 32 bytes of data:
Reply from 192.168.0.3: bytes=32 time<1ms TTL=128
Reply from 192.168.0.3: bytes=32 time=1ms TTL=128
Reply from 192.168.0.3: bytes=32 time<1ms TTL=128
Reply from 192.168.0.3: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Realtime
Simulation


Cisco Packet Tracer
Logical
Physical
1
Fa0
Fa1
Fa0
Fa0
Fa1
5
Fa1
Fa1
Fa0
2
3
1
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=7ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 1ms
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=31ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 31ms, Average = 9ms
C:\>
Top
Realtime
Simulation
Copper Cross-Over


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig 192.168.0.1 255.255.255.0
C:\>ping 192.168.0.4
Ping request could not find host 192.168.0.4. Please check the name and try again.
C:\>ping 192.168.0.6
Pinging 192.168.0.6 with 32 bytes of data:
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time=1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.0.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Switch4
Fa1/1
Fa0/1
Fa0/1
Fa0/1
Switch5
Fa3/1
Fa2/1
Fa1/1
Fa3/1
Fa0
Fa0
Fa0
Fa0
Fa0
Fa0
PC4
PC5
PC6
PC7
PC8
PC9
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Top
Fa0
Fa0/1
Fa2/1
Fa3/1
Fa1/1
Fa2/1
Fa0/1
Fa3/1
Fa0
Fa1/1
Fa1/1
Fa3/1
Fa0/4
Fa0
Fa0/1
Fa2/1
Fa0/2
Fa0/1
Fa0
PC1
PC2
PC0
PC3
Switch2
Switch3
Switch4
Realtime
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC4
PC5
PC6
Fa0
Fa1/1
Fa2/1
Fa0/1
Fa1/1
Switch5
Switch6
Switch7
Fa4/1
Fa3/1
Switch3
PC1
PC2
Fa2/1
Fa0/4
Fa0/2
Switch2
Switch4
PC0
PC3
PC1
Reply from 192.168.0.3: bytes=32 time=8ms TTL=128
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 8ms, Average = 8ms
C:\>ping 192.168.0.6
Pinging 192.168.0.6 with 32 bytes of data:
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.0.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Time: 00:45:41
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete

Rubrics for Experiment Assessment:

Rubrics	**Good**	**Normal**	**Poor**	**Marks**
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

RESULT:

Thus, The Study and Design of all Network Topologies has been done successfully.

Ex.No:	3a
Name of the Experi ment	Study of Network Commands
Date	

AIM

To study, analyze, and execute various basic networking and troubleshooting commands using the Windows Command Prompt.

HARDWARE & SOFTWARE REQUIREMENTS

1. **Software:** Windows Operating System (Command Prompt / PowerShell)

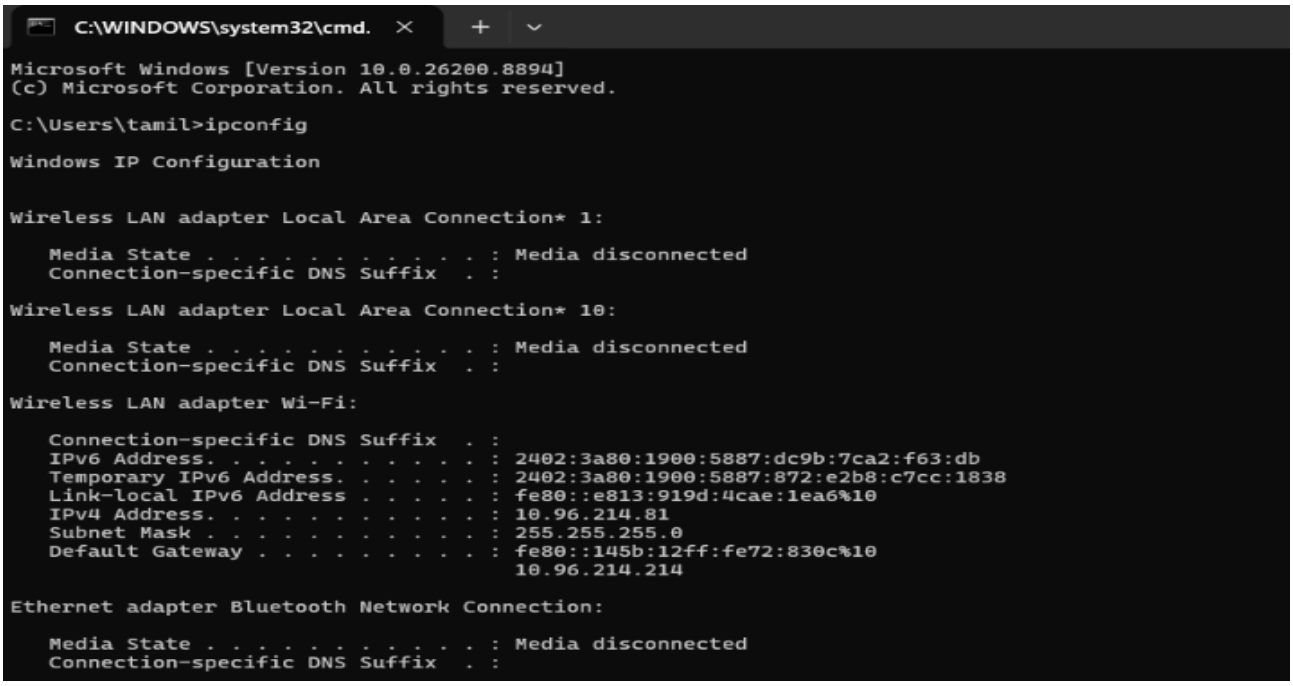
2. **Hardware / Devices:** 1 x PC/Laptop with active Ethernet or Internet Services.

PROCEDURE

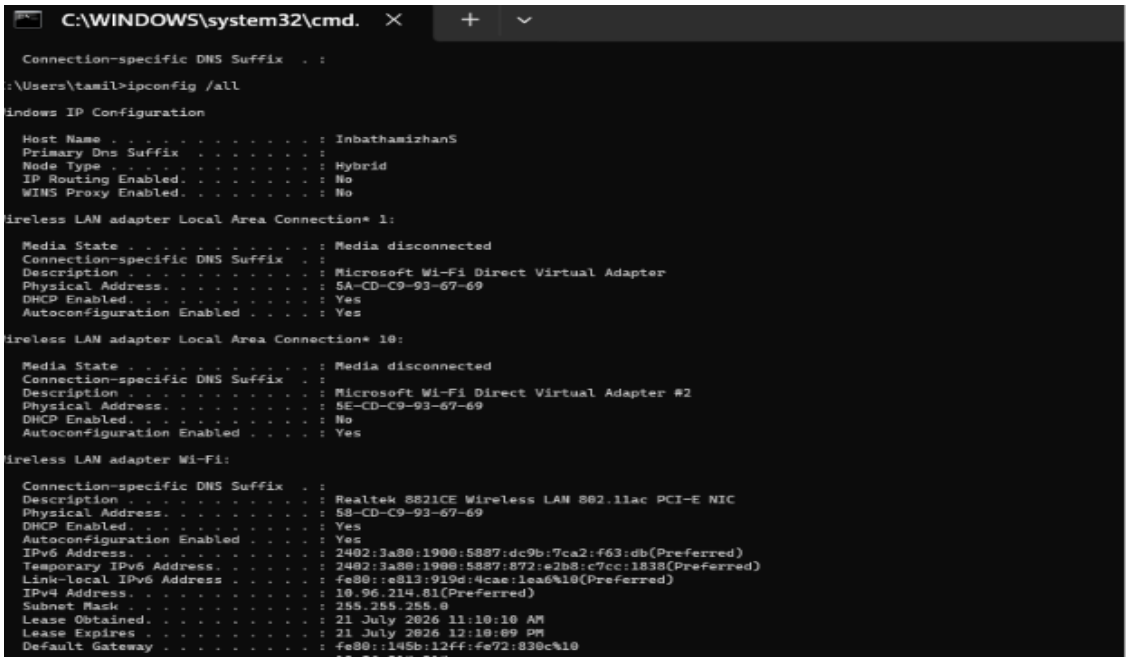
1. **Open Command Prompt:** Press Windows + R, type cmd, and press Enter to open the Command Line Interface.
2. **Execute Diagnostic Commands:** Run various configuration and diagnostic commands (ipconfig, hostname, systeminfo, nslookup, ping, tracert, pathping, netstat, getmac, arp, and route) to view network settings, check routing paths, and analyze active network connections.
3. **Analyze Outputs:** Record and verify the outputs for each command to understand device configurations and network connectivity status.

Command Execution Details

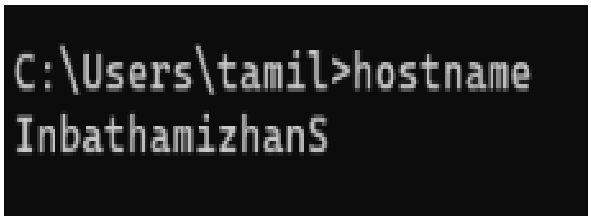
- **ipconfig**: Displays basic IP configuration details such as IPv4 address, Subnet Mask, and Default Gateway.
- *Syntax:* C:\Users\KARE> ipconfig



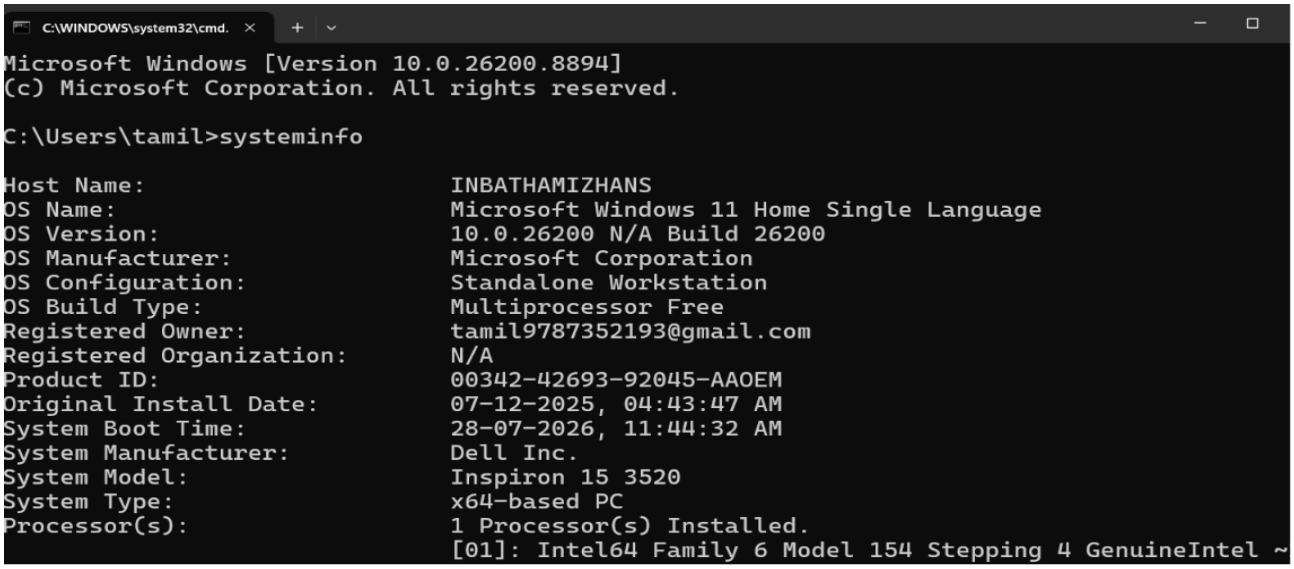
2. **ipconfig /all**: Provides detailed configuration info including the physical (MAC) address and advanced adapter states.
   - *Syntax:* C:\Users\KARE> ipconfig /all



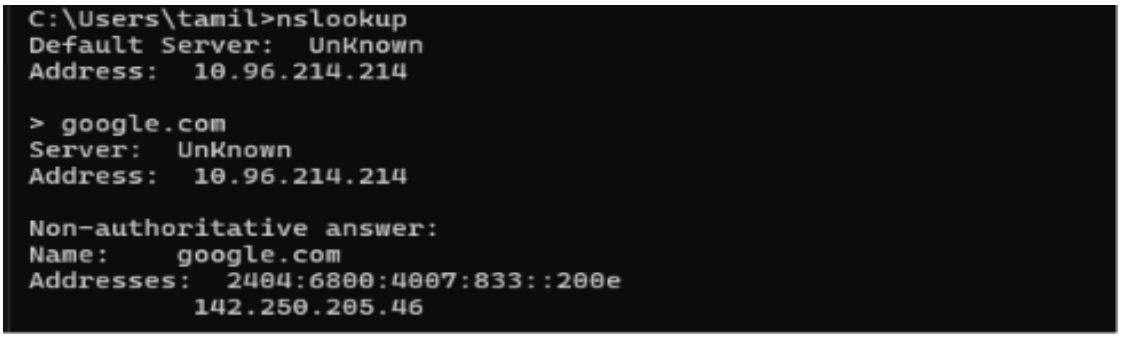
3. **hostname**: Displays the host name of the current computer system.
    - *Syntax:* C:\Users\KARE> hostname



4. **systeminfo**: Displays comprehensive system configuration information, OS version, processor, and network card details.
    - *Syntax:* C:\Users\KARE> systeminfo



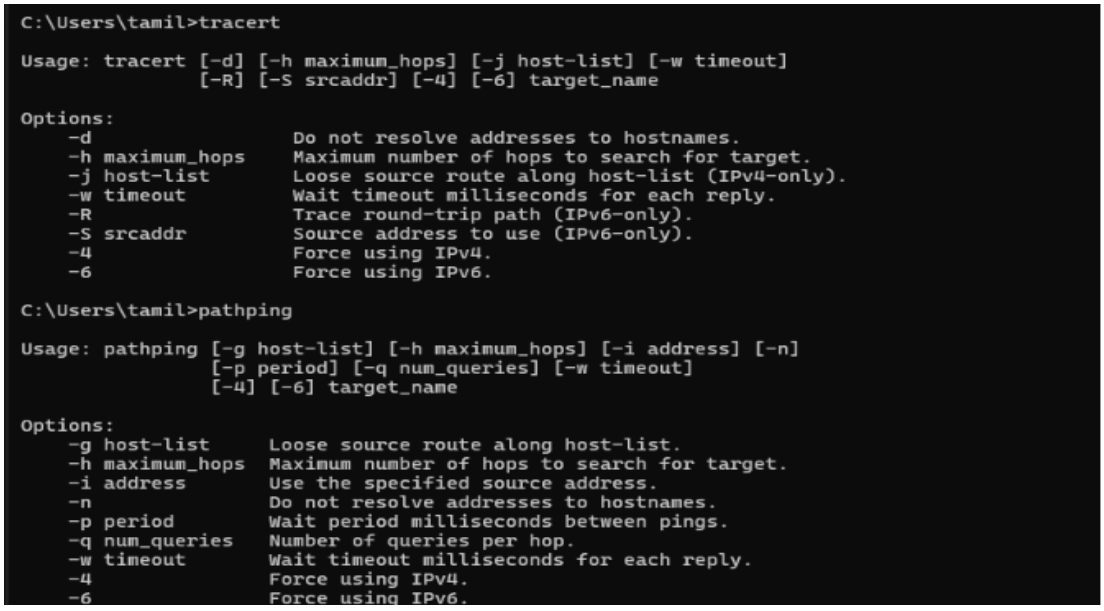
5. **nslookup**: Resolves domain names to their corresponding IP addresses using DNS.
    - *Syntax:* C:\Users\KARE> nslookup <Hostname_or_IP>



6. **ping**: Tests end-to-end reachability to a target destination IP or hostname using ICMP echo request packets.
    - *Syntax:* C:\Users\KARE> ping <IP_Address_or_Hostname>



7. **tracert**: Tracks the network route and lists all router hops taken by packets to reach a destination.
    - *Syntax:* C:\Users\KARE> tracert <IP_Address_or_Hostname>



8. **pathping**: Combines features of ping and tracert to compute packet loss on a per-hop basis along the route.
    - *Syntax:* C:\Users\KARE> pathping <IP_Address_or_Hostname>

```
C:\Users\tamil>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

9. **netstat**: Displays active network connections, listening ports, local/foreign addresses, and connection states.
    - *Syntax:* C:\Users\KARE> netstat

```
C:\Users\tamil>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.96.214.81:53390     155.133.224.22:27037   ESTABLISHED
  TCP    10.96.214.81:53486     135.149.172.176:https  ESTABLISHED
  TCP    10.96.214.81:53610     10.96.214.214:domain   TIME_WAIT
  TCP    10.96.214.81:53612     172.178.240.161:https  TIME_WAIT
  TCP    10.96.214.81:53624     a23-195-74-43:http     TIME_WAIT
  TCP    10.96.214.81:53625     52.168.117.175:https   TIME_WAIT
  TCP    10.96.214.81:53627     a104-94-18-50:http     TIME_WAIT
```

10. **getmac**: Displays the Media Access Control (MAC) addresses for all network adapters.

    - *Syntax:* C:\Users\KARE> getmac

```
C:\Users\tamil>getmac

Physical Address    Transport Name
=================== ============================================================
58-CD-C9-93-67-6A   Media disconnected
58-CD-C9-93-67-69   \Device\Tcpip_{92EBF1F4-D856-44AC-974E-DAC58E2BF303}
```

11. **arp**: Displays and modifies the IP-to-MAC address translation tables.

    - *Syntax:* C:\Users\KARE> arp -a

```
C:\Users\tamil>ARP

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
```

12. **route**: Manages and displays manual entries in the network routing tables.

    - *Syntax:* C:\Users\KARE> route print

```
C:\WINDOWS\system32\cmd.

(c) Microsoft Corporation. All rights reserved.

C:\Users\tamil>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.
```

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

RESULT:

Thus the Study of Networking commands has been done successfully.

Ex.No:	4a
Name of the Experi ment	Configuration of Intra VLAN network
Date	

OBJECTIVE(S):

To design and implement Intra VLAN using switch configuration

INTRODUCTION:

A VLAN is a group of devices on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments. Because VLANs are based on logical instead of physical connections, they are extremely flexible. VLANs define broadcast domains in a Layer 2 network. A broadcast domain is the set of all devices that will receive broadcast frames originating from any device within the set. Broadcast domains are typically bounded by routers because routers do not forward broadcast frames. Layer 2 switches create broadcast domains based on the configuration of the switch. Switches are multiport bridges that allow you to create multiple broadcast domains. Each broadcast domain is like a distinct virtual bridge within a switch.

HARDWARE & SOFTWARE REQUIREMENTS

- **Software:** Cisco Packet Tracer Simulation Tool
- **Devices:**
    a. **End Devices:** PC1, PC2, PC3, PC4, PC5, PC6, PC7, PC8
    b. **Networking Devices:** PT Switch (2 Units: S1 and S2)
    c. **Connections:** Copper Straight-Through cables, Copper Cross-Over cable
    d. **Cables**: Copper Straight-Through Wires

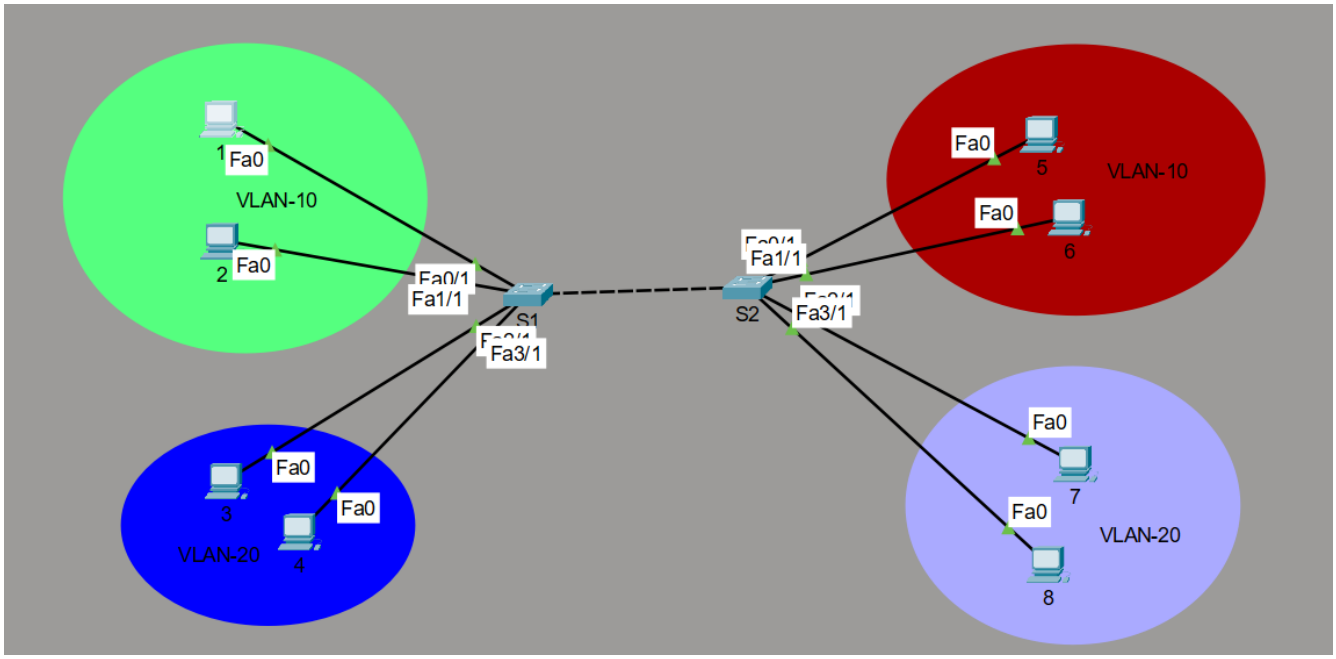
CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask
PC 1	FastEthernet0	192.168.10.1	255.255.255.0
PC 2	FastEthernet0	192.168.10.2	255.255.255.0
PC 3	FastEthernet0	192.168.20.1	255.255.255.0
PC 4	FastEthernet0	192.168.20.2	255.255.255.0
PC 5	FastEthernet0	192.168.10.3	255.255.255.0
PC 6	FastEthernet0	192.168.10.4	255.255.255.0
PC 7	FastEthernet0	192.168.20.3	255.255.255.0
PC 8	FastEthernet0	192.168.20.4	255.255.255.0
S1	Fa0/1 to Fa4/1	*Layer 2 (No IP)*	*N/A*
S2	Fa0/1 to Fa4/1	*Layer 2 (No IP)*	*N/A*

PROCEDURE

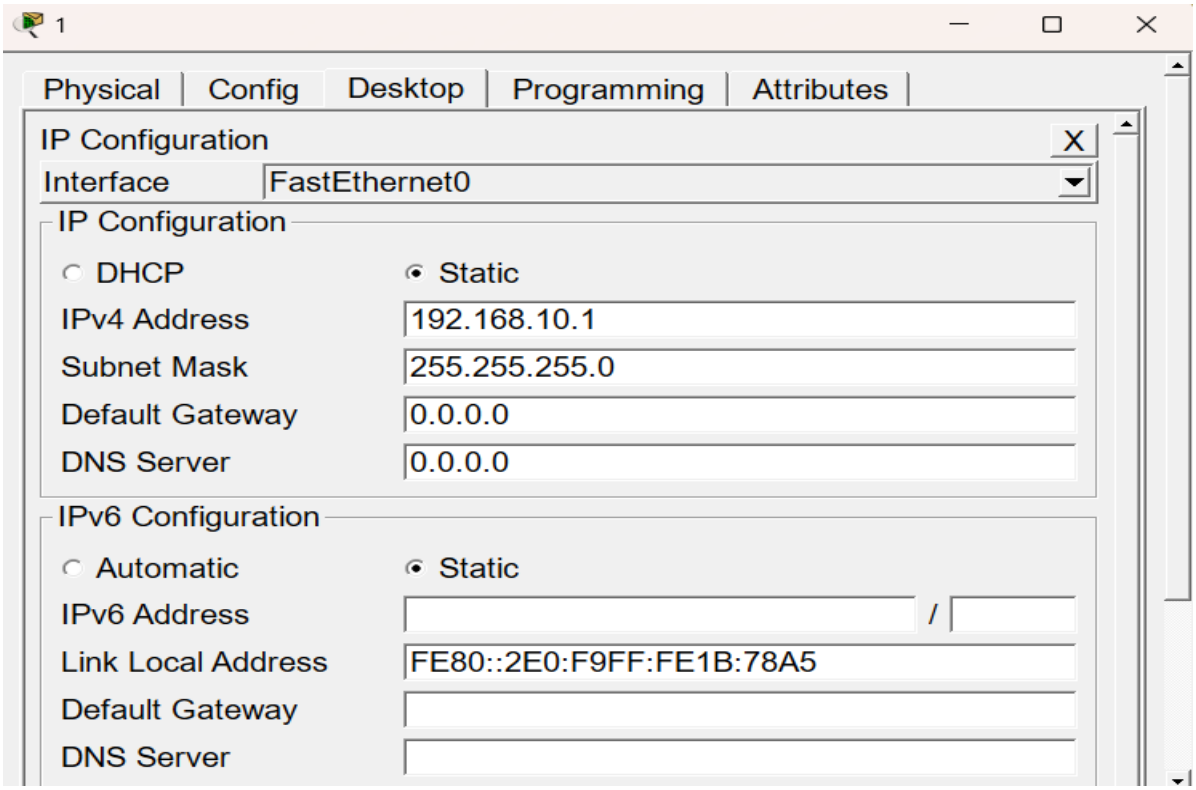
1. **Building the Topology:**
    - Launch Cisco Packet Tracer.
    - Drag and drop eight Generic PCs (PC1 through PC8) and two PT Switches (S1 and S2) onto the workspace.
    - Connect PC1, PC2 (VLAN 10), and PC3, PC4 (VLAN 20) to Switch S1 using Copper Straight-Through cables on interfaces FastEthernet0/1 through FastEthernet3/1.
    - Connect PC5, PC6 (VLAN 10), and PC7, PC8 (VLAN 20) to Switch S2 using Copper Straight-Through cables on interfaces FastEthernet0/1 through FastEthernet3/1.

- Connect Switch S1 to Switch S2 using a Copper Cross-Over cable on interface FastEthernet4/1.

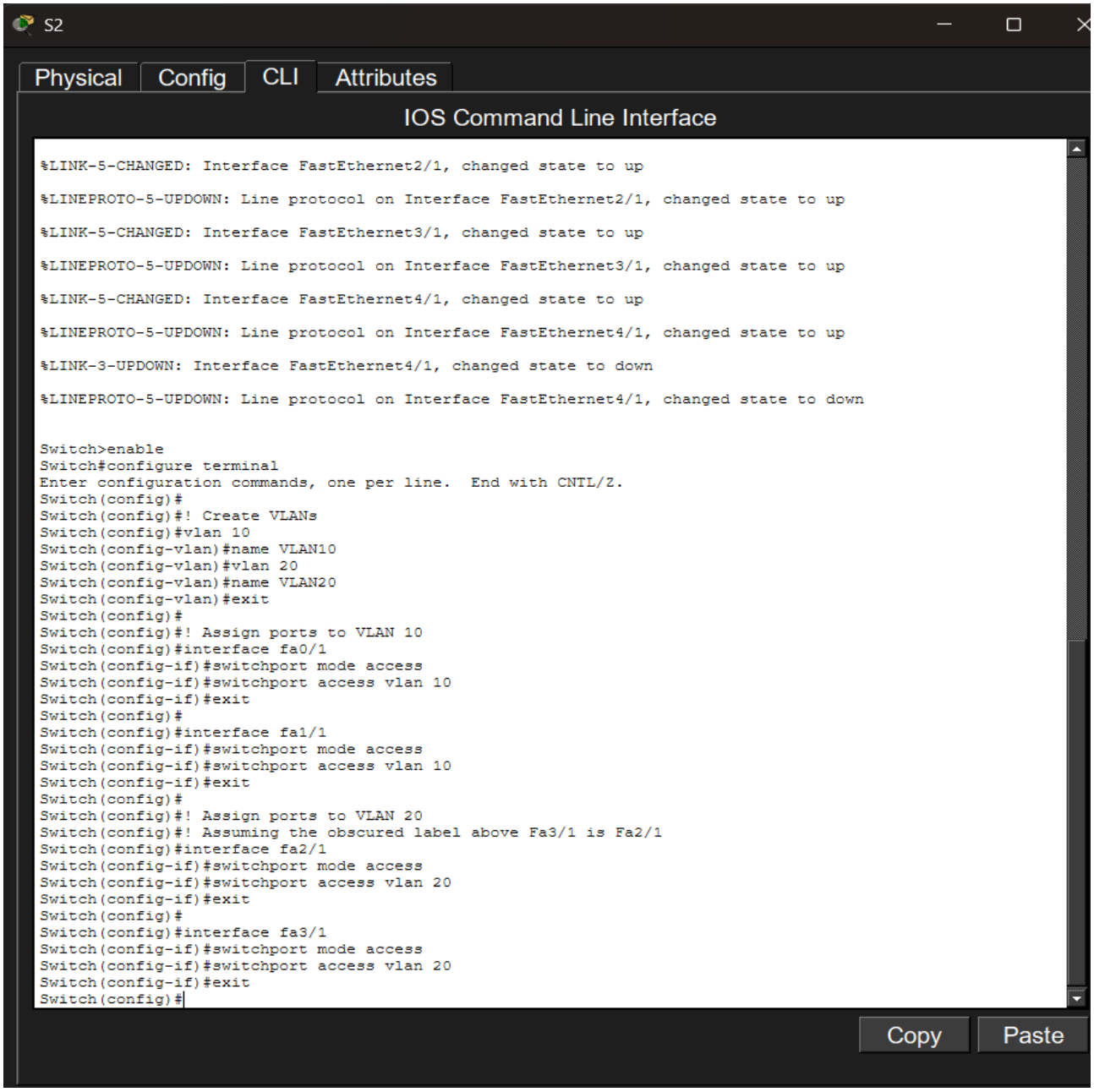


2. IP Configuration:

3. Click on PC1 -> Navigate to Desktop -> Open IP Configuration. Select Static and enter IPv4 Address: 192.168.10.1 and Subnet Mask: 255.255.255.0.



- Repeat this step for PC2 (192.168.10.2), PC5 (192.168.10.3), and PC6 (192.168.10.4) for VLAN 10.
- Click on PC3 -> Navigate to Desktop -> Open IP Configuration. Select Static and enter IPv4 Address: 192.168.20.1 and Subnet Mask: 255.255.255.0.
- Repeat this step for PC4 (192.168.20.2), PC7 (192.168.20.3), and PC8 (192.168.20.4) for VLAN 20.



4. **Step-by-Step Configuration Steps (Executed on both S1 and S2):**

- **Create VLANs:** Open the CLI on the switch and enter Global Configuration Mode to create the VLAN database.

Cisco CLI

Switch> enable

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name VLAN10

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name VLAN20

Switch(config-vlan)# exit

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa1/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa2/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan20
                                         ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa3/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchportaccess vlan 20
                              ^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#
```

- **Configure Interfaces:** Assign the specific switch ports connected to the PCs to their respective VLANs.

  Cisco CLI

  ```
  ! Configuring VLAN 10 Access Ports
  Switch(config)# interface fa0/1
  Switch(config-if)# switchport mode access
  Switch(config-if)# switchport access vlan 10
  Switch(config-if)# exit

  Switch(config)# interface fa1/1
  Switch(config-if)# switchport mode access
  Switch(config-if)# switchport access vlan 10
  Switch(config-if)# exit

  ! Configuring VLAN 20 Access Ports
  Switch(config)# interface fa2/1
  Switch(config-if)# switchport mode access
  Switch(config-if)# switchport access vlan 20
  Switch(config-if)# exit

  Switch(config)# interface fa3/1
  Switch(config-if)# switchport mode access
  Switch(config-if)# switchport access vlan 20
  Switch(config-if)# exit
  ```

- **Configure Trunking:** Configure the interface connecting the two switches (Fa4/1) to carry multiple VLANs.

  Cisco CLI

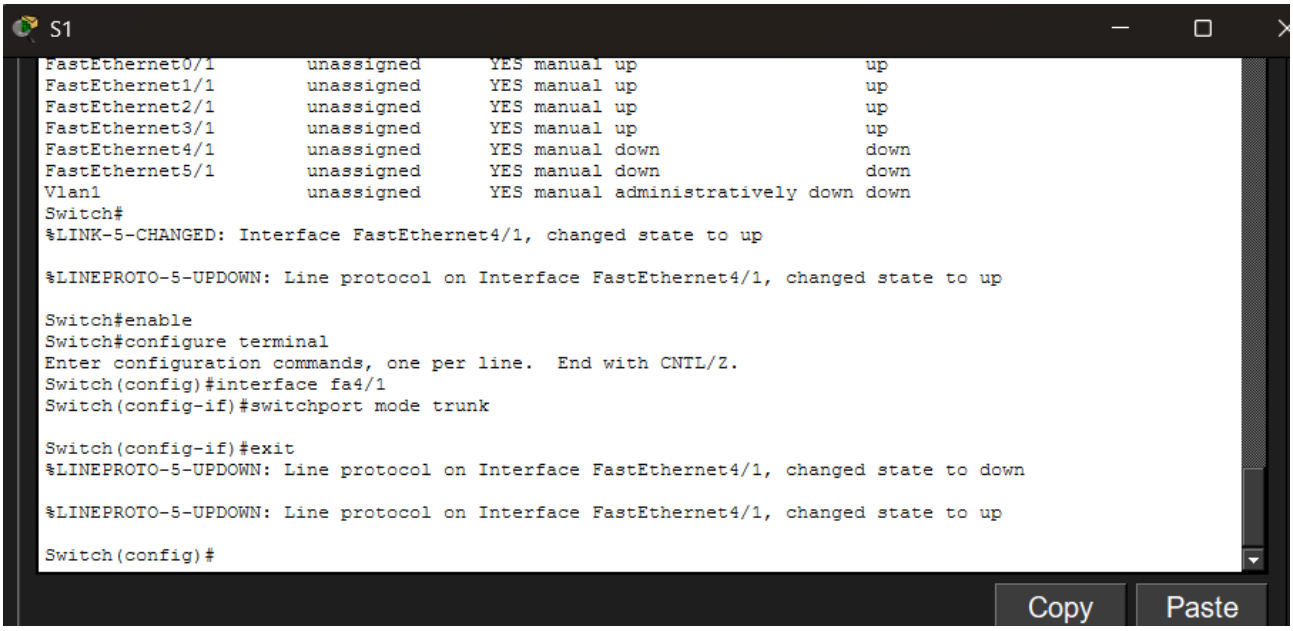
  Switch(config)# interface fa4/1

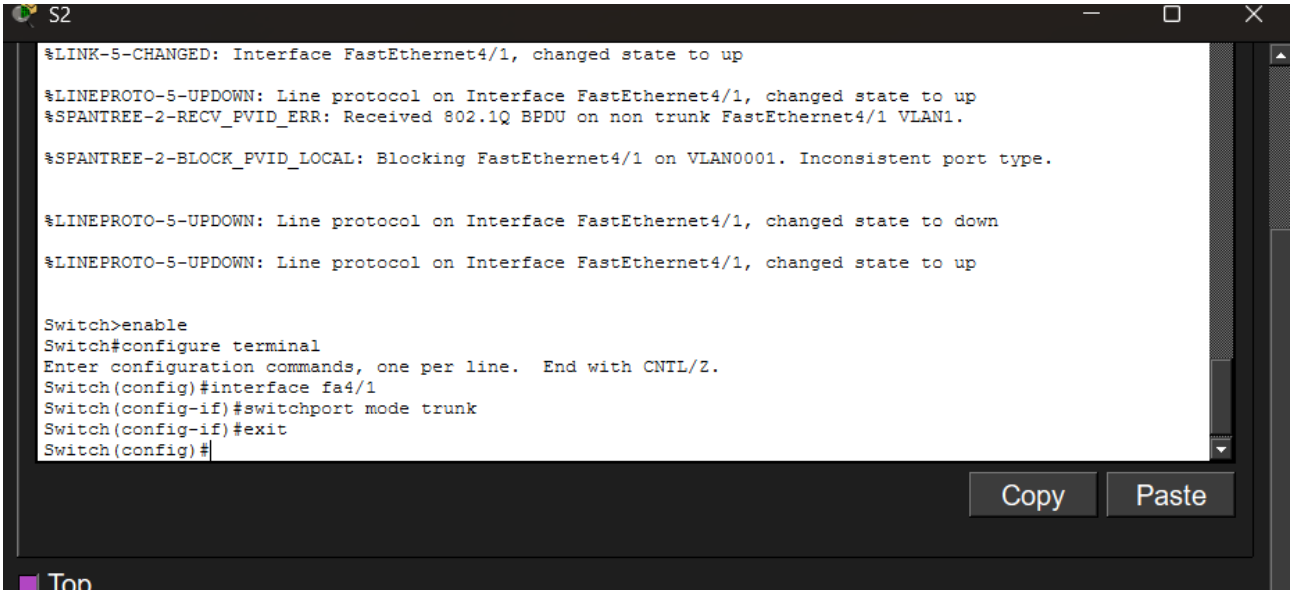
  Switch(config-if)# switchport mode trunk

  Switch(config-if)# exit

  Switch(config)# exit

  Switch# copy running-config startup-config



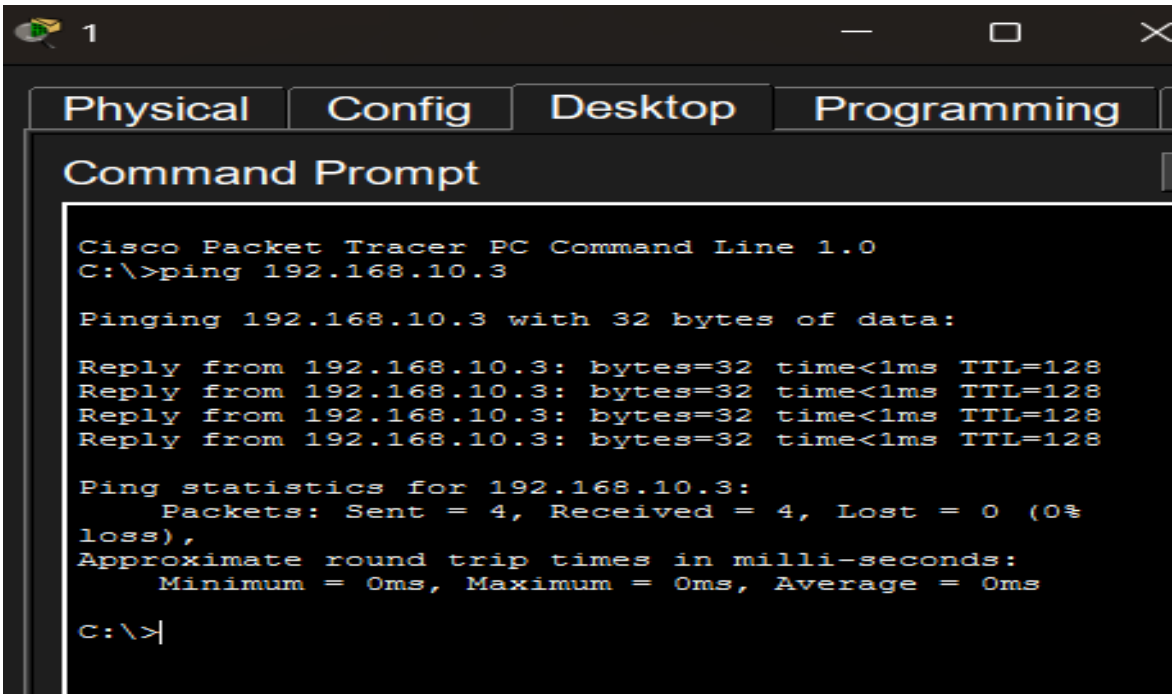


5. **Executing Commands & Connectivity Verification:**

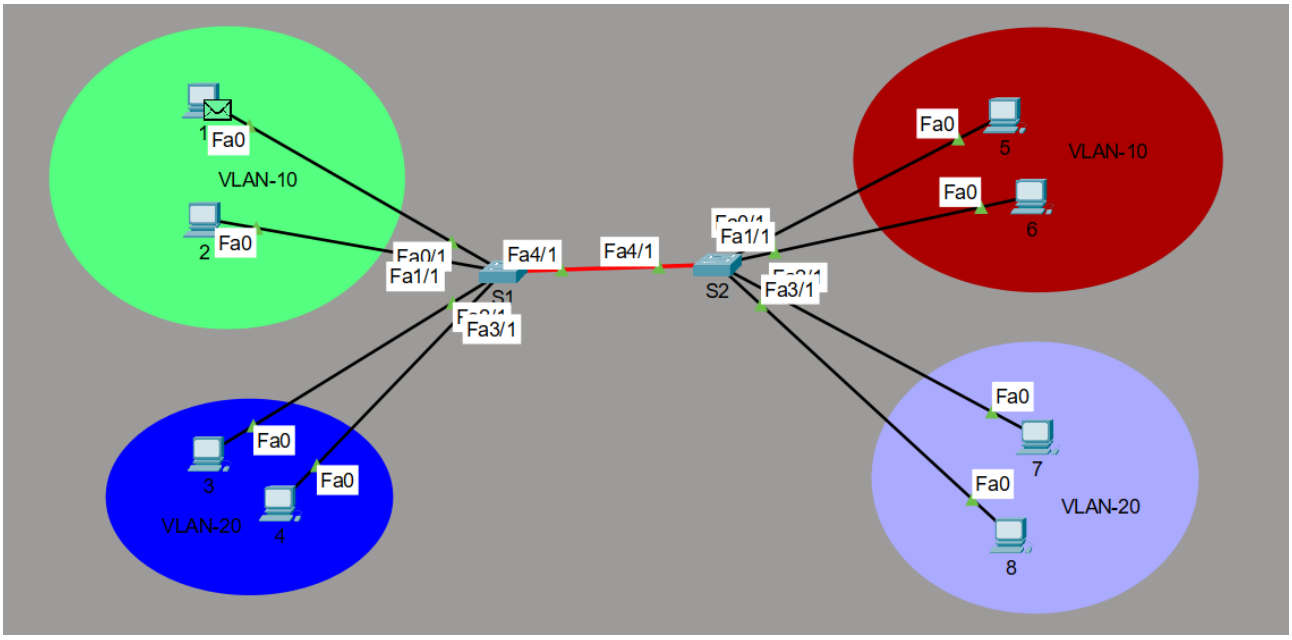
- Open the Command Prompt on PC1 (VLAN 10).
- Run ipconfig to verify local IP settings.
- Run ping 192.168.10.3 to verify ICMP reachability to PC5 across the trunk link.
- Open the Command Prompt on PC3 (VLAN 20).
- Run ping 192.168.20.4 to verify ICMP reachability to PC8 across the trunk link.

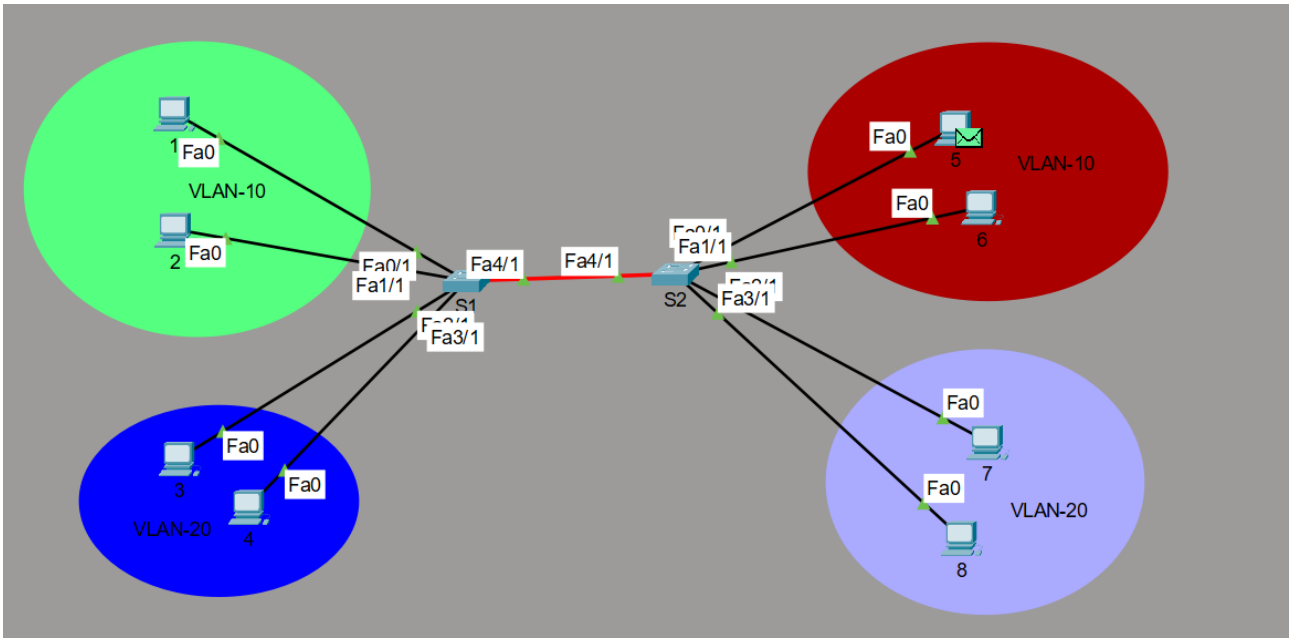
COMMAND EXECUTION DETAILS

- vlan : Creates a Virtual Local Area Network with the specified ID on the switch.
- switchport mode access: Puts the interface into permanent non-trunking mode, meant for connecting end devices.
- switchport access vlan : Assigns the specific access port to a designated VLAN.
- switchport mode trunk: Configures the interface as a trunk link, allowing traffic from multiple VLANs to pass between switches.
- ping : Sends ICMP Echo Request packets to check network-layer connectivity with another host.



- ipconfig: Displays the IP address, Subnet Mask, and Default Gateway of the active interface.
- show vlan brief: Displays the current VLAN database and the ports assigned to each VLAN.
- show interfaces trunk: Displays information about all trunk ports on the switch and the VLANs allowed across them.





Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION:

The study concludes that Cisco Packet Tracer is a robust and user-friendly tool for configuration of Intra VLAN networks. It plays a critical role in preparing individuals for industry certifications and real-world network management challenges. Its capabilities, combined with its accessibility, make it a cornerstone in the toolkit of network engineers and educators.

RESULT:

Thus, the design and configuration of the Intra VLAN network has been done successfully.

Ex.No:	4b
Name of the Experi ment	a) Configuration of Inter VLAN network using L3 switch
Date	

OBJECTIVE(S):

To design and implement Inter VLAN using switch configuration

INTRODUCTION:

Normally, Routers are used to divide the broadcast domain and switches (at layer 2) Operate in a single broadcast domain but Switches can also divide the broadcast domain by using the concept of VLAN (Virtual LAN).

VLAN is the logical grouping of devices in the same or different broadcast domains. By default, all the switch ports are in VLAN 1. As the single broadcast domain is divided into multiple broadcast domains, Routers or layer 3 switches are used for intercommunication between the different VLANs.The process of intercommunication of the different Vlans is known as Inter Vlan Routing (IVR).

Suppose we have made 2 logical groups of devices (VLAN) named sales and finance. If a device in the sales department wants to communicate with a device in the finance department, inter-VLAN routing has to be performed. These can be performed by either router or layer 3 switches.

Switch Virtual Interface (SVI): SVI is a logical interface on a multilayer switch that provides layer 3 processing for packets to all switch ports associated with that VLAN. A single SVI can be created for a VLAN. SVI on the layer 3 switch provides both management and routing services while SVI on layer 2 switch provides only management services like creating VLANs or telnet/SSH services.

Process of Inter Vlan Routing by Layer 3 Switch: The SVI created for the respective VLAN acts as a default gateway for that VLAN just like the sub-interface of the router (in the process of Router On a stick). If the packet is to be delivered to different VLANs i.e inter VLAN Routing is to be performed on the layer 3 switch then first the packet is delivered to the layer 3 switch and then to the destination just like in the process of the router on a stick.

HARDWARE & SOFTWARE REQUIREMENTS

- **Software:** Cisco Packet Tracer Simulation Tool

- **Devices:**
  - **End Devices:** PC1, PC2, PC3, PC4, PC5, PC6, PC7, PC8
  - **Networking Devices:** PT Switch (2 Units: S1 and S2)
  - **Connections:** Copper Straight-Through cables, Copper Cross-Over cable

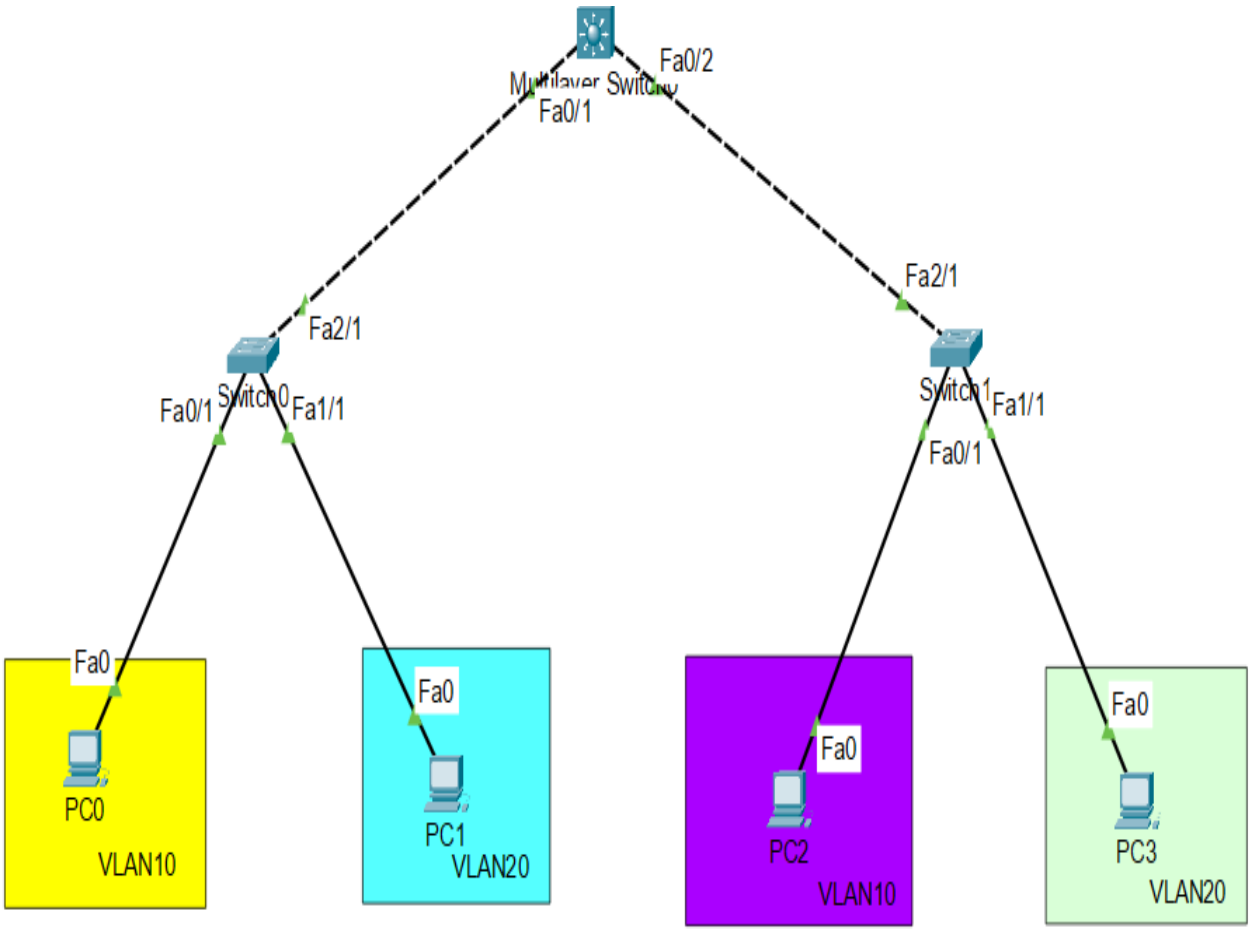
CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask
PC0	fa 0/1	192.168.10.1	255.255.255.0
PC1	fa 0/2	192.168.20.1	255.255.255.0
PC2	fa 0/1	192.168.10.2	255.255.255.0
PC3	fa 0/2	192.168.20.2	255.255.255.0
Switch0	Fa0/1, Fa1/1, Fa2/1	*Layer 2 (No IP)*	*N/A*
Switch1	Fa0/1, Fa1/1, Fa2/1	*Layer 2 (No IP)*	*N/A*
Multilayer Switch0	Fa0/1, Fa0/2	Trunk Links	N/A

PROCEDURE

1. Building the Topology:

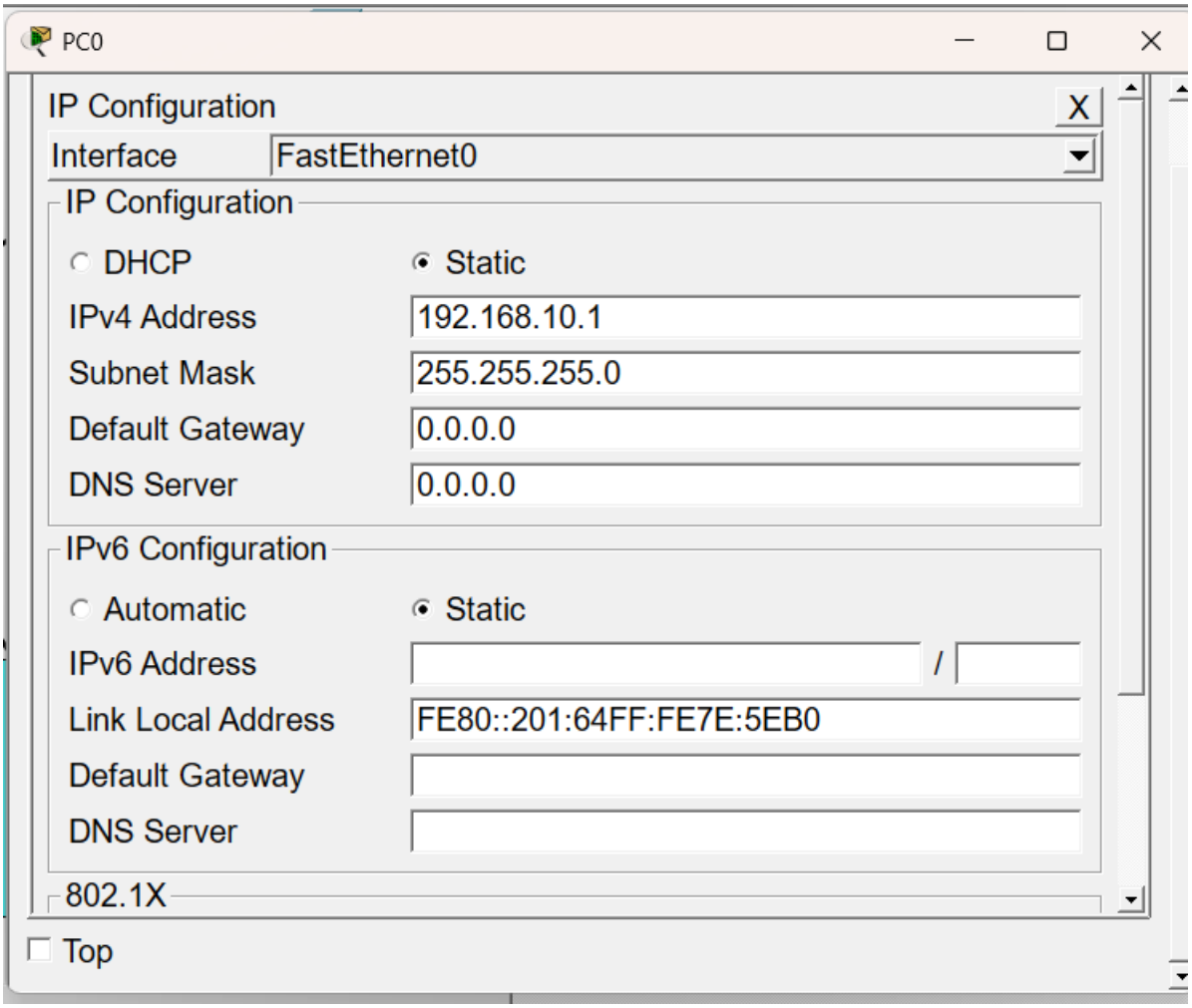
- Drag and drop four Generic PCs (PC0, PC1, PC2, PC3), two Layer 2 Switches (Switch0, Switch1), and one Multilayer Switch onto the workspace.
- Connect PC0 (Fa0) to Switch0 (Fa0/1) using a Copper Straight-Through cable.
- Connect PC1 (Fa0) to Switch0 (Fa1/1) using a Copper Straight-Through cable.
- Connect PC2 (Fa0) to Switch1 (Fa0/1) using a Copper Straight-Through cable.
- Connect PC3 (Fa0) to Switch1 (Fa1/1) using a Copper Straight-Through cable.
- Connect Switch0 (Fa2/1) to Multilayer Switch (Fa0/1) using a Copper Cross-Over cable.
- Connect Switch1 (Fa2/1) to Multilayer Switch (Fa0/2) using a Copper Cross-Over cable.



2. IP Configuration:

- Click on PC0 -> Desktop -> IP Configuration. Select Static and enter IPv4 Address: 192.168.10.1, Subnet Mask: 255.255.255.0.

- Click on PC1 -> Desktop -> IP Configuration. Select Static and enter IPv4 Address: 192.168.20.1, Subnet Mask: 255.255.255.0.
- Click on PC2 -> Desktop -> IP Configuration. Select Static and enter IPv4 Address: 192.168.10.2, Subnet Mask: 255.255.255.0.
- Click on PC3 -> Desktop -> IP Configuration. Select Static and enter IPv4 Address: 192.168.20.2, Subnet Mask: 255.255.255.0.



3. Step-by-Step Configuration:

SWITCH-0

Cisco CLI

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#interface fa0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#exit

Switch(config)#interface fa0/2

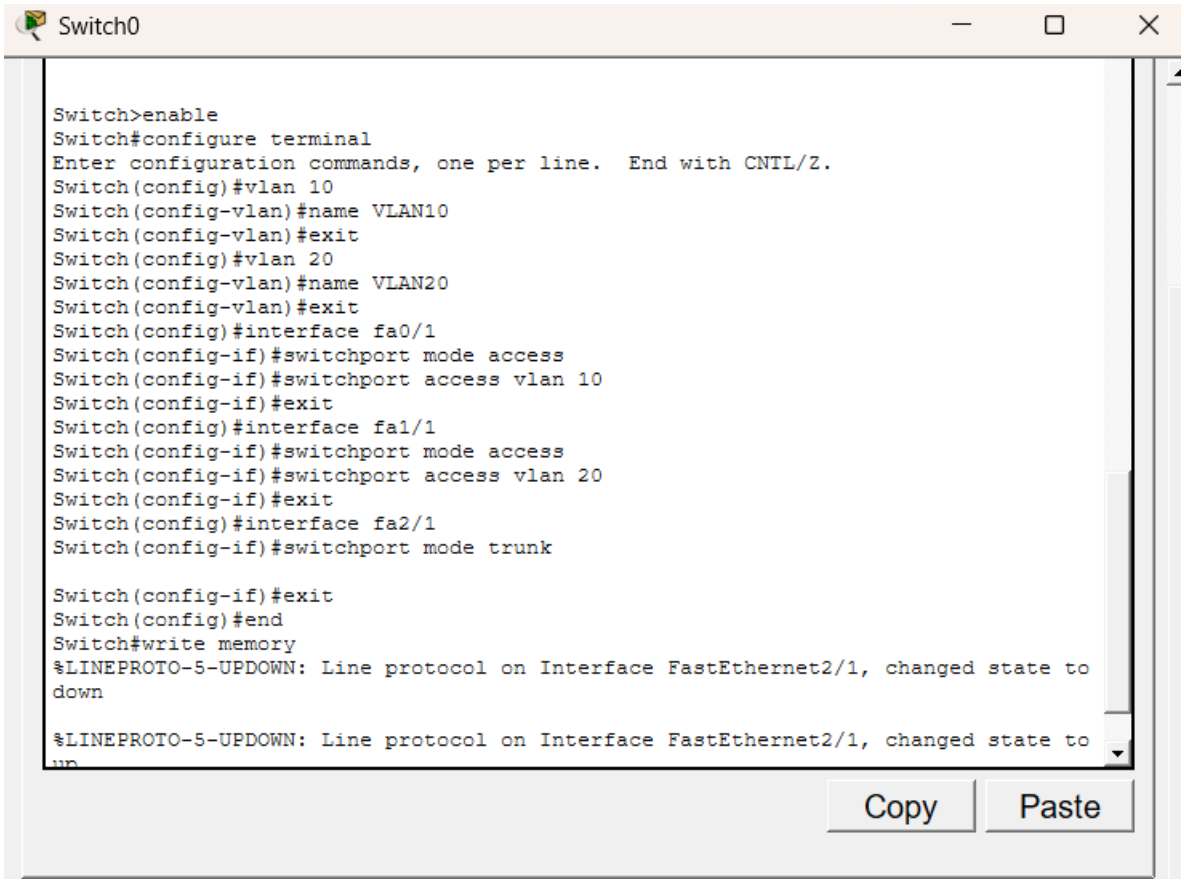
Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#interface fa0/3

Switch(config-if)#switchport mode trunk



SWITCH-1

Cisco CLI

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#interface fa0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#exit

Switch(config)#interface fa0/2

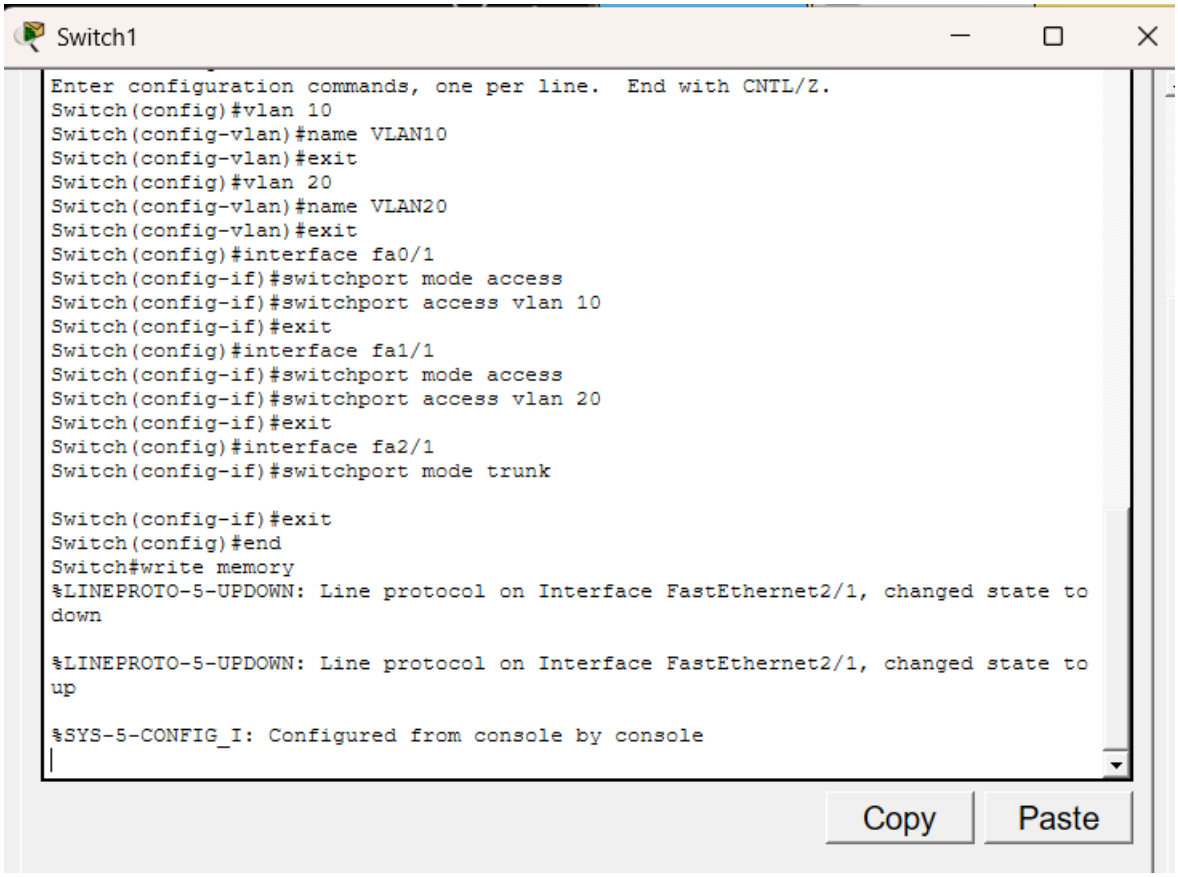
Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#interface fa0/3

Switch(config-if)#switchport mode trunk



MULTI LAYER SWITCH 0

Cisco CLI

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#interface fa0/1

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

Switch(config)#interface fa0/2

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

Switch(config)#ip routing

Switch(config)#interface vlan 10

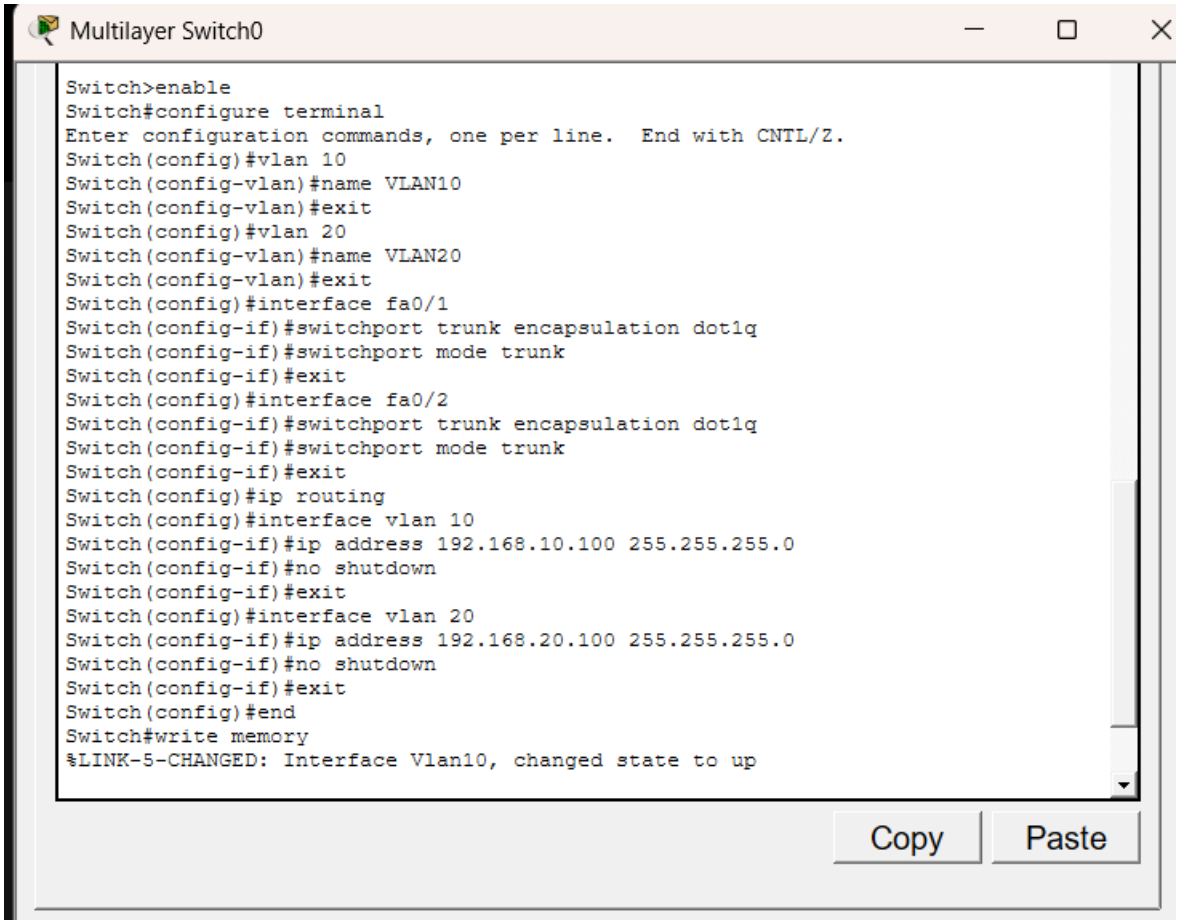
Switch(config-if)#ip address 192.168.10.100 255.255.255.0

Switch(config-if)#exit

Switch(config)#interface vlan 20

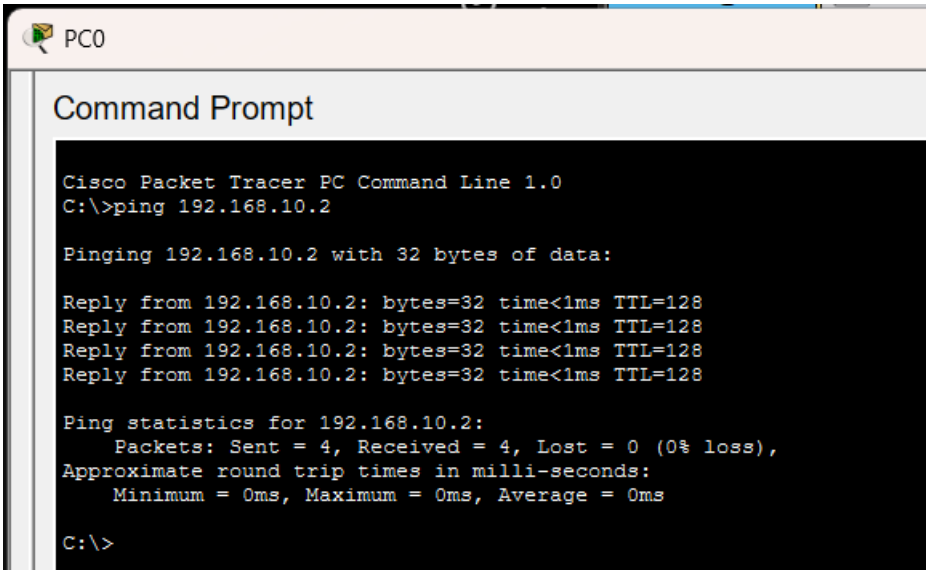
Switch(config-if)#ip address 192.168.20.100 255.255.255.0

Switch(config-if)#exit



4. Connectivity Verification:

- Open the Command Prompt on PC0.
- Run ping 192.168.10.2 to verify ICMP reachability to PC2 (within VLAN 10).



COMMAND EXECUTION DETAILS

- vlan <number>: Creates a Virtual Local Area Network with the specified ID on the switch.
- switchport mode access: Puts the interface into permanent non-trunking mode, meant for connecting end devices.
- switchport access vlan <number>: Assigns the specific access port to a designated VLAN.
- switchport trunk encapsulation dot1q: Specifies IEEE 802.1Q encapsulation for the trunk link on a multilayer switch. This must be set before changing the mode to trunk on older L3 switches.
- switchport mode trunk: Configures the interface as a trunk link, allowing traffic from multiple VLANs to pass between switches.
- ip routing: Enables Layer 3 IP routing functions on the multilayer switch for Inter-VLAN communication.
- interface vlan <number>: Creates a logical Switch Virtual Interface (SVI) to act as the default gateway for a specific VLAN.
- ip address <ip> <mask>: Assigns the IP address and subnet mask to an interface or SVI.
- ipconfig: Displays the IP address, Subnet Mask, and Default Gateway of the active interface on an end device.
- ping <IP>: Sends ICMP Echo Request packets to check network-layer connectivity with another host across different VLANs.

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION:

A Layer 3 switch for Inter-VLAN routing provides a cost-effective and highly efficient way to enable communication between different VLANs within a network without relying on a separate external router. Implementing VLANs allows for the logical segmentation of the network infrastructure, which enhances security and significantly reduces unnecessary broadcast traffic. Ultimately, the experiment successfully demonstrates that Cisco Packet Tracer is a robust and user-friendly simulation tool for designing, configuring, and verifying complex Inter-VLAN routing topologies.

RESULT:

Thus the implement of configuration of intervalence network using multi layer- 3 speech has been executed successfully in Cisco Packet

Ex.No:	4b
Name of the Experi ment	b) Configuration of Inter VLAN using Router on a stick method
Date	

OBJECTIVE(S):

To design and implement Inter VLAN routing using the Router on a Stick method.

INTRODUCTION:

'Router on a Stick' allows routing between VLANs with only one interface. Each VLAN represents a different Subnet. In general, routers can take traffic from only one subnet and transfer it to another subnet. And we can assign only one IP Address to a router interface. 'Router on a stick' allow us to create sub-interfaces, and assign IP Addresses to those subinterfaces. To make it work, we have to create a truck connection between the switch and a router so that traffic from multiple VLANs can be sent to the router.

If we create a route between VLANs without the 'Router on a Stick' method, then we have to waste interfaces on the switches and routers. And if we enable routing between multiple VLANs then it will become practically inefficient as the switches and the routers will use those multiple interfaces.

The network consists of multiple VLANs (Virtual Local Area Networks) interconnected through switches and a central router. Each VLAN is represented by a distinct color, with devices assigned specific IP addresses in their respective VLAN subnets.

1. VLAN Segmentation: VLANs are used to logically separate devices within the same physical network, improving security and reducing broadcast traffic. Each VLAN (e.g., VLAN 10, 20, 30) is assigned its subnet range.

2. Switches: The switches connect devices within the same VLAN. Trunk links (dashed lines) between switches allow traffic from multiple VLANs to pass through.

3. Router-on-a-Stick Configuration: The central router facilitates inter-VLAN communication by using sub-interfaces for each VLAN, configured with appropriate IP addresses and VLAN tagging.

4. Devices: PCs are connected to specific switches within VLANs and assigned IP addresses corresponding to their VLAN subnet (e.g., 192.168.x.x).

5. Connectivity: The dashed lines represent connections between switches and the router, indicating trunk links that handle traffic for multiple VLANs. Access ports are used for connecting end devices (e.g., PCs) to specific VLANs.

In the below lab, we will configure 'Router on a Stick' that would allow routing between the VLANs. Some of the important concepts in this lab are – to create sub-interfaces, use encapsulation dot1Q command to encapsulate the traffic, and mentioning the VLAN number to ascertain that for which VLAN the sub-interface should respond.

HARDWARE & SOFTWARE REQUIREMENTS

- **Software:** Cisco Packet Tracer Simulation Tool

- **Devices:**
  - **End Devices**: PC0, PC1, PC2, PC3, PC4, PC5, PC6, PC7, PC8
  - **Networking Devices**: PT Switch (4 Units: Switch 0, Switch 1, Switch 2, Switch 3), PT Router (1 Unit: Router 0)
  - **Connections**: Copper Straight-Through cables, Copper Cross-Over cables

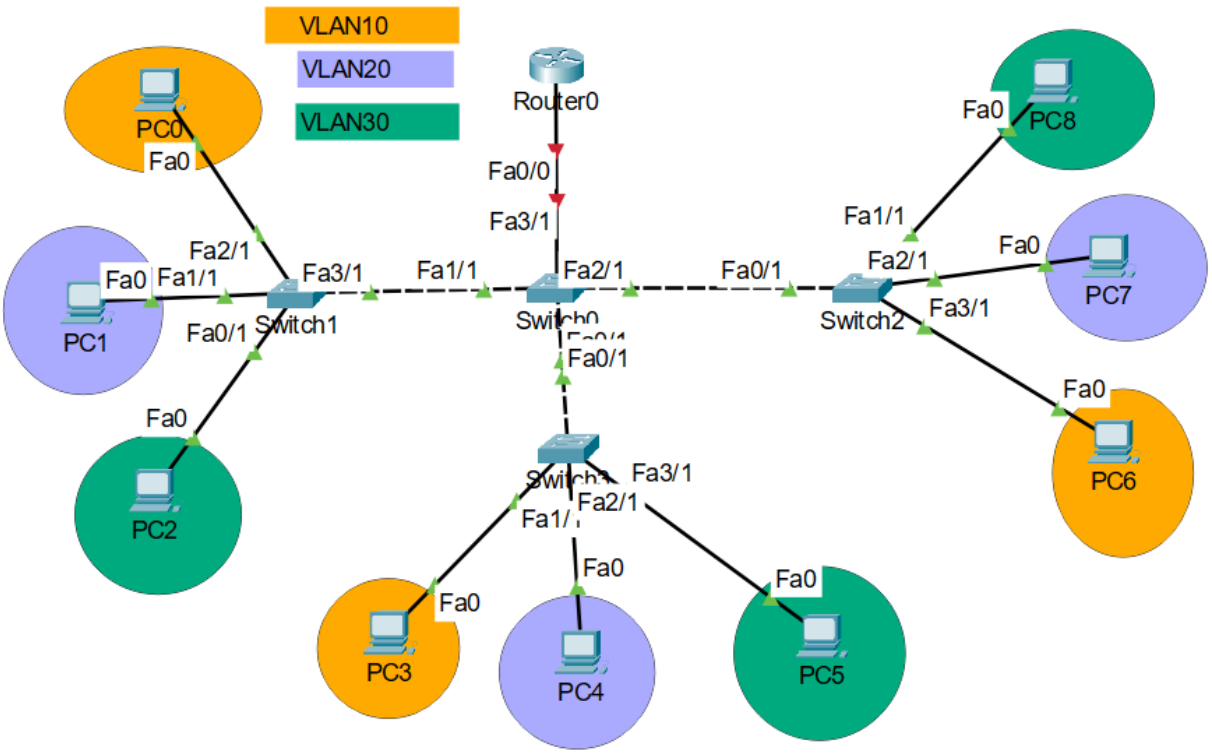
CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask	Default Gateway
PC0 (VLAN 10)	FastEthernet0	192.168.10.1	255.255.255.0	192.168.10.100
PC1 (VLAN 20)	FastEthernet0	192.168.20.1	255.255.255.0	192.168.20.100
PC2 (VLAN 30)	FastEthernet0	192.168.30.1	255.255.255.0	192.168.30.100
PC3 (VLAN 10)	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.100
PC4 (VLAN 20)	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.100
PC5 (VLAN 30)	FastEthernet0	192.168.30.2	255.255.255.0	192.168.30.100
PC6 (VLAN 10)	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.100
PC7 (VLAN 20)	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.100
PC8 (VLAN 30)	FastEthernet0	192.168.30.3	255.255.255.0	192.168.30.100
Switch 1, 2 & 3(Edge)	fa0/1 to fa0/4	Layer 2 (No IP)	N/A	N/A
Switch 0 (Center)	fa0/1 to fa0/3, gig0/1	Layer 2 (No IP)	N/A	N/A
Router 0	gig0/0.10 (Sub-int)	192.168.10.100	255.255.255.0	N/A
Router 0	gig0/0.20 (Sub-int)	192.168.20.100	255.255.255.0	N/A
Router 0	gig0/0.30 (Sub-int)	192.168.30.100	255.255.255.0	N/A

PROCEDURE

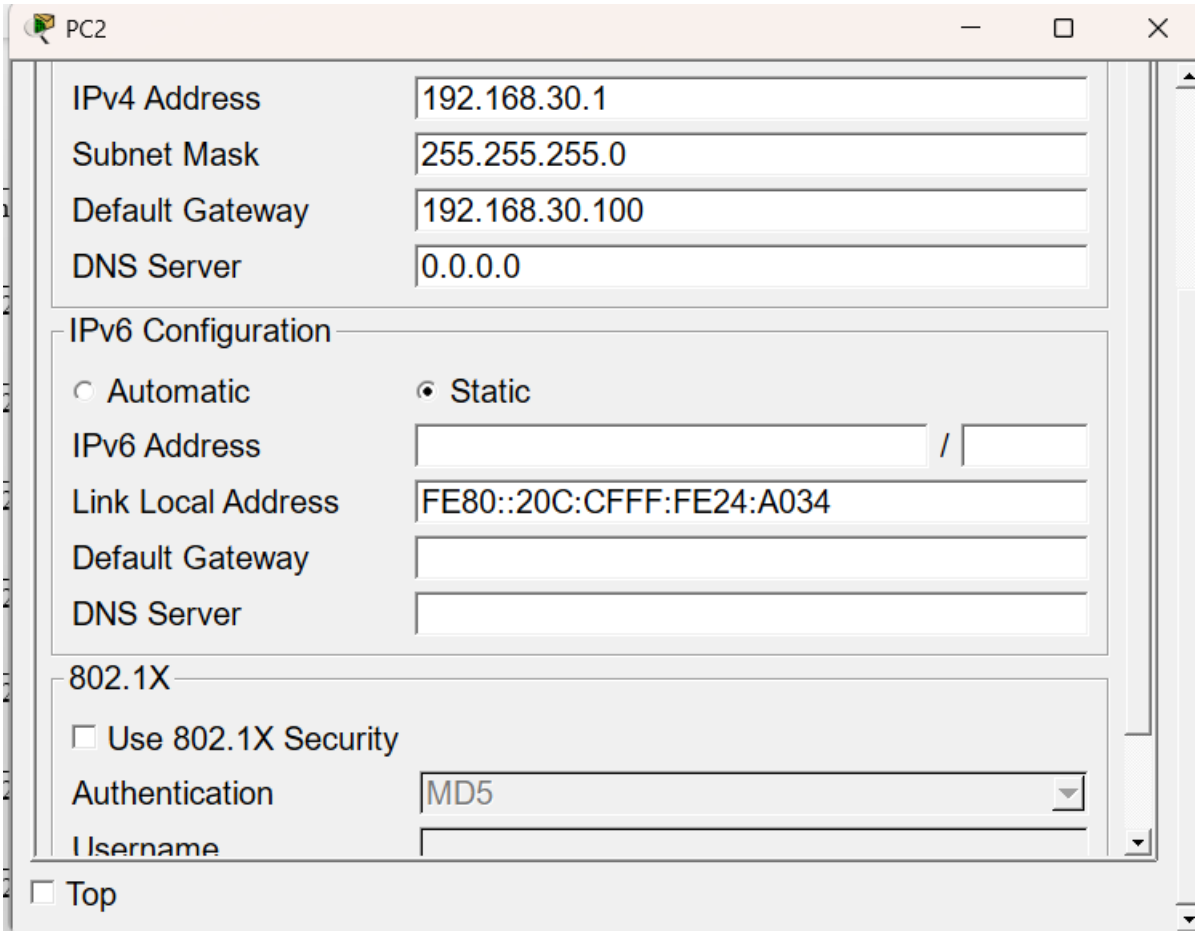
1. Building the Topology:

1. Open Cisco Packet Tracer and drag one Router (Router0), four Layer 2 Switches (Switch0, Switch1, Switch2, Switch3), and nine PCs (PC0 to PC8) onto the workspace.
2. Connect Router0 (Fa0/0) to the center switch, Switch0 (Fa3/1), using a Copper Straight-Through cable.
3. Connect the center Switch0 to the edge switches using Copper Cross-Over cables:
    - Switch0 (Fa1/1) to Switch1 (Fa3/1)
    - Switch0 (Fa2/1) to Switch2 (Fa0/1)
    - Switch0 (Fa0/1) to Switch3 (Fa0/1)
4. Connect the PCs to the edge switches using Copper Straight-Through cables based on their designated VLANs:
    - **Switch1:** PC0 (Fa0) to Fa2/1, PC1 (Fa0) to Fa1/1, PC2 (Fa0) to Fa0/1
    - **Switch2:** PC6 (Fa0) to Fa3/1, PC7 (Fa0) to Fa2/1, PC8 (Fa0) to Fa1/1
    - **Switch3:** PC3 (Fa0) to Fa1/1, PC4 (Fa0) to Fa2/1, PC5 (Fa0) to Fa3/1



2. IP Configuration:

1. Click on PC0 and navigate to **Desktop** -> **IP Configuration**.
2. Select **Static** and input the IP Address (192.168.10.1) and Subnet Mask (255.255.255.0).
3. Input the Default Gateway (192.168.10.100).
4. Repeat steps 1-3 for PC1 through PC8, applying the respective IP addresses, subnet masks, and default gateways based on their assigned VLAN configuration table.



3. Command Line Execution

Router0

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface fa0/0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)#interface fa0/0.10

Router(config-subif)#encapsulation dot1Q 10

Router(config-subif)#ip address 192.168.10.100 255.255.255.0

Router(config-subif)#exit

Router(config)#interface fa0/0.20

Router(config-subif)#encapsulation dot1Q 20

Router(config-subif)#ip address 192.168.20.100 255.255.255.0

Router(config-subif)#exit

Router(config)#interface fa0/0.30

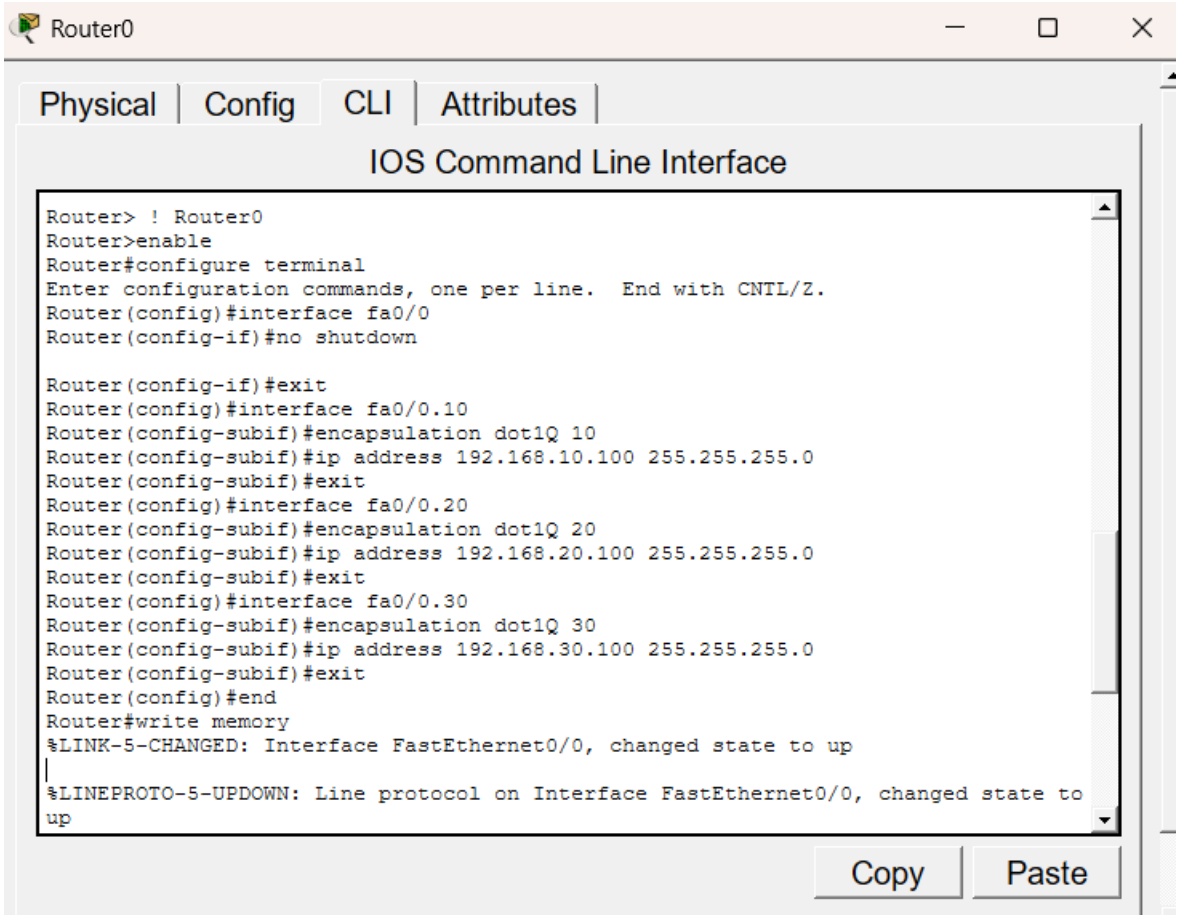
Router(config-subif)#encapsulation dot1Q 30

Router(config-subif)#ip address 192.168.30.100 255.255.255.0

Router(config-subif)#exit

Router(config)#end

Router#write memory



Switch0 (Center)

Cisco CLI

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#vlan 30

Switch(config-vlan)#exit

Switch(config)#interface fa0/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

Switch(config)#interface fa1/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

Switch(config)#interface fa2/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

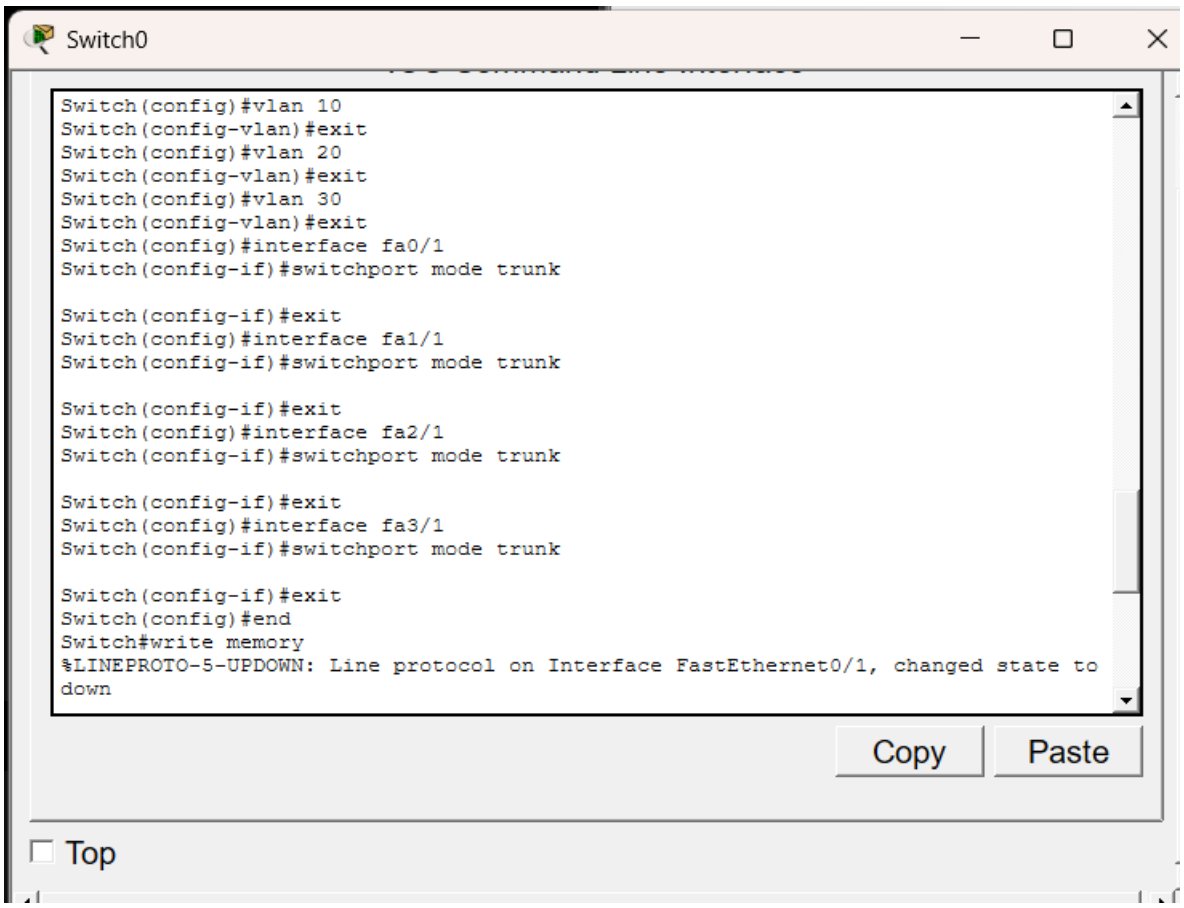
Switch(config)#interface fa3/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

Switch(config)#end

Switch#write memory



Switch1 (Left)

Cisco CLI

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#vlan 30

Switch(config-vlan)#exit

Switch(config)#interface fa2/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#exit

Switch(config)#interface fa1/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#interface fa0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 30

Switch(config-if)#exit

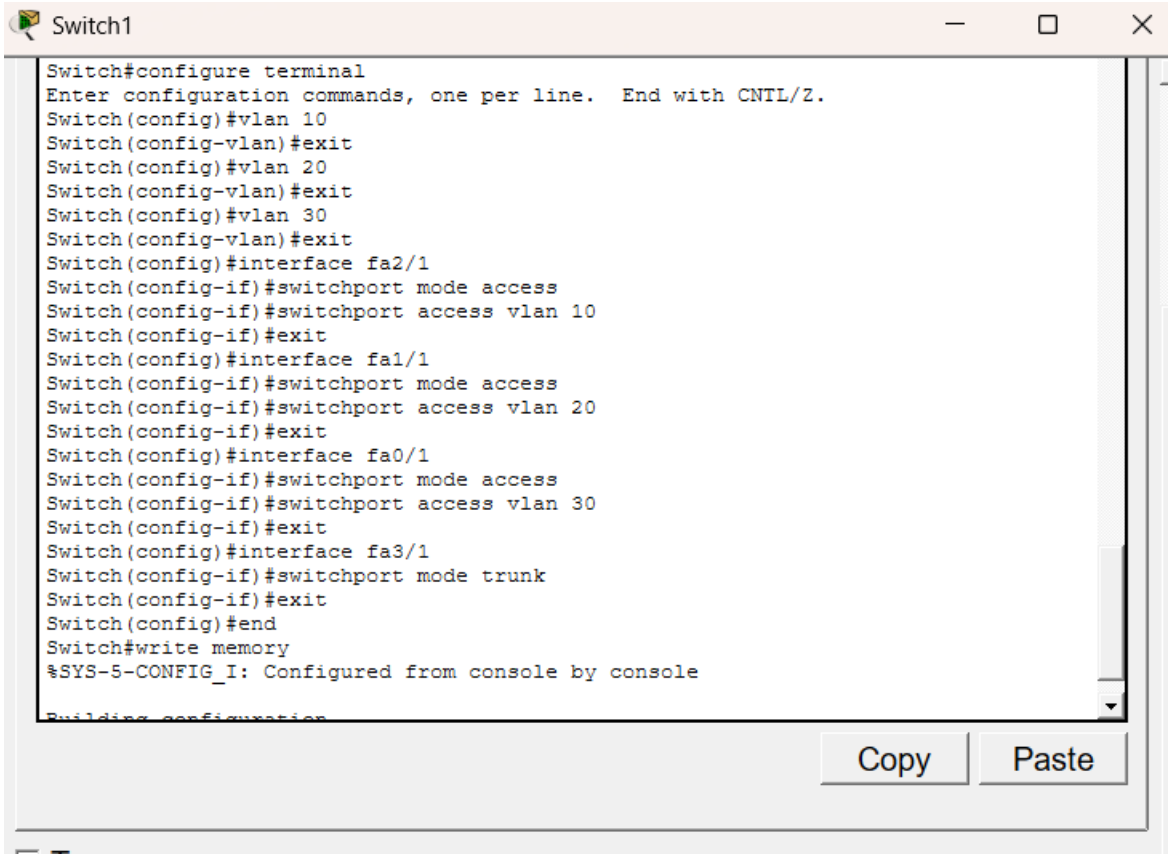
Switch(config)#interface fa3/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

Switch(config)#end

Switch#write memory



Switch2 (Right)

Cisco CLI

Switch>enable

Switch#configure terminal

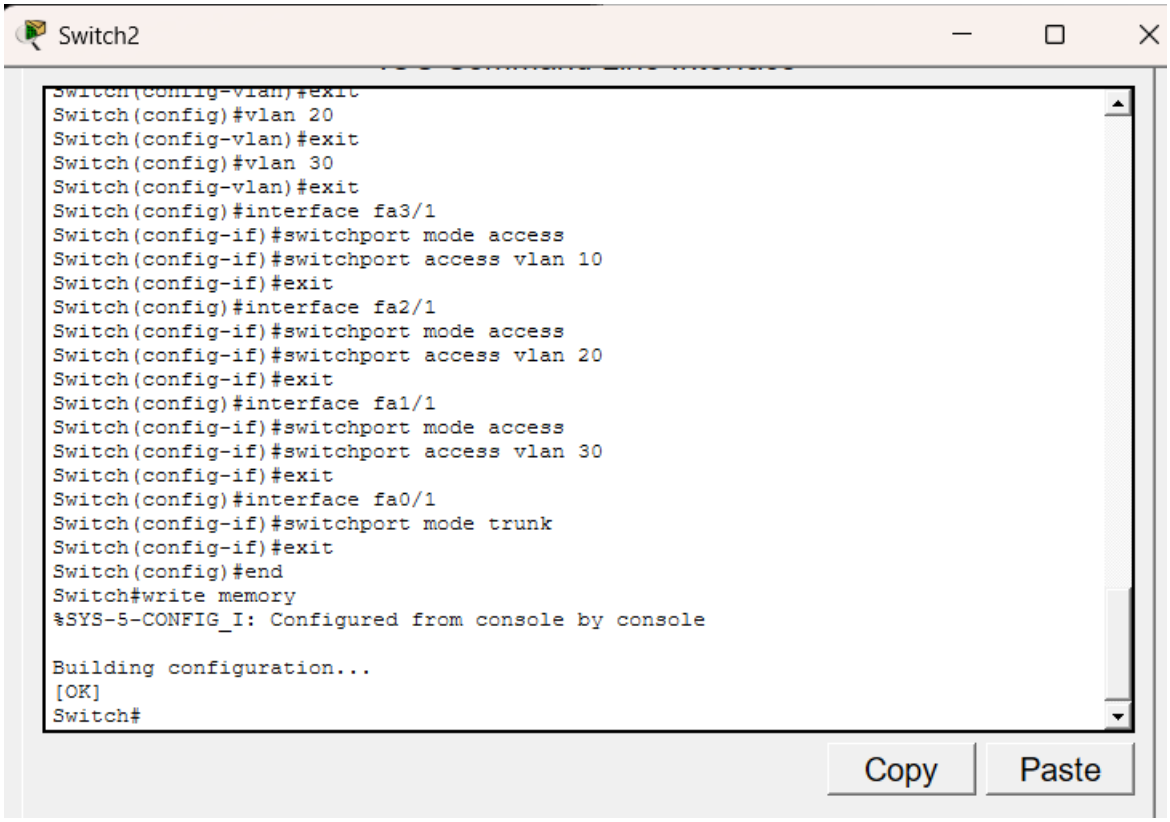
Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

```
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#interface fa3/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa2/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa1/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#write memory
```



Switch3 (Bottom)

Cisco CLI

Switch>enable

Switch#configure terminal

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#vlan 30

Switch(config-vlan)#exit

Switch(config)#interface fa1/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#exit

Switch(config)#interface fa2/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#interface fa3/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 30

Switch(config-if)#exit

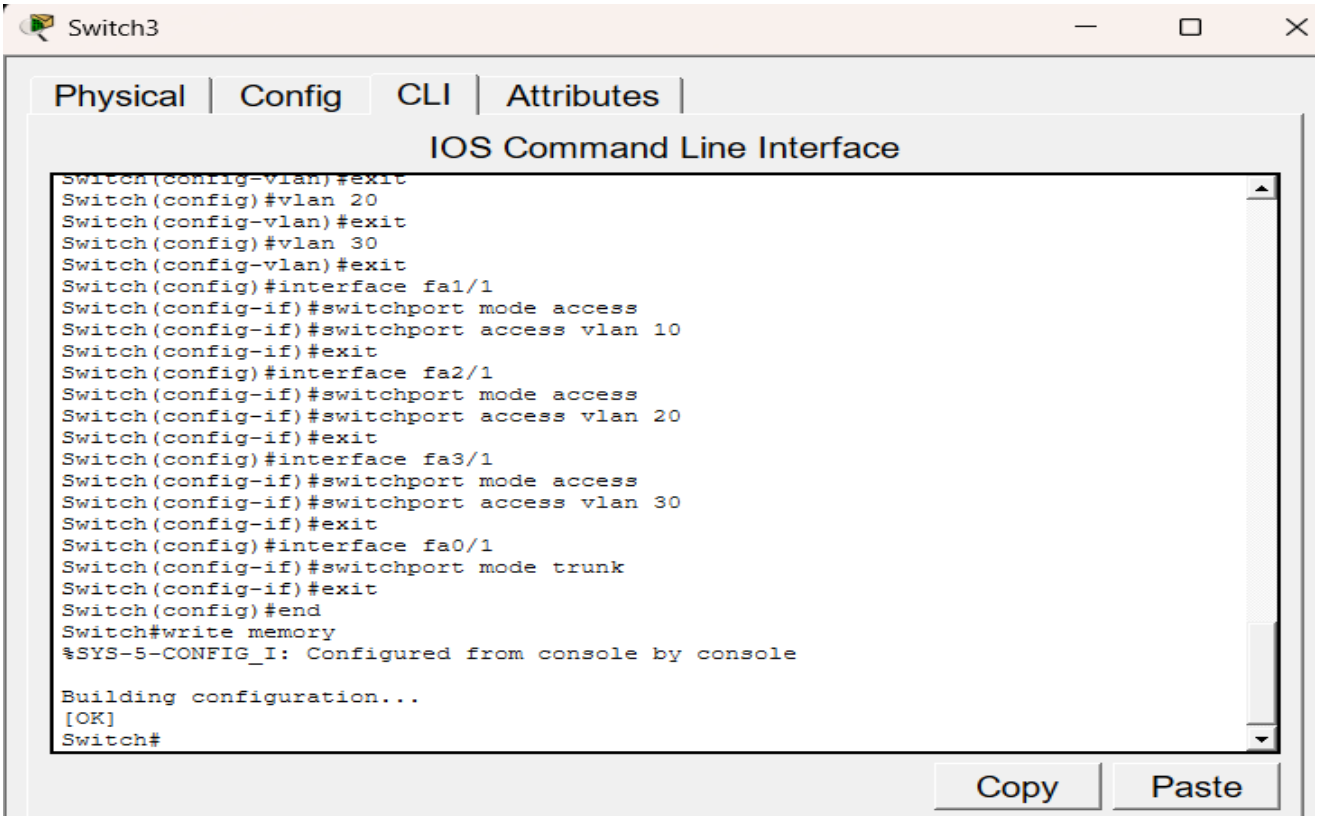
Switch(config)#interface fa0/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#exit

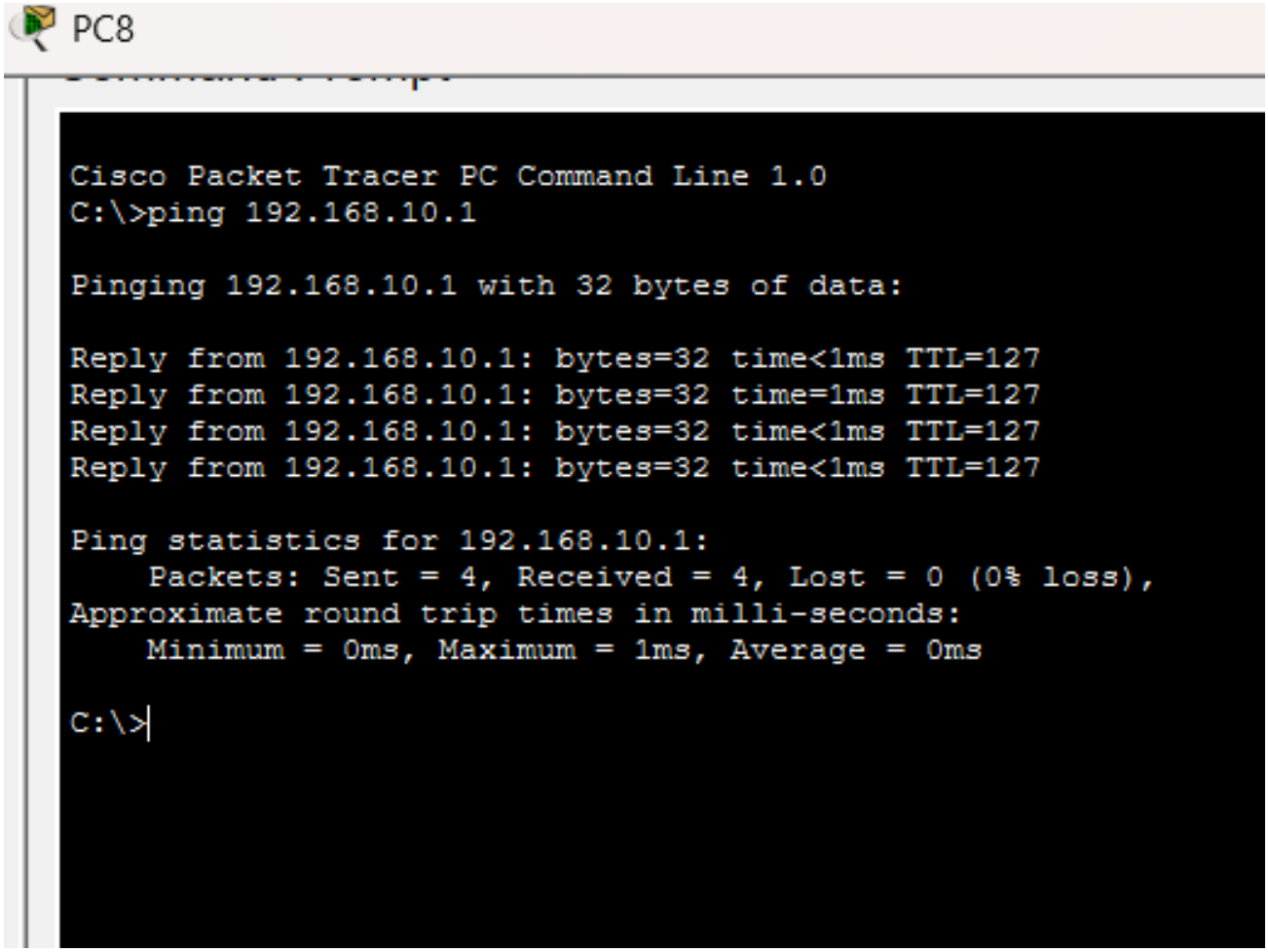
Switch(config)#end

Switch#write memory



.4 Connectivity Verification:

1. Click on PC8, navigate to **Desktop**, and open the **Command Prompt**.
2. Type ping 192.168.10.1 and press Enter to verify successful Inter-VLAN communication with PC1.



COMMAND EXECUTION DETAILS

- vlan <number>: Creates a Virtual Local Area Network with the specified ID on the switch.
- switchport mode access: Puts the interface into permanent non-trunking mode, meant for connecting end devices.
- switchport access vlan <number>: Assigns the specific access port to a designated VLAN.
- switchport trunk encapsulation dot1q: Specifies IEEE 802.1Q encapsulation for the trunk link on a multilayer switch. This must be set before changing the mode to trunk on older L3 switches.
- switchport mode trunk: Configures the interface as a trunk link, allowing traffic from multiple VLANs to pass between switches.
- ip routing: Enables Layer 3 IP routing functions on the multilayer switch for Inter-VLAN communication.
- interface vlan <number>: Creates a logical Switch Virtual Interface (SVI) to act as the default gateway for a specific VLAN.
- ip address <ip> <mask>: Assigns the IP address and subnet mask to an interface or SVI.
- ipconfig: Displays the IP address, Subnet Mask, and Default Gateway of the active interface on an end device.
- ping <IP>: Sends ICMP Echo Request packets to check network-layer connectivity with another host across different VLANs.

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION:

A router is used in Inter-VLAN routing to enable communication between different VLANs by acting as a Layer 3 device that directs traffic based on IP addresses using the Router-on-a-Stick configuration.

RESULT:

Thus the implementation of configuration of Inter VLAN network using router monistic method has been executed successfully in Cisco Packet.

Ex.No:	5a
Name of the Experi ment	Link State Routing
Date	

OBJECTIVE(S):

To design and implement Link state routing using packet tracer.

INTRODUCTION:

Link State Routing Protocols used to select the path for data packet in an internetwork. Link state routing protocols uses link state routers to share information of connected network devices. This is a learning process. By learning process each router maintain the routing table to select the shortest path for data packet transmission. Each router update the network topology to nearby router only. Link state routing protocols are also known as shortest path first protocol. Link state protocols allow routers to share the information about network connected to it. This information passed to neighbour router only. An accurate information of network topology around the router updated in routing table. By help of the routing table better routing path selected by the router. The information passes by router is known as link state advertisements(LSAs). In distance vector the information message passes in a fix time interval. Link state advertisements shared only when any changes done in the network topology. The bandwidth less consumed by link state routing protocol. The time of convergence is less than in distance vector protocol.

Function of link state routing protocol Important terms of link state routing protocol are link state packet, database, algorithm, routing table etc. Link state packets contains the routing information and sent to neighbour only when any changes occurs in connected network. Link state packets update the routing table in nearby routers. The information collected by link state packets stored in link state database.

HARDWARE & SOFTWARE REQUIREMENTS

- **Software:** Cisco Packet Tracer Simulation Tool

- **Devices:**
  - **End Devices:** PC0, PC1
  - **Networking Devices:** PT Switch (2 Units) and Router (3 units)
  - **Connections:** Copper Straight-Through cables, Copper Cross-Over cable

CONFIGURATION DETAILS

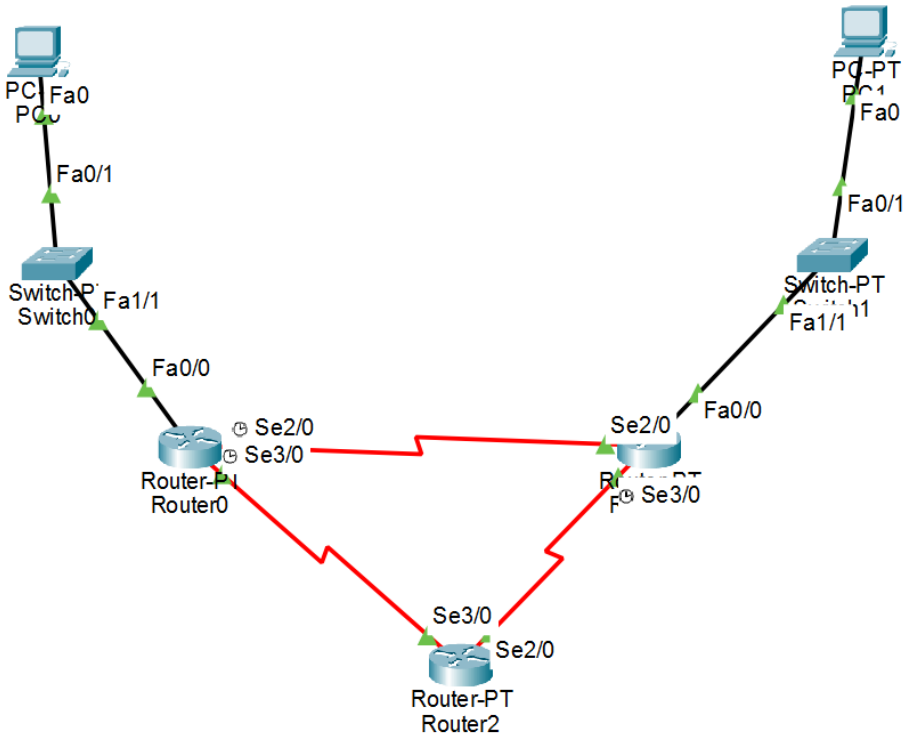
Device Name	Interface Name	IP Address	Subnet Mask
PC0	Fa0/1	10.0.0.2	255.0.0.0
PC1	Fa0/1	20.20.20.2	255.0.0.0
Rotuer0	Fa0/0	*N/A*	*N/A*
Rotuer1	Se 2-3/0	*N/A*	*N/A*
Router2	Se 2-3/0	*N/A*	*N/A*
Switch0	Fa0/2	*N/A*	*N/A*
Switch1	Fa0/2	*N/A*	*N/A*

PROCEDURE

1. Building the Topology:

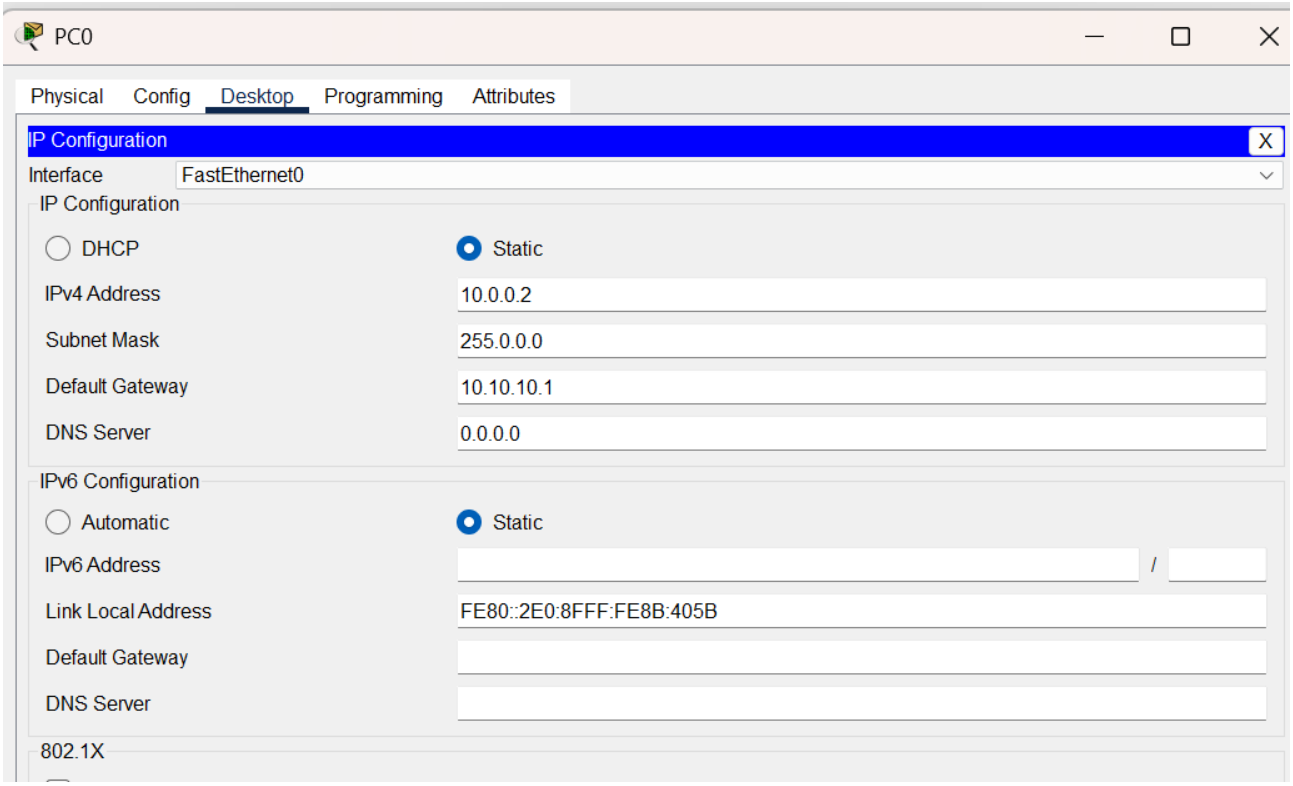
- Open Cisco Packet Tracer and drag three Routers (Router0, Router1, Router2), two Layer 2 Switches (Switch0, Switch1), and two PCs (PC0, PC1) onto the workspace.
- Click on each Router, navigate to the Physical tab, power off the device, drag the HWIC-2T module into an empty slot, and power it back on to enable serial connections.
- Connect the PCs to the Switches using Copper Straight-Through cables:
  - PC0 (Fa0) to Switch0 (Fa0/1)

    - PC1 (Fa0) to Switch1 (Fa0/1)
- Connect the Switches to the Routers using Copper Straight-Through cables:
    - Switch0 (Fa0/2) to Router0 (Fa0/0)
    - Switch1 (Fa0/2) to Router1 (Fa0/0)
- Connect the Routers using Serial DCE cables:
    - Router0 (Se2/0) to Router1 (Se2/0)
    - Router0 (Se3/0) to Router2 (Se3/0)
    - Router1 (Se3/0) to Router2 (Se2/0)



2. IP Configuration:

- Click on PC0 and navigate to Desktop -> IP Configuration.
- Select Static and input the IP Address (10.0.0.2) and Subnet Mask (255.0.0.0).
- Input the Default Gateway (10.10.10.1).
- Repeat the steps for PC1, applying the IP Address (20.20.20.2), Subnet Mask (255.0.0.0), and Default Gateway (20.20.20.1).



3. Command Line Execution

Router0 CLI

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#ip address 10.10.10.1 255.0.0.0

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 30.30.30.1 255.0.0.0

Router(config-if)#exit

Router(config)#interface Serial3/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 40.40.40.1 255.0.0.0

Router(config-if)#exit

Router(config)#router ospf 1

Router(config-router)#network 10.0.0.0 0.255.255.255 area 0

Router(config-router)#network 30.0.0.0 0.255.255.255 area 0

Router(config-router)#network 40.0.0.0 0.255.255.255 area 0

Router(config-router)#end

Router#write memory

Router1 CLI

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#ip address 20.20.20.1 255.0.0.0

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#no shutdown

Router(config-if)#ip address 30.30.30.3 255.0.0.0

Router(config-if)#exit

Router(config)#interface Serial3/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 50.50.50.1 255.0.0.0

Router(config-if)#exit

Router(config)#router ospf 1

Router(config-router)#network 20.0.0.0 0.255.255.255 area 0

Router(config-router)#network 30.0.0.0 0.255.255.255 area 0

Router(config-router)#network 50.0.0.0 0.255.255.255 area 0

Router(config-router)#end

Router#write memory

Router2 CLI

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface Serial3/0

Router(config-if)#no shutdown

Router(config-if)#ip address 40.40.40.2 255.0.0.0

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#no shutdown

Router(config-if)#ip address 50.50.50.2 255.0.0.0

Router(config-if)#exit

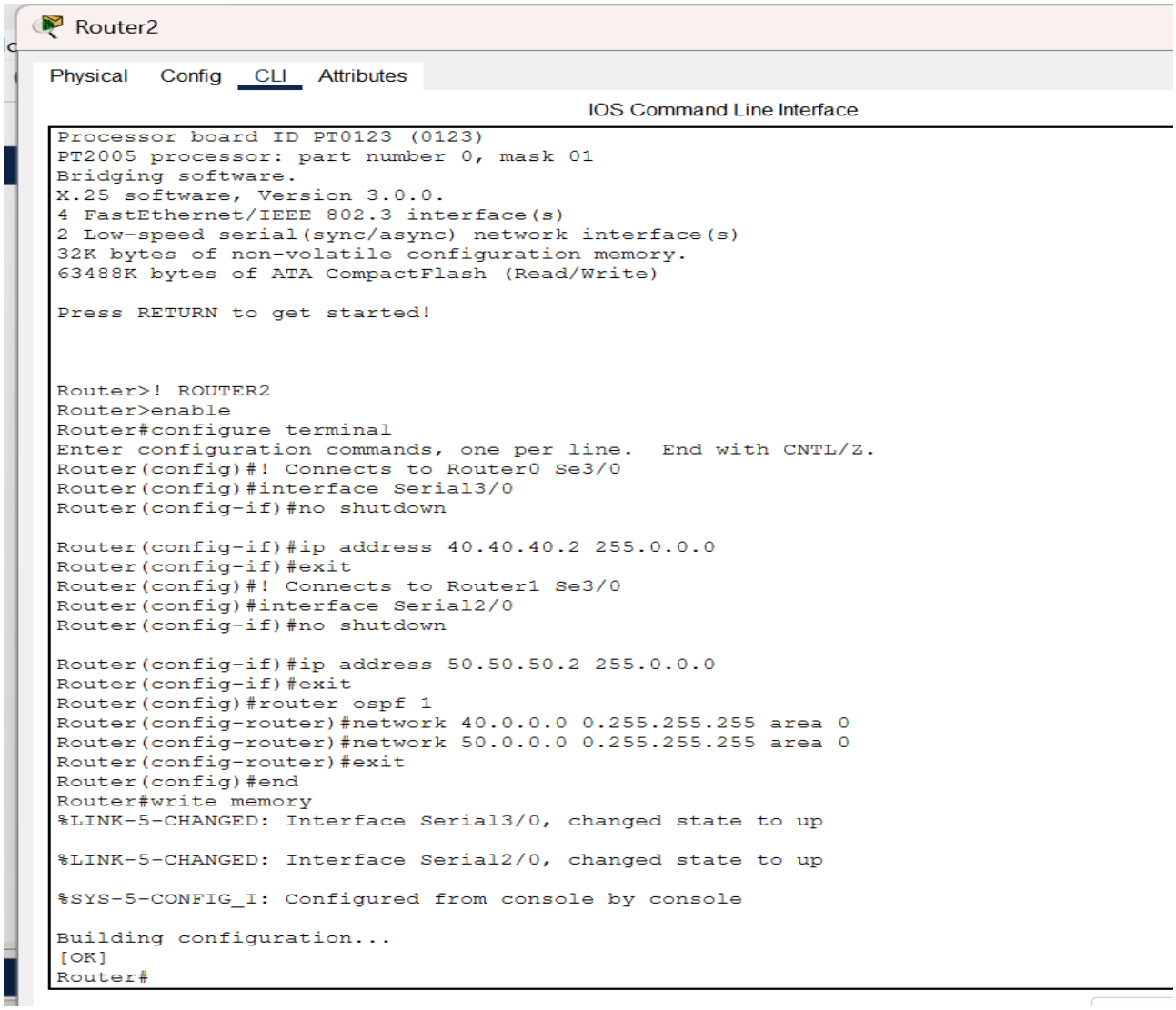
Router(config)#router ospf 1

Router(config-router)#network 40.0.0.0 0.255.255.255 area 0

Router(config-router)#network 50.0.0.0 0.255.255.255 area 0

Router(config-router)#end

Router#write memory



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>! ROUTER0
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown

Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#exit
Router(config)#! Connects to Router1 Se2/0
Router(config)#interface Serial2/0
Router(config-if)#no shutdown

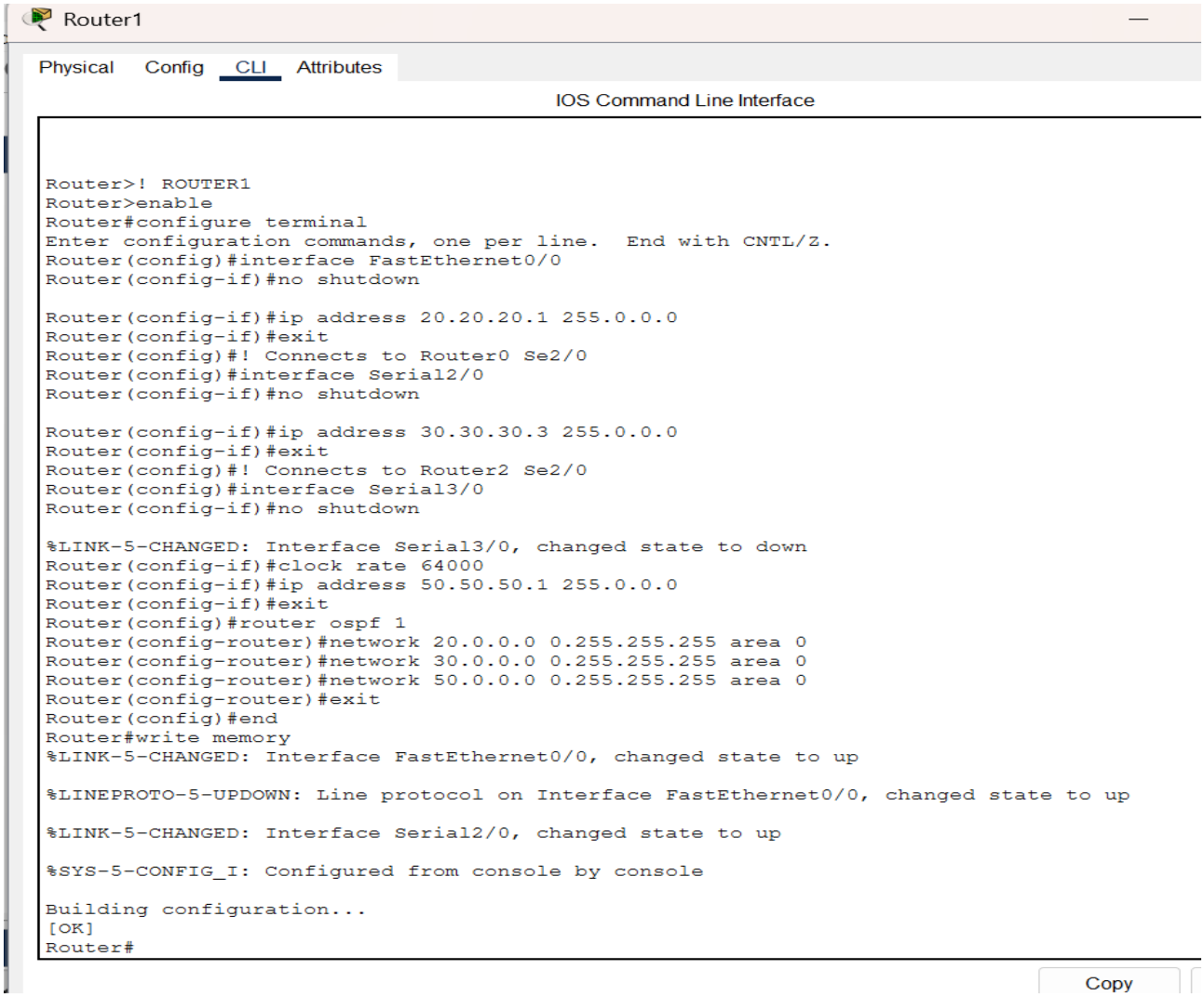
%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#clock rate 64000
Router(config-if)#ip address 30.30.30.1 255.0.0.0
Router(config-if)#exit
Router(config)#! Connects to Router2 Se3/0
Router(config)#interface Serial3/0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#clock rate 64000
Router(config-if)#ip address 40.40.40.1 255.0.0.0
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#network 40.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#end
Router#write memory
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```



4. Connectivity Verification:

- Click on PC0, navigate to Desktop, and open the Command Prompt.
- Type ping 20.20.20.2 and press Enter to verify successful OSPF routing communication with PC1.
- Open the Router0 CLI and type show ip route to confirm that routes marked with O (OSPF) are present in the routing table.

COMMAND EXECUTION DETAILS

- clock rate <value>: Sets the clocking rate for the serial DCE interfaces to synchronize the serial connection between routers.
- router ospf <process-id>: Enables the OSPF (Open Shortest Path First) routing process on the router with the specified local process ID.
- network <network-address> <wildcard-mask> area <area-id>: Identifies the interfaces on which OSPF is active and assigns them to a specific OSPF area (Area 0 for single-area OSPF).

PC0

Physical Config Desktop Programming Attributes

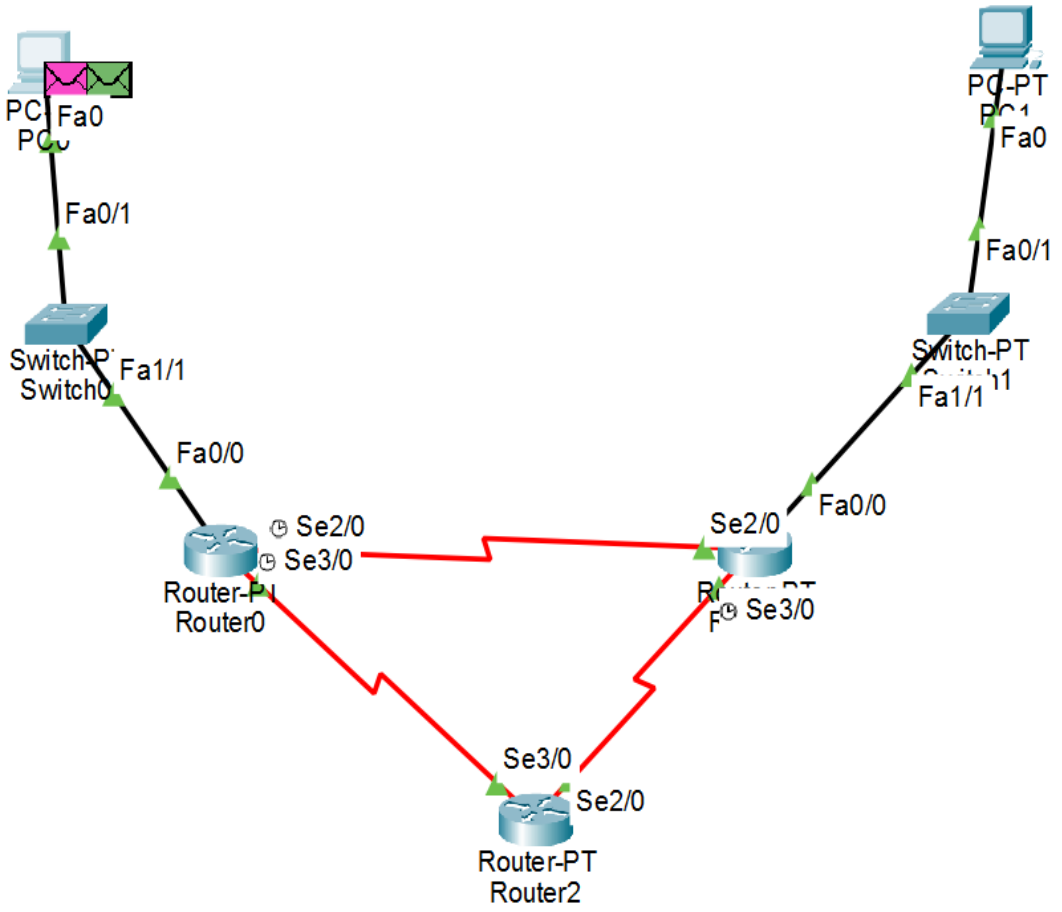
Command Prompt

```
C:\>ping 20.20.20.2

Pinging 20.20.20.2 with 32 bytes of data:

Reply from 20.20.20.2: bytes=32 time=8ms TTL=126
Reply from 20.20.20.2: bytes=32 time=1ms TTL=126
Reply from 20.20.20.2: bytes=32 time=2ms TTL=126
Reply from 20.20.20.2: bytes=32 time=11ms TTL=126

Ping statistics for 20.20.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 5ms
```



Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION:

In this experiment, we configured Link-State Routing, which uses a dynamic routing protocol like OSPF (Open Shortest Path First) to determine the best paths in a network. This method ensures faster convergence, scalability, and efficient use of network resources.

RESULT:

Thus, the design and implementation of Link State Routing using Cisco Packet Tracer was executed and verified successfully.

Ex.No:	5b
Name of the Experi ment	Distance Vector Routing
Date	

OBJECTIVE(S):

To design and implement Distance Vector routing using packet tracer

INTRODUCTION:

Distance-Vector routing protocols select the best path for data packets. Here distance is reference of hop in network. Distance-Vector protocols calculate the distance between source and destination on the basis of hop count. Suppose there are two path available for data packet from source and destination. Distance- Vector protocol select the path in which the number of hopes are less. RIP and IGRP are example of Distance-Vector routing protocol. Distance vector routing protocols manage the selection of best path for data packets by routers. Routing table of all routers update by sharing the information on the network. The destination network path defines by hop count up to destination network. Distance vector routing protocols generally known as DVRP. Distance vector routing protocols is mostly used protocol in present scenario. DVRP sent the data packets over the internet protocol. There are two terms in DVRP. The first term is distance and second is vector. Distance is number of hop or step to send the data packets up to destination network. Path selection for a data packet is depends on the hop count. Minimum hope count path selected by the Distance vector routing protocols. The term vector refers to the propagating of the packet on a given set of network nodes. Routers broadcast the information of remote network to next router. Every router does the same thing so the routing table of all routers updated automatically. All router informs about the connected networks to next router then router update its own routing table.

Network topology changes time to time. Adding or removing a router in a network is very common phenomena. Any change in network should be updated in all router's routing table. Doing this manually is very critical work. Distance vector routing protocols do this job automatically. The process of broadcasting any update in routing table and updation in all routing tables is known as convergence. The algorithm distance vector routing protocol find the routes on a internetwork. The

other algorithm used to select the best path for data packets is Link State routing protocols. DVRP algorithm allow routers to exchange the routing tables with each other. Each router received the routing table from neighbour router, update own routing table and share the updated table to next neighbour router. This process repeat after a fix predefined time interval. By repeating this process all devices connected in the network maintain the routing table which allow the flow of data packets efficiently.

HARDWARE & SOFTWARE REQUIREMENTS

- **Software:** Cisco Packet Tracer Simulation Tool

- **Devices:**
  - **End Devices:** PC0, PC1
  - **Networking Devices:** PT Switch (2 Units) and Router (3 units)
  - **Connections:** Copper Straight-Through cables, Copper Cross-Over cable

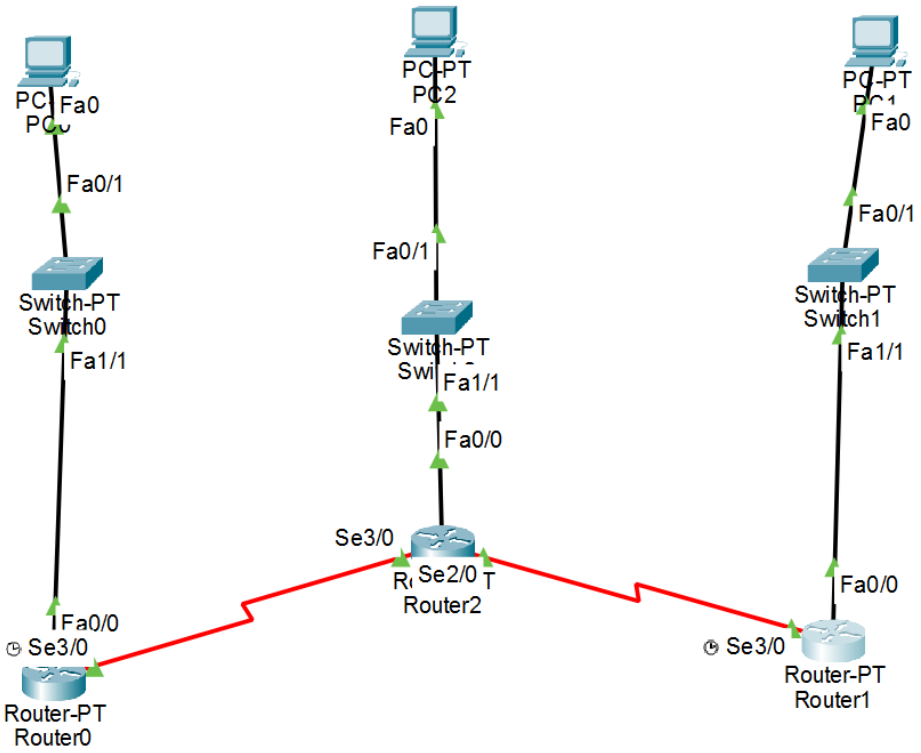
CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask
PC0	Fa0/1	192.168.10.2	255.255.255.0
PC1	Fa0/1	192.168.20.2	255.255.255.0
PC2	Fa0/1	192.168.30.2	255.255.255.0
Rotuer0	Fa0/0	*N/A*	*N/A*
Rotuer1	Se 2-3/0	*N/A*	*N/A*
Router2	Se 2-3/0	*N/A*	*N/A*
Switch0	Fa0/2	*N/A*	*N/A*
Switch1	Fa0/2	*N/A*	*N/A*
Switch2	Fa0/2	*N/A*	*N/A*

PROCEDURE

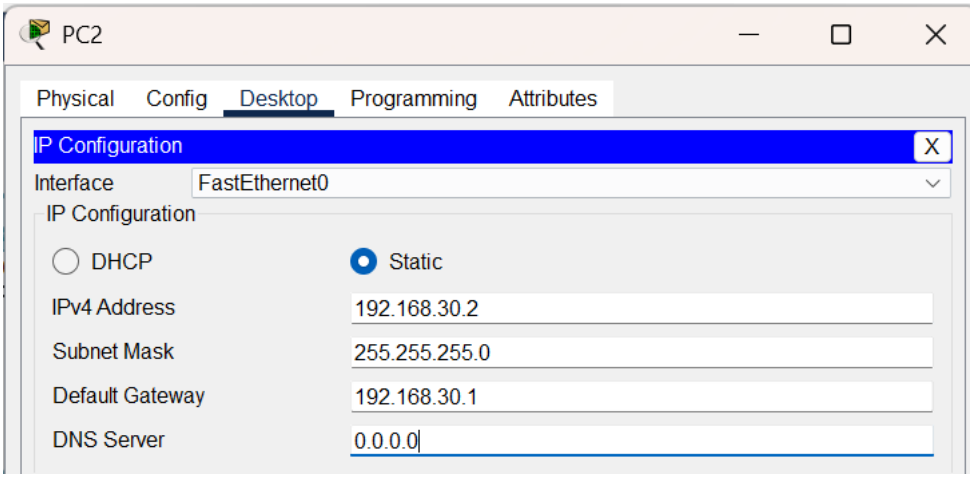
1. Building the Topology:

- Open Cisco Packet Tracer and drag three Routers (Router0, Router1, Router2), three Layer 2 Switches (Switch0, Switch1, Switch2), and three PCs (PC0, PC1, PC2) onto the workspace.
- Click on each Router, navigate to the Physical tab, power off the device, drag the HWIC-2T module into an empty slot, and power it back on to enable serial connections.
- Connect the PCs to the Switches using Copper Straight-Through cables:
    - PC0 (Fa0) to Switch0 (Fa0/1)
    - PC1 (Fa0) to Switch1 (Fa0/1)
    - PC2 (Fa0) to Switch2 (Fa0/1)
- Connect the Switches to the Routers using Copper Straight-Through cables:
    - Switch0 (Fa1/1) to Router0 (Fa0/0)
    - Switch1 (Fa1/1) to Router1 (Fa0/0)
    - Switch2 (Fa1/1) to Router2 (Fa0/0)
- Connect the Routers using Serial DCE cables:
    - Router0 (Se3/0) to Router2 (Se3/0)
    - Router2 (Se2/0) to Router1 (Se3/0)



2. IP Configuration:

- Click on PC0 and navigate to Desktop -> IP Configuration.
- Select Static and input the IP Address (192.168.10.2) and Subnet Mask (255.255.255.0). Input the Default Gateway (192.168.10.1).
- Repeat the steps for PC1, applying the IP Address (192.168.20.2), Subnet Mask (255.255.255.0), and Default Gateway (192.168.20.1).
- Repeat the steps for PC2, applying the IP Address (192.168.30.2), Subnet Mask (255.255.255.0), and Default Gateway (192.168.30.1).



3. Command Line Execution

Router0 CLI

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#ip address 192.168.10.1 255.255.255.0

Router(config-if)#exit

Router(config)#interface Serial3/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#exit

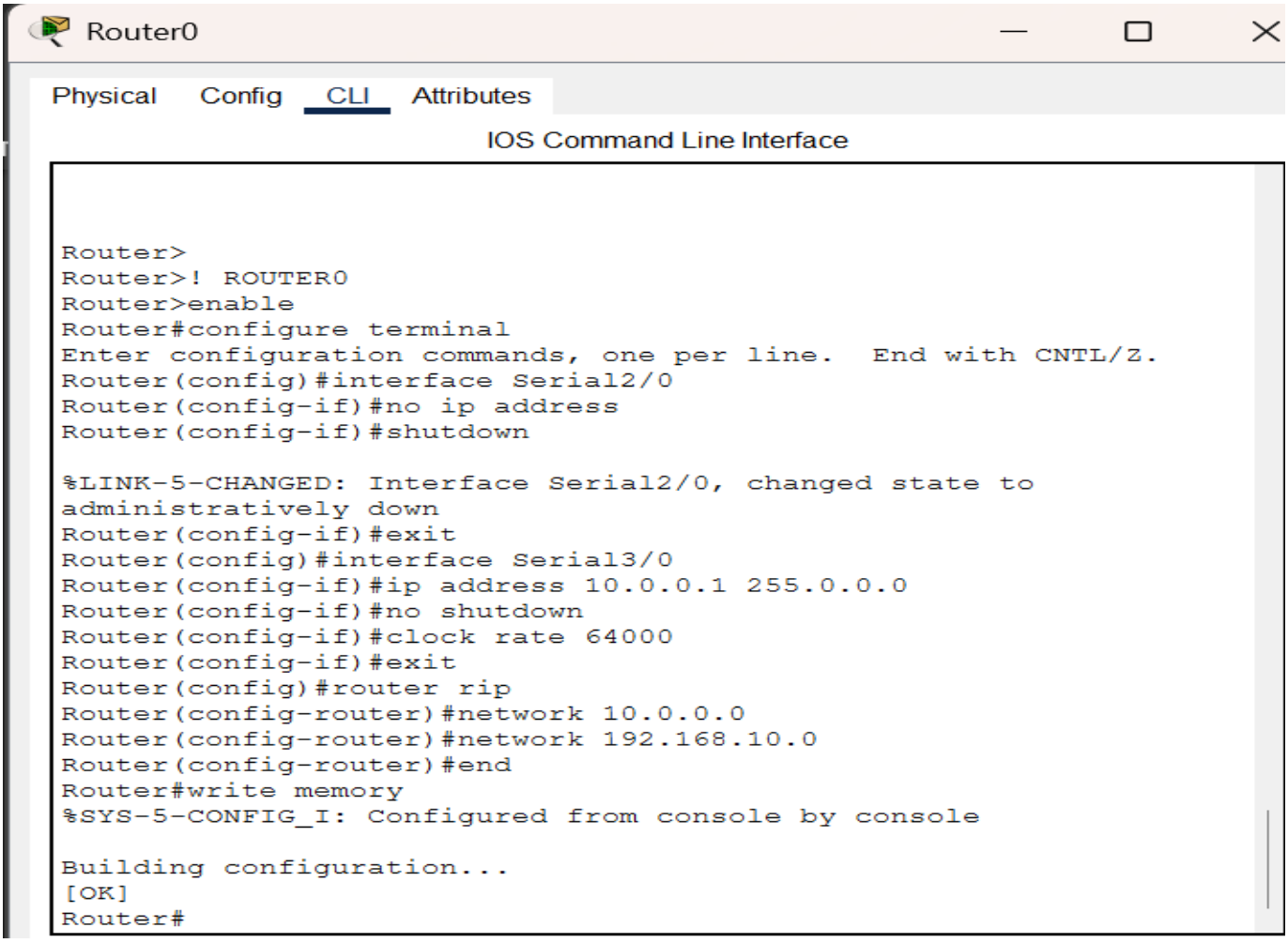
Router(config)#router rip

Router(config-router)#network 10.0.0.0

Router(config-router)#network 192.168.10.0

Router(config-router)#end

Router#write memory



Router2 CLI

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#ip address 192.168.30.1 255.255.255.0

Router(config-if)#exit

Router(config)#interface Serial3/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 10.0.0.2 255.0.0.0

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 20.0.0.1 255.0.0.0

Router(config-if)#exit

Router(config)#router rip

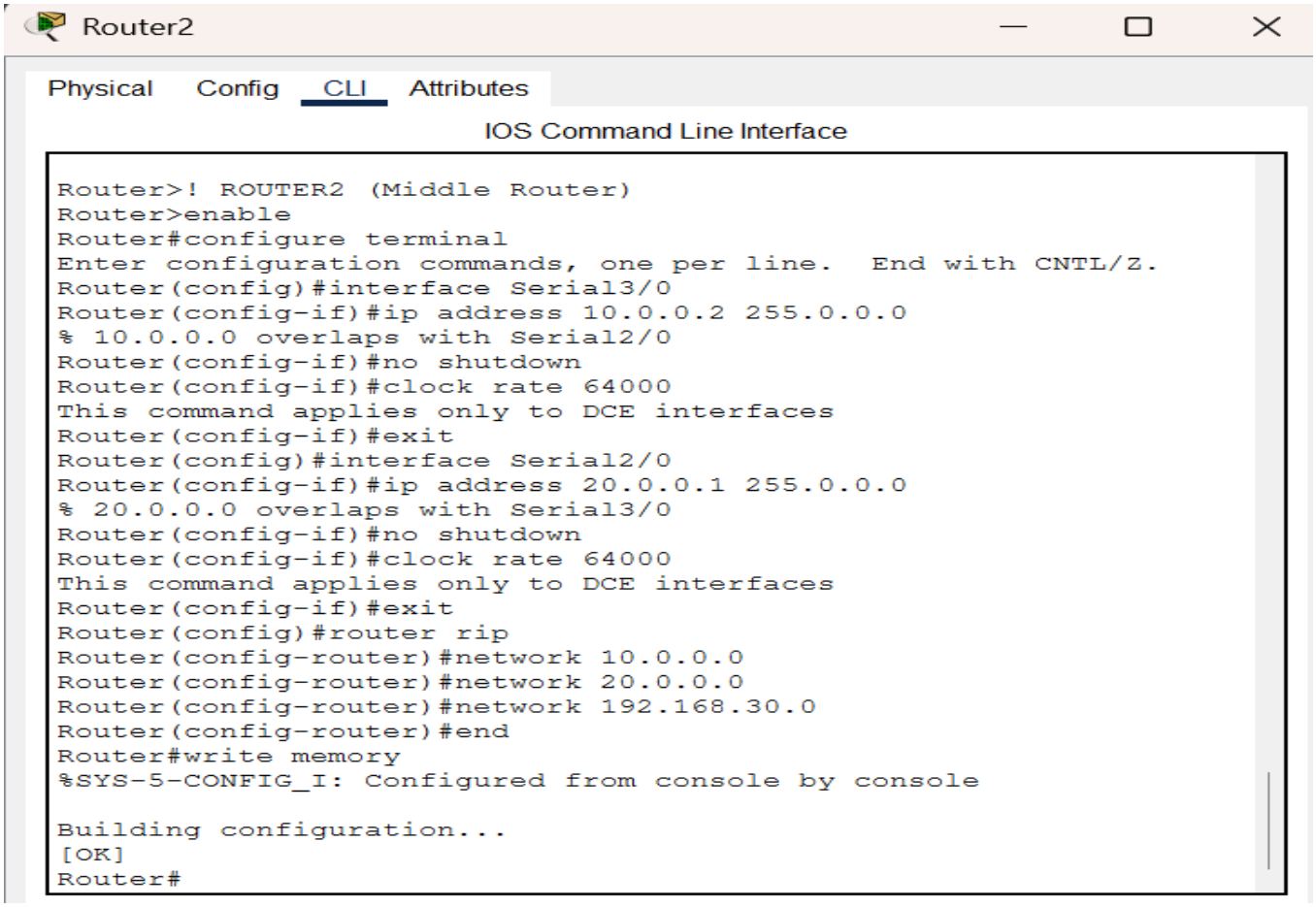
Router(config-router)#network 10.0.0.0

Router(config-router)#network 20.0.0.0

Router(config-router)#network 192.168.30.0

Router(config-router)#end

Router#write memory



Router1 CLI

Cisco CLI

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#ip address 192.168.20.1 255.255.255.0

Router(config-if)#exit

Router(config)#interface Serial3/0

Router(config-if)#no shutdown

Router(config-if)#clock rate 64000

Router(config-if)#ip address 20.0.0.2 255.0.0.0

Router(config-if)#exit

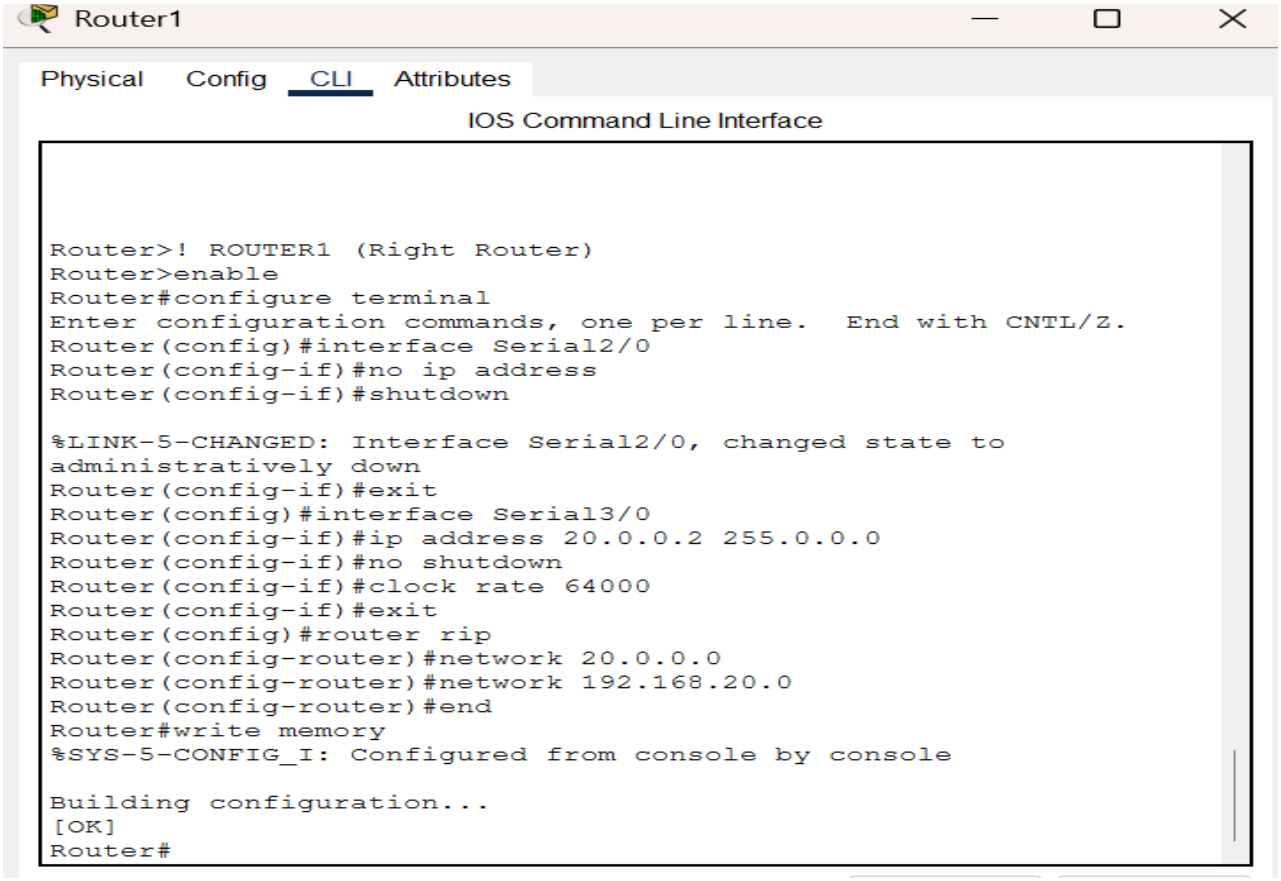
Router(config)#router rip

Router(config-router)#network 20.0.0.0

Router(config-router)#network 192.168.20.0

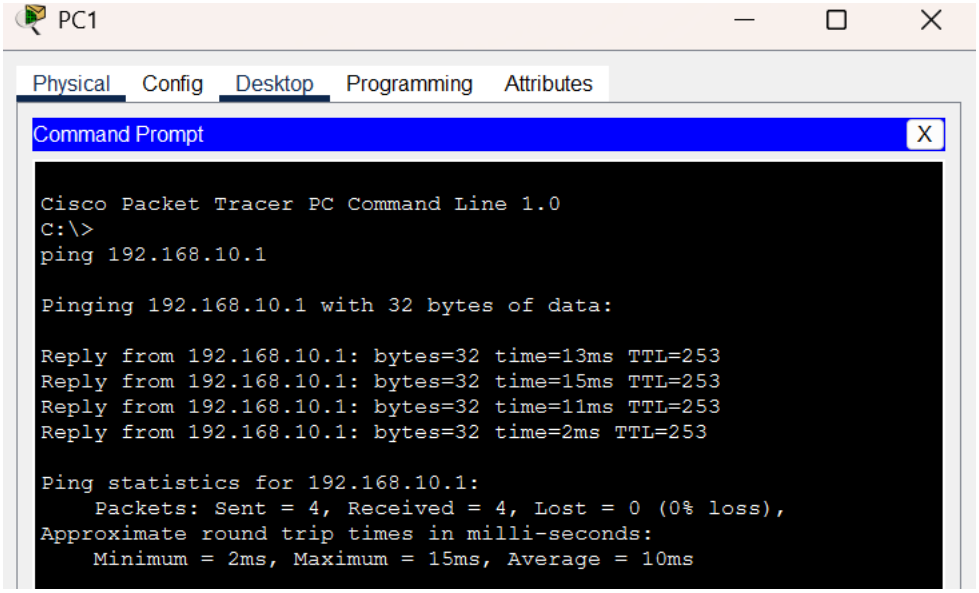
Router(config-router)#end

Router#write memory



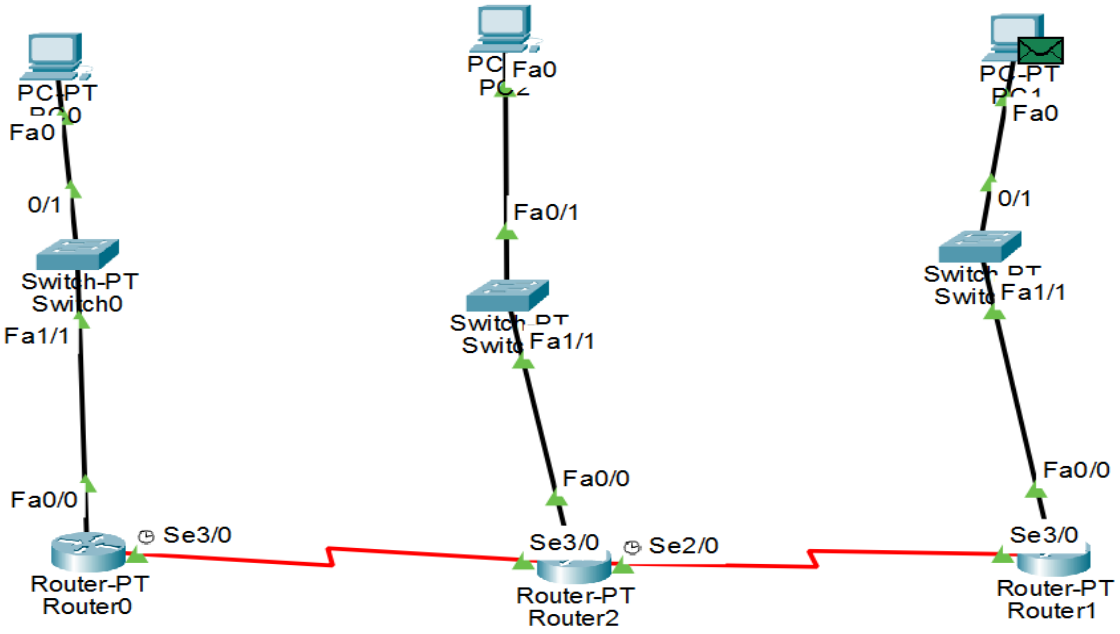
4. Connectivity Verification:

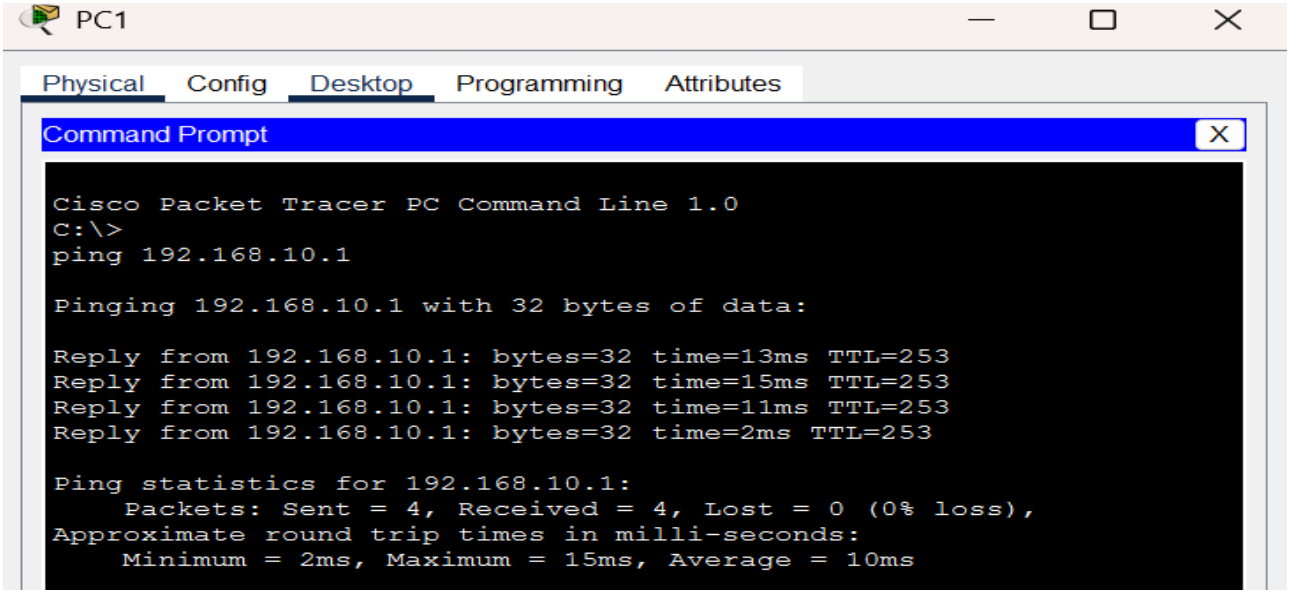
- Click on PC1, navigate to Desktop, and open the Command Prompt.
- Type ping 192.168.10.1 and press Enter to verify successful RIP routing communication with PC0.



COMMAND EXECUTION DETAILS

- router rip: Enables the Routing Information Protocol (RIP) routing process on the router.
- network <network-address>: Identifies the directly connected networks that will participate in the RIP routing process. This command allows the router to broadcast RIP updates on the interfaces associated with these networks.
- clock rate <value>: Sets the clocking rate for the serial DCE interfaces to synchronize the serial connection between routers.
- show ip route: Displays the current status of the routing table, indicating directly connected networks (C) and networks learned dynamically via RIP (R).


Fa0
0/1
Fa0/1
Fa1/1
Fa0/0
Se3/0
Se2/0
Router-PT
Router0
Router-PT
Router2
Router-PT
Router1


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=13ms TTL=253
Reply from 192.168.10.1: bytes=32 time=15ms TTL=253
Reply from 192.168.10.1: bytes=32 time=11ms TTL=253
Reply from 192.168.10.1: bytes=32 time=2ms TTL=253
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 15ms, Average = 10ms

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION:

In this experiment, we configured Distance Vector Routing using the Routing Information Protocol (RIP). We observed how routers dynamically learn network paths by exchanging routing tables with their neighbours and selecting the best route based on the lowest hop count. This demonstrated the automatic convergence of the network topology and the successful transmission of data packets across multiple networks.

RESULT:

Thus, the design and implementation of Distance Vector Routing using Cisco Packet Tracer was executed and verified successfully.

Ex.No:	6
Name of the Experi ment	Subnetting
Date	

OBJECTIVE(S):

To design and implement Subnetting configuration using packet tracer.

INTRODUCTION:

Subnetting is the process of dividing a large network into smaller networks called “subnets.” Subnets provide each group of devices with their own space to communicate, which ultimately helps the network to work easily. This also boosts security and makes it easier to manage the network, as each subnet can be monitored and controlled separately.

An IP address is made up of different parts, each serving a specific purpose in identifying a device on a network. An IPv4 address consists of four parts called “octets,” separated by dots (e.g., 192.168.1.1). It has two main sections:

- Network Portion: Identifies the network the device belongs to.
- Host Portion: Uniquely identifies a device within the network.

IPv4 addresses are divided into classes based on the length of the network and host portions:

- Class A: 8-bit network ID, 24-bit host ID.
- Class B: 16-bit network ID, 16-bit host ID.
- Class C: 24-bit network ID, 8-bit host ID.

HARDWARE & SOFTWARE REQUIREMENTS

Software:

- Cisco Packet Tracer Simulation Tool

Devices:

- End Devices: 6 PCs (PC0, PC1, PC2, PC3, PC4, PC5)
- Networking Devices: 2 Switches (Switch 0, Switch 1) and 1 Router (Router 0)
- Connections: Copper Straight-Through cables

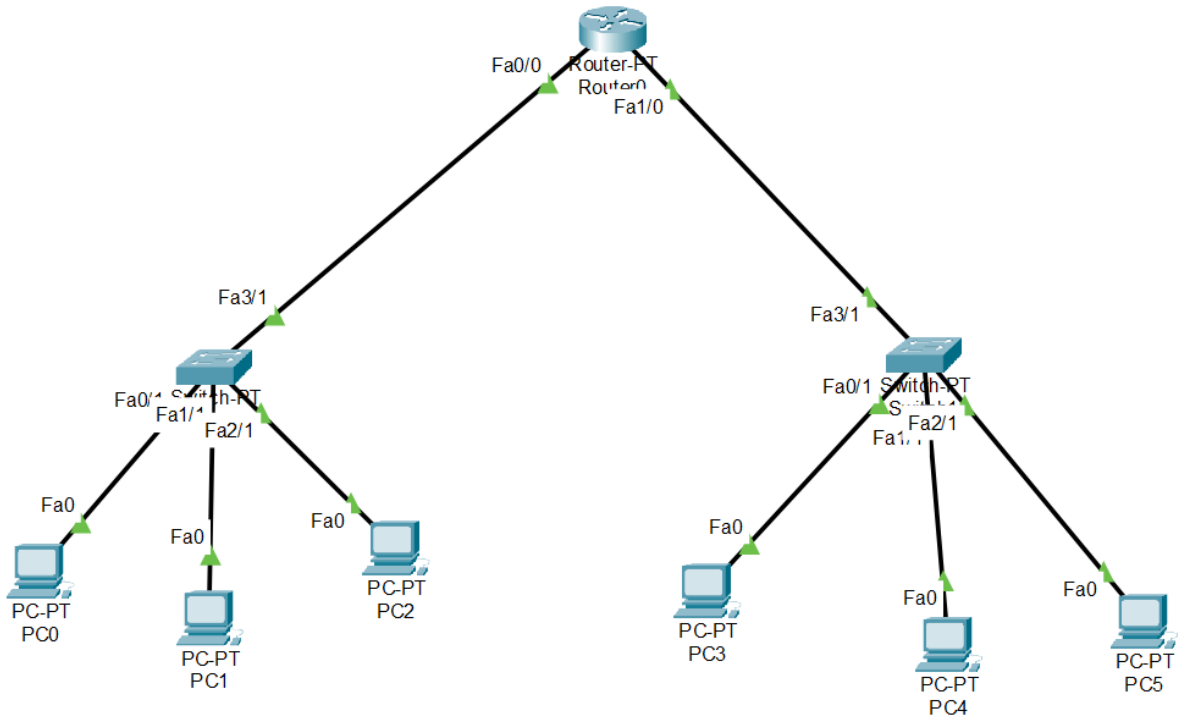
CONFIGURATION DETAILS

Device Name	Interface	IP Address	Subnet Mask	Default Gateway
PC0	Fa0	192.168.10.1	255.255.255.128	192.168.10.4
PC1	Fa0	192.168.10.2	255.255.255.128	192.168.10.4
PC2	Fa0	192.168.10.3	255.255.255.128	192.168.10.4
PC3	Fa0	192.168.10.129	255.255.255.128	192.168.10.132
PC4	Fa0	192.168.10.130	255.255.255.128	192.168.10.132
PC5	Fa0	192.168.10.131	255.255.255.128	192.168.10.132
Router 0	Fa0/0	192.168.10.4	255.255.255.128	N/A
Router 0	Fa1/0	192.168.10.132	255.255.255.128	N/A
Switch 0	Fa3/1	N/A	N/A	N/A
Switch 1	Fa2/1	N/A	N/A	N/A

PROCEDURE

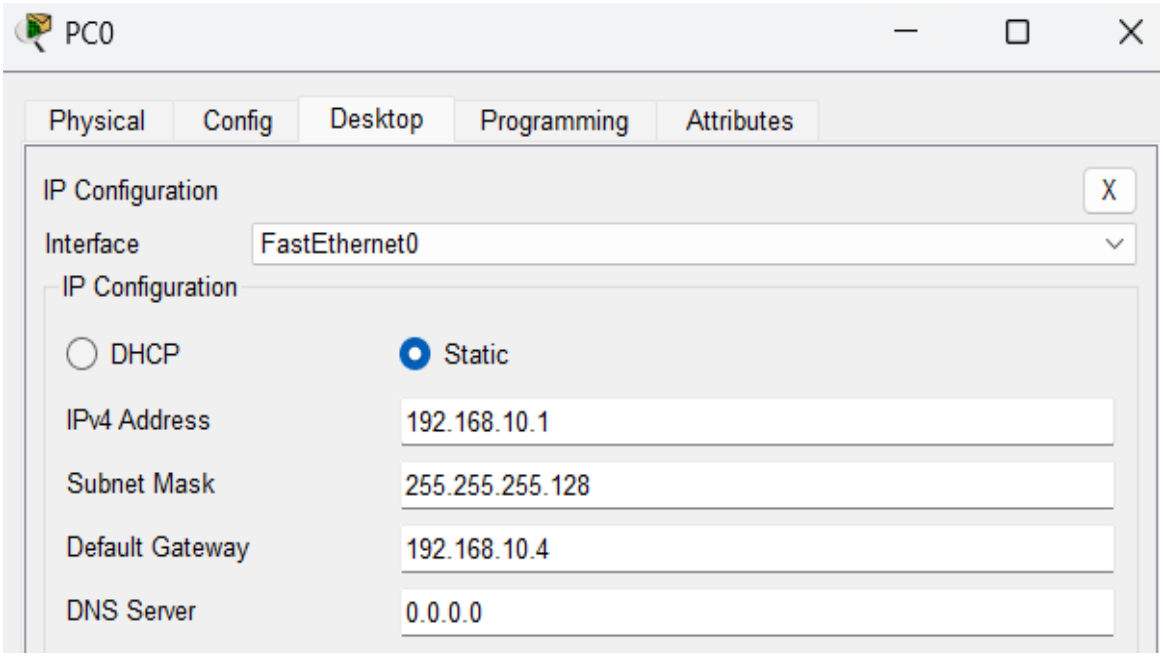
1. Building the Topology:

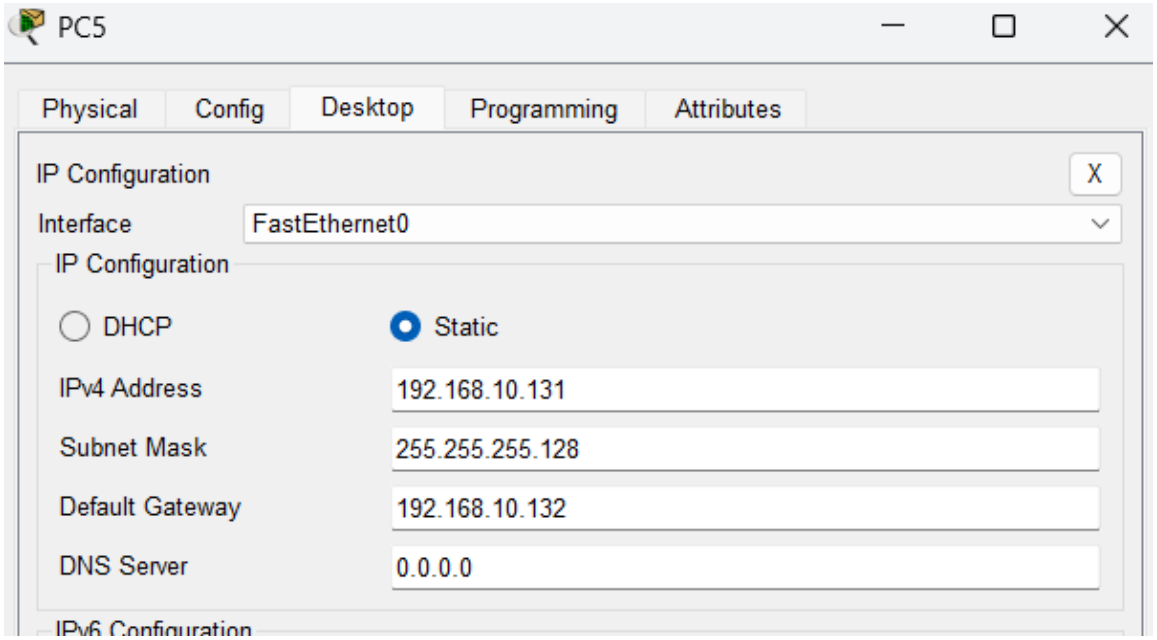
- Open Cisco Packet Tracer and drag one Router (Router-PT), two Layer 2 Switches (Switch-PT: Switch0, Switch1), and six PCs (PC0 to PC5) onto the workspace.
- Connect the PCs to the Switches using Copper Straight-Through cables:
    - PC0 (Fa0) to Switch0 (Fa0/1)
    - PC1 (Fa0) to Switch0 (Fa1/1)
    - PC2 (Fa0) to Switch0 (Fa2/1)
    - PC3 (Fa0) to Switch1 (Fa0/1)
    - PC4 (Fa0) to Switch1 (Fa1/1)
    - PC5 (Fa0) to Switch1 (Fa2/1)
- Connect the Switches to the Router using Copper Straight-Through cables:
    - Switch0 (Fa3/1) to Router 0 (Fa0/0)
    - Switch1 (Fa2/1) to Router 0 (Fa1/0)



2. IP Configuration:

- Click on **PC0** and navigate to **Desktop** -> **IP Configuration**. Select Static and input the IP Address (192.168.10.1), Subnet Mask (255.255.255.128), and Default Gateway (192.168.10.4).
- Repeat the steps for **PC1**, applying IP Address (192.168.10.2), Subnet Mask (255.255.255.128), and Default Gateway (192.168.10.4).
- Repeat the steps for **PC2**, applying IP Address (192.168.10.3), Subnet Mask (255.255.255.128), and Default Gateway (192.168.10.4).
- Repeat the steps for **PC3**, applying IP Address (192.168.10.129), Subnet Mask (255.255.255.128), and Default Gateway (192.168.10.132).
- Repeat the steps for **PC4**, applying IP Address (192.168.10.130), Subnet Mask (255.255.255.128), and Default Gateway (192.168.10.132).
- Repeat the steps for **PC5**, applying IP Address (192.168.10.131), Subnet Mask (255.255.255.128), and Default Gateway (192.168.10.132).





3. Command Line Execution

Router 0 CLI

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#no shutdown

Router(config-if)#ip address 192.168.10.4 255.255.255.128

Router(config-if)#exit

Router(config)#interface FastEthernet1/0

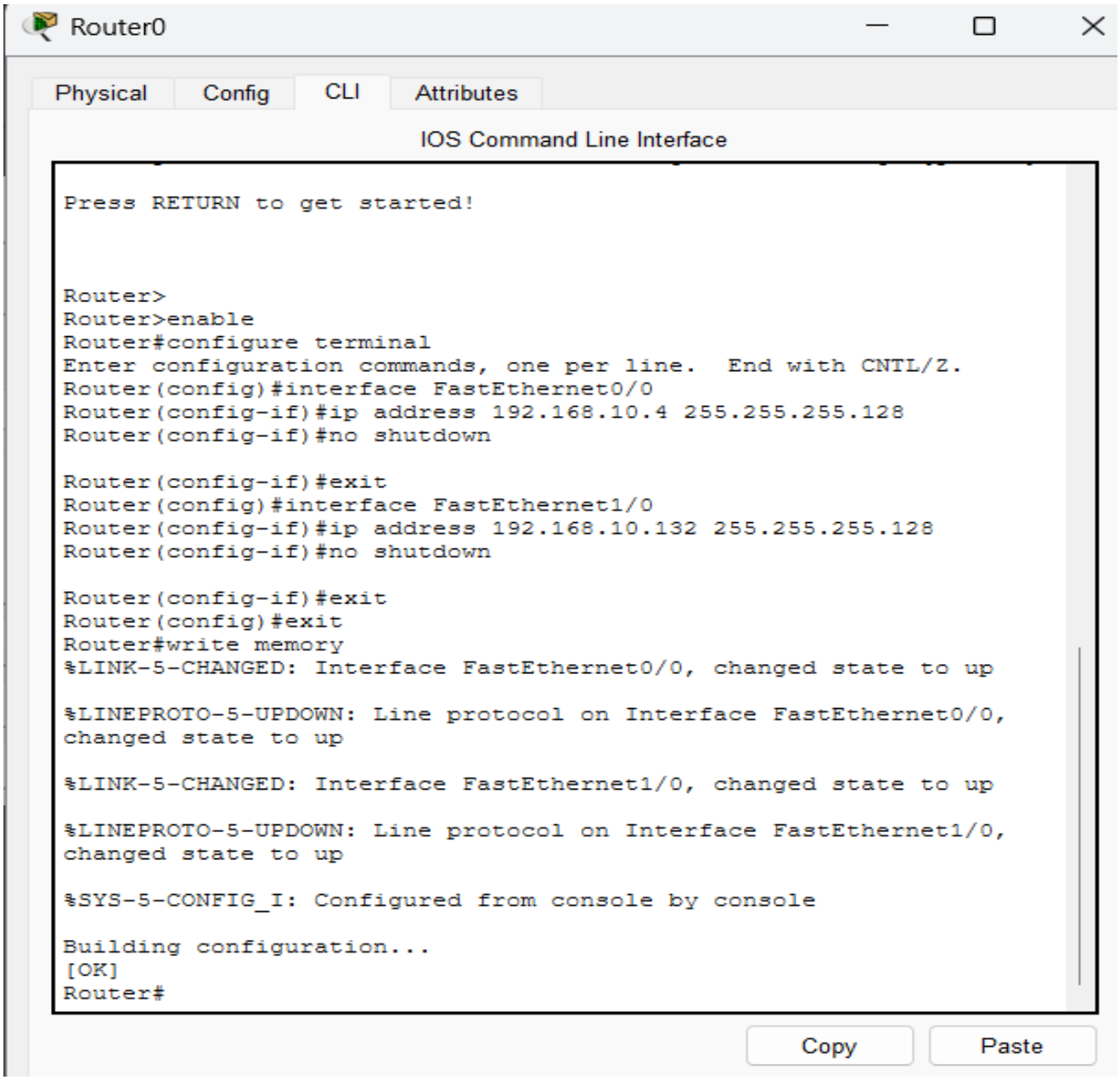
Router(config-if)#no shutdown

Router(config-if)#ip address 192.168.10.132 255.255.255.128

Router(config-if)#exit

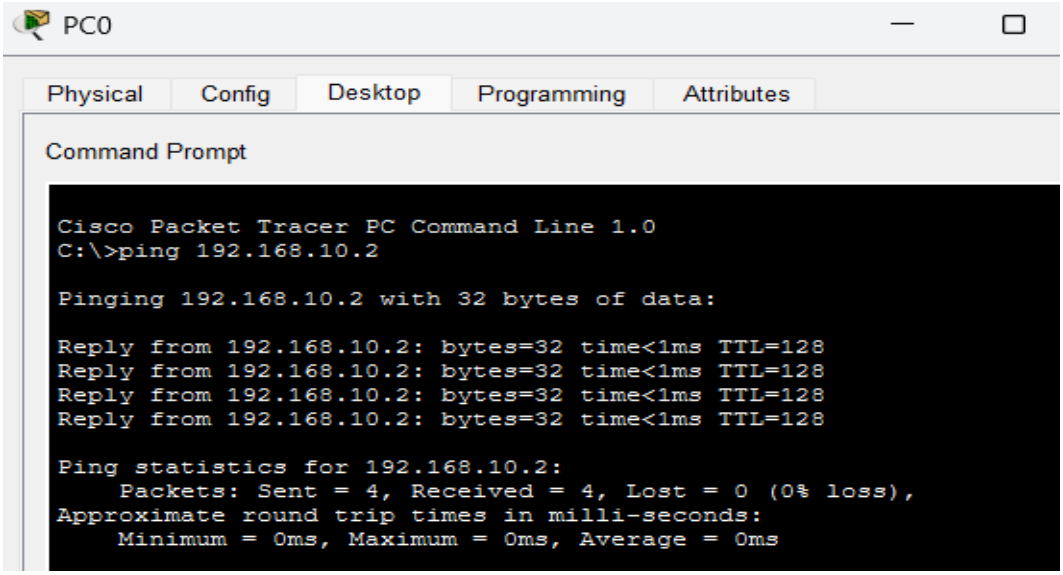
Router(config)#exit

Router#write memory



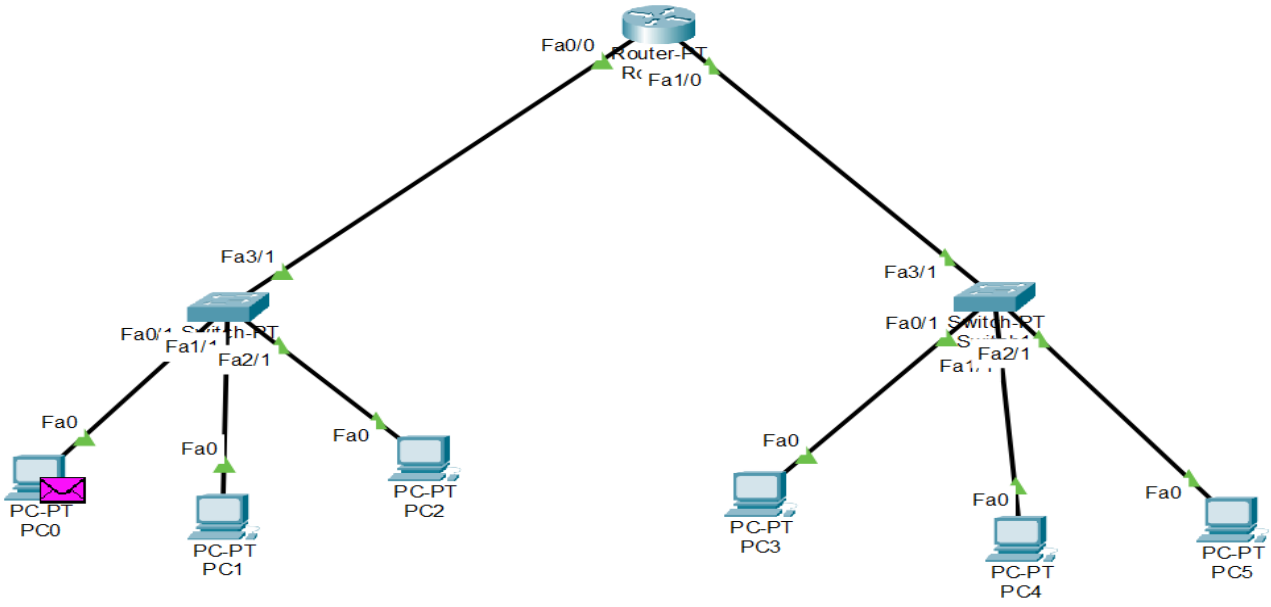
4. Connectivity Verification:

- Click on **PC0**, navigate to **Desktop**, and open the **Command Prompt**.
- Type ping 192.168.10.2 and press Enter to verify intra-subnet communication with **PC1**.
- Type ping 192.168.10.129 and press Enter to verify inter-subnet routing communication with **PC3**.



COMMAND EXECUTION DETAILS

- **interface :** Enters the specific interface configuration mode (e.g., FastEthernet0/0) to apply IP and status settings.
- **ip address :** Assigns the designated IP address and subnet mask to the router interface, acting as the Default Gateway for that subnet.



Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION:

In this experiment, we configured Subnetting using Class C network division (/25 CIDR prefix) on Cisco Packet Tracer. We observed how a single IP network (192.168.10.0/24) was split into two distinct logical subnets using custom subnet masks (255.255.255.128). This demonstrated how a central router facilitates inter-subnet routing while maintaining separate broadcast domains across multiple end devices.

RESULT:

Thus, the design and implementation of Subnetting configuration using Cisco Packet Tracer was executed and verified successfully.

Ex.No:	7
Name of the Experi ment	DHCP Configuration
Date	

OBJECTIVE(S):

To design and implement DHCP configuration using packet tracer.

INTRODUCTION:

In this activity, you will continue to configure the Cisco 1841 ISR router for the customer network by configuring the DHCP service. The customer has several workstations that need to be automatically configured with IP addresses on the local subnet and appropriate DHCP options to allow access to the Internet. The DHCP pool will use the 192.168.1.0/24 network but the first 49 addresses are excluded. The default gateway and DNS server also need to be configured as 192.168.1.1 and 192.168.1.10. For this activity, both the user and privileged EXEC passwords are cisco. Note: Packet Tracer does not currently support the domain name and lease period options. These options are not used in this activity.

HARDWARE & SOFTWARE REQUIREMENTS

Software:

- Cisco Packet Tracer Simulation Tool

Devices:

- **End Devices:** 8 PCs (PC1, PC2, PC3, PC4, PC5, PC6, PC7, PC8)
- **Networking Devices:** 2 Switches (Switch 0, Switch 1) and 1 Router (Router 2)
- **Connections:** Copper Straight-Through cables

CONFIGURATION DETAILS

Device Name	Interface	IP Address	Subnet Mask	Default Gateway
PC0	Fa0	192.168.10.2 (DHCP)	255.255.255.0	192.168.10.1
PC1	Fa0	192.168.10.7 (DHCP)	255.255.255.0	192.168.10.1
PC2	Fa0	192.168.10.8 (DHCP)	255.255.255.0	192.168.10.1
PC3	Fa0	192.168.10.9 (DHCP)	255.255.255.0	192.168.10.1
PC4	Fa0	192.168.20.2 (DHCP)	255.255.255.0	192.168.20.1
PC5	Fa0	192.168.20.3 (DHCP)	255.255.255.0	192.168.20.1
PC6	Fa0	192.168.20.4 (DHCP)	255.255.255.0	192.168.20.1
PC7	Fa0	192.168.20.5 (DHCP)	255.255.255.0	192.168.20.1
Router0	**Fa4/0**	192.168.10.1	255.255.255.0	N/A
Router0	**Fa5/0**	192.168.20.1	255.255.255.0	N/A

PROCEDURE

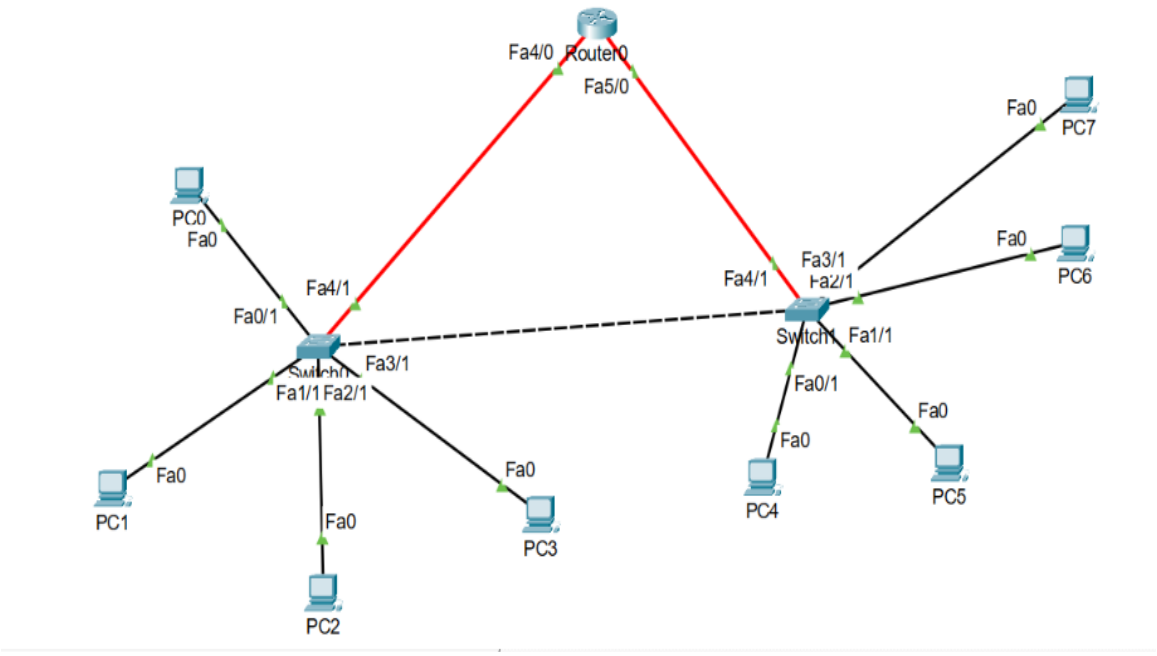
1. Building the Topology:

- Open Cisco Packet Tracer and place one Router (Router-PT), two Switches (Switch-PT: Switch0, Switch1), and eight PCs (PC0 to PC7) onto the workspace.
- Connect the PCs to the Switches using Copper Straight-Through cables:
    - PC0 (Fa0) to Switch0 (Fa0/1)
    - PC1 (Fa0) to Switch0 (Fa1/1)
    - PC2 (Fa0) to Switch0 (Fa2/1)
    - PC3 (Fa0) to Switch0 (Fa3/1)
    - PC4 (Fa0) to Switch1 (Fa0/1)

- PC5 (Fa0) to Switch1 (Fa1/1)
- PC6 (Fa0) to Switch1 (Fa2/1)
- PC7 (Fa0) to Switch1 (Fa3/1)

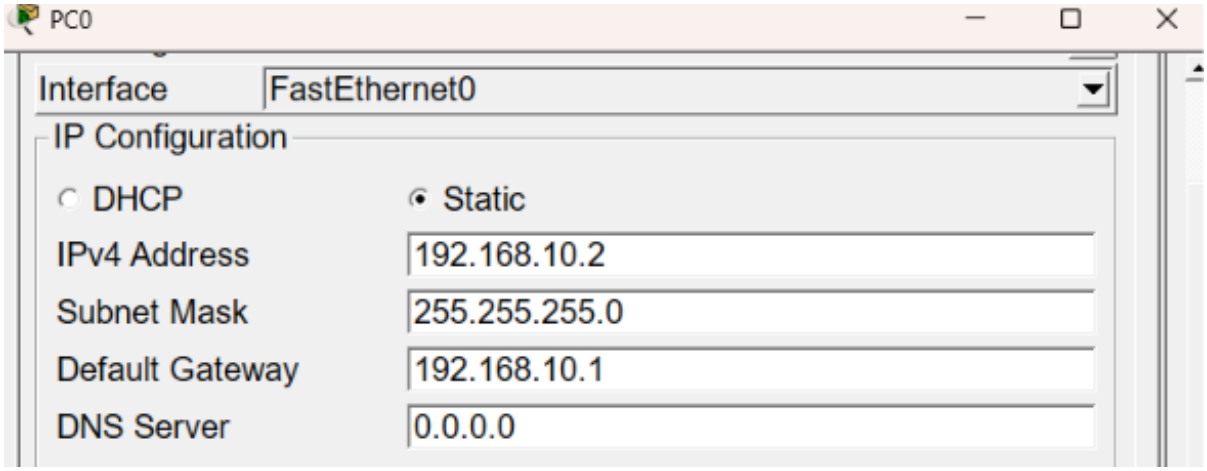
- Connect the Switches to the Router using Copper Straight-Through cables:

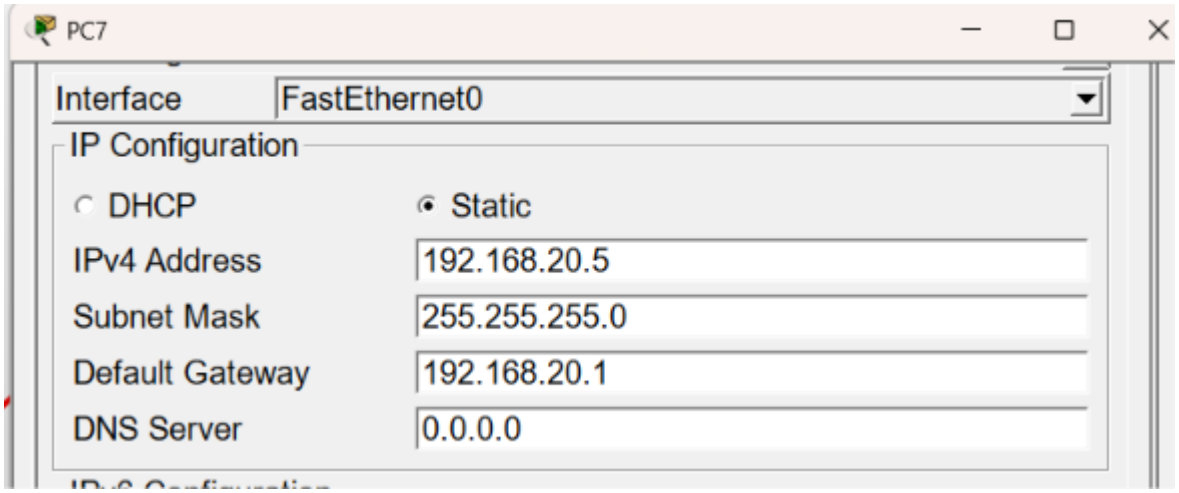
  - Switch0 (Fa4/1) to Router0 (Fa4/0)
  - Switch1 (Fa4/1) to Router0 (Fa5/0)



2. IP Configuration:

- Click on **PC0** and navigate to **Desktop** -> **IP Configuration**. Select **DHCP** to automatically obtain the IP Address (192.168.10.2), Subnet Mask (255.255.255.0), and Default Gateway (192.168.10.1).
- Repeat the steps for **PC1**, **PC2**, and **PC3**, selecting **DHCP** to acquire IP parameters within the 192.168.10.0/24 subnet (skipping the excluded addresses 192.168.10.3 to 192.168.10.6).
- Repeat the steps for **PC4**, **PC5**, **PC6**, and **PC7**, selecting **DHCP** to acquire IP parameters within the 192.168.20.0/24 subnet.





3. Command Line Execution:

Router 0 CLI

Plaintext

Router>enable

Router#configure terminal

Router(config)#interface FastEthernet4/0

Router(config-if)#ip address 192.168.10.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)#interface FastEthernet5/0

Router(config-if)#ip address 192.168.20.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)#ip dhcp pool 10

Router(dhcp-config)#network 192.168.10.0 255.255.255.0

Router(dhcp-config)#default-router 192.168.10.1

Router(dhcp-config)#exit

Router(config)#ip dhcp excluded-address 192.168.10.3 192.168.10.6

Router(config)#ip dhcp pool 20

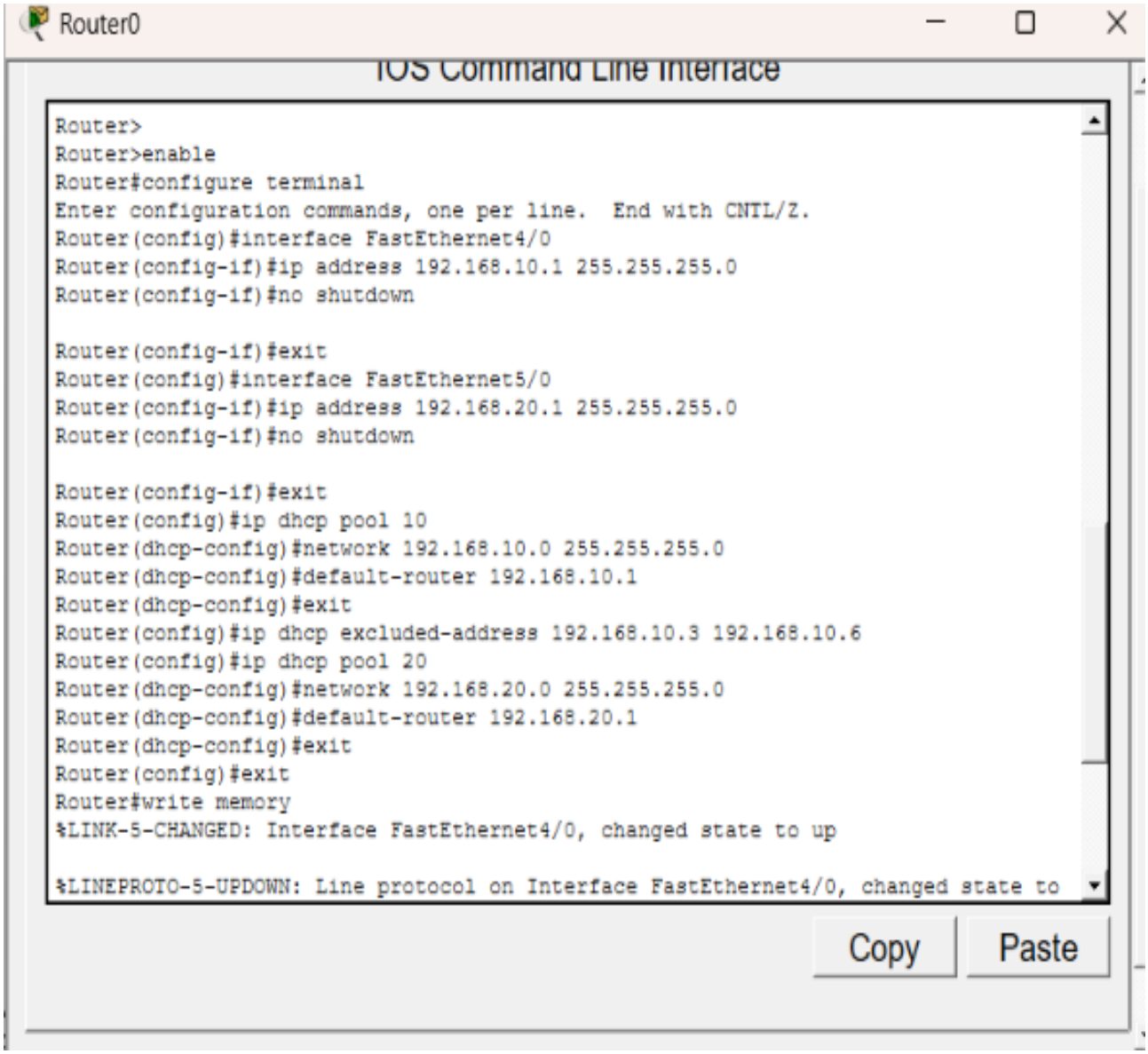
Router(dhcp-config)#network 192.168.20.0 255.255.255.0

Router(dhcp-config)#default-router 192.168.20.1

Router(dhcp-config)#exit

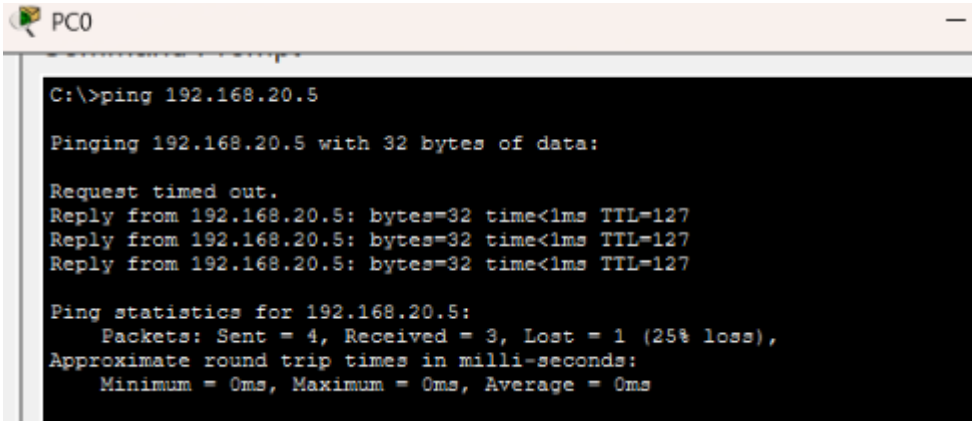
Router(config)#exit

Router#write memory



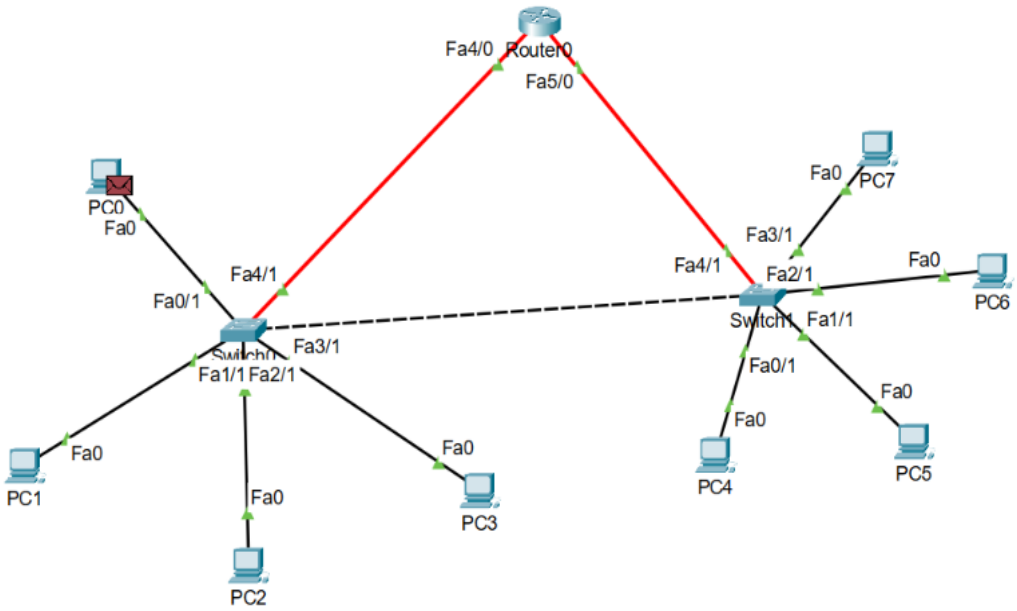
4. Connectivity Verification:

- Click on **PC0**, navigate to **Desktop**, and open the **Command Prompt**.
- Type ping 192.168.10.7 and press Enter to verify intra-subnet communication with **PC1**.
- Type ping 192.168.20.2 and press Enter to verify inter-subnet routing communication with **PC4**.



COMMAND EXECUTION DETAILS

- **interface :** Enters configuration mode for the specified router interface.
- **ip address :** Assigns the IP address and subnet mask to the router interface, configuring it as the default gateway.
- **ip dhcp pool :** Defines a named DHCP pool to allocate IP parameters dynamically.



Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION: In this experiment, we configured Dynamic Host Configuration Protocol (DHCP) on a Cisco router using Cisco Packet Tracer. We observed how the router dynamically allocated IP parameters to end devices across two distinct subnets (192.168.10.0/24 and 192.168.20.0/24) while respecting excluded IP address ranges. This demonstrated how centralized DHCP service eliminates manual client configuration and simplifies IP address management across local networks.

RESULT: Thus, the design and implementation of DHCP configuration using Cisco Packet Tracer was executed and verified successfully.

Ex.No:	8
Name of the Experi ment	Capture and Anlyze ICMP packet
Date	

OBJECTIVE(S):

To capture and analyse ICMP Request Response packet using Wireshark.

INTRODUCTION:

The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet protocol suite. It is used by network devices, including routers, to send error messages and operational information which indicates that a requested service is not available or that a host or router could not be reached. It is layer 3 i.e. network layer protocol used by the ping command for sending a message through ICMP payload which is encapsulated with IP Header Packet. According to MTU the size of the ICMP packet cannot be greater than 1500 bytes.

ICMP Packet at Network Layer

IP Header	**ICMP Header**	**ICMP Payload Size**	**Total MTU**
20 bytes	8 bytes	1472 bytes (maximum)	1500 bytes (20 + 8 + 1472)

ICMP Packet at Data Link Layer

Ethernet Header	**IP Header**	**ICMP Header**	**ICMP Payload Size**	**Total MTU**
14 bytes	20 bytes	8 bytes	1472 bytes (maximum)	1514 bytes (14 + 20 + 8 + 1472)

ICMP Message codes:

ICMP messages contain two types of codes: Query and Error.

- **Query**: Information received from a router or another destination host (e.g., Type 0 = Echo Reply, Type 8 = Echo Request, Type 9 = Router Advertisement).
- **Error**: Reports problems which a router or destination host may generate (e.g., Type 3 = Destination Unreachable, Type 4 = Source Quench, Type 11 = Time Exceeded).

HARDWARE & SOFTWARE REQUIREMENTS

Software:

- Wireshark Packet Analyzer,
- Windows Command Prompt (cmd)

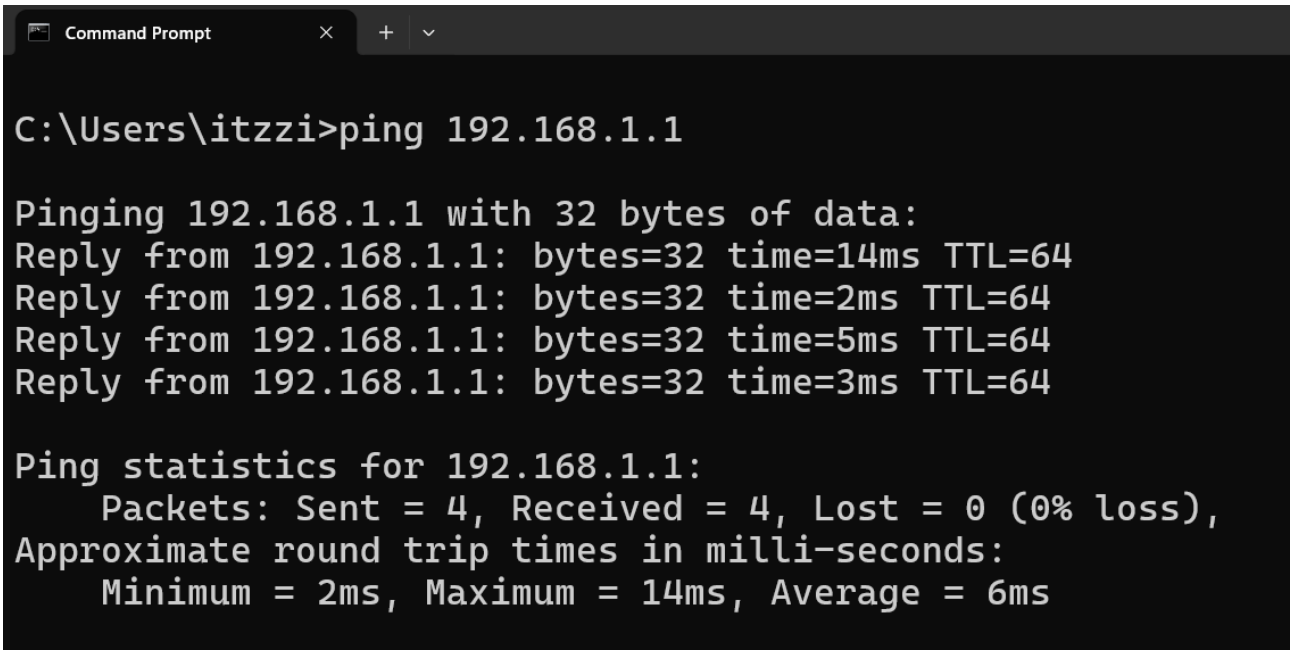
Devices:

- PC with Internet connectivity

PROCEDURE

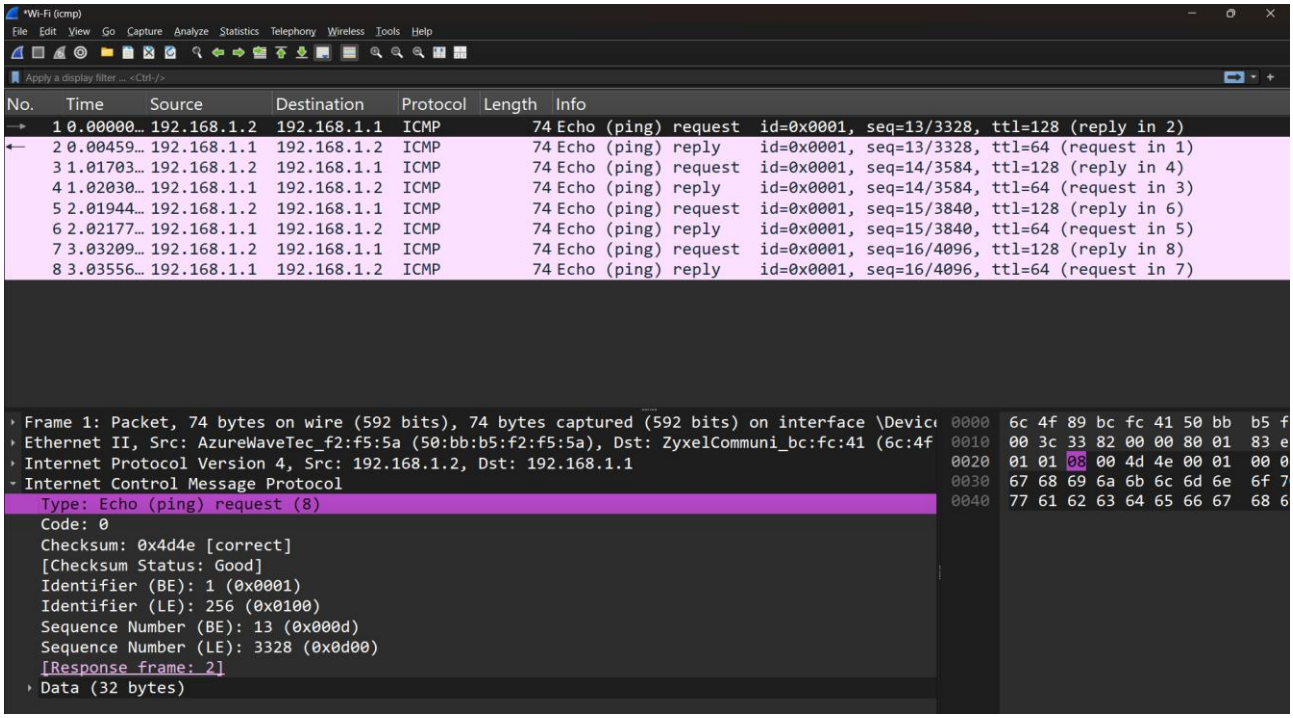
1. Basic ICMP Echo Request and Reply Analysis

1. Open Wireshark and start capturing packets on your active network interface.
2. Open the Command Prompt and execute the standard ping command: ping 192.168.1.1
3. Stop the Wireshark capture and filter by icmp.
4. Observe that the default size of the payload sent is 32 bytes, the TTL is 128 (Windows default), and 8 total packets were captured (4 requests, 4 replies).



2. Identifying ICMP Query Codes in Wireshark

1. In Wireshark, select the first captured packet from the source machine.
2. Expand the Internet Control Message Protocol section in the packet details pane.
3. Observe the highlighted text showing the ICMP query code: Type 8 (Echo (ping) request).
4. Select corresponding reply packet and observe the ICMP query: Type 0 (Echo (ping) reply).



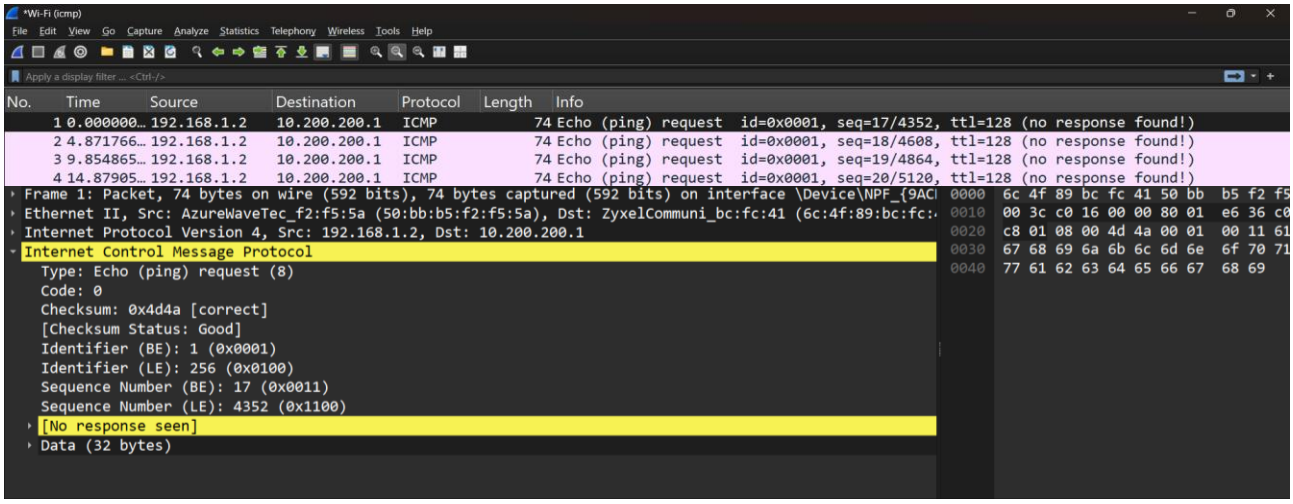
3. Simulating and Identifying ICMP Error Codes

1. Ping a non-existent or blocked IP address to generate an error.
2. In Wireshark, analyze the returned packet.
3. Observe the ICMP error code: Type 3 (Destination unreachable).

```
C:\Users\itzzi>ping 10.200.200.1

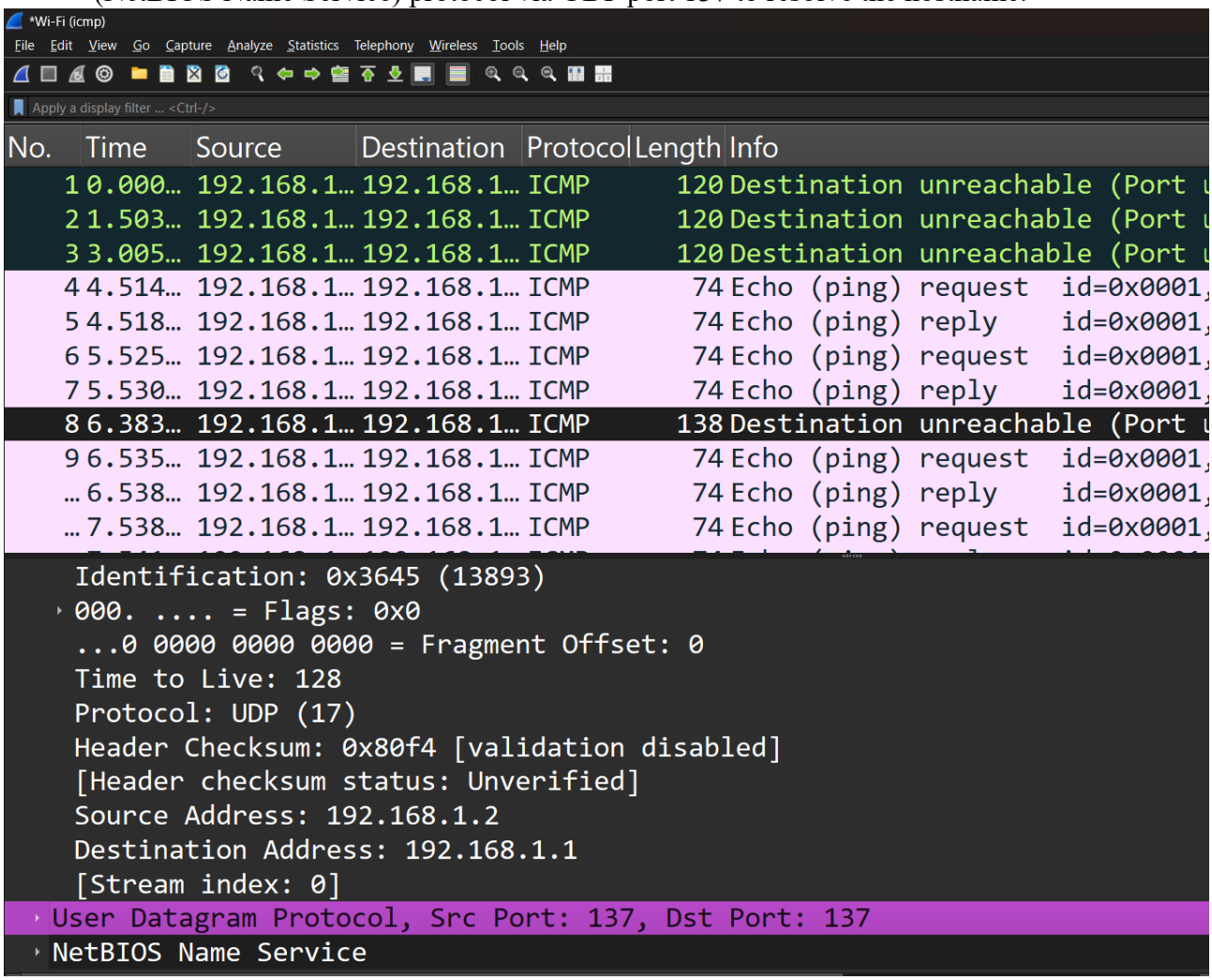
Pinging 10.200.200.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.200.200.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



4. Advanced Ping Parameters and NBNS Protocol

1. Execute reverse name resolution by typing: ping -a 192.168.1.1
2. Observe in Wireshark that instead of ICMP, the request may be sent through NBNS (NetBIOS Name Service) protocol via UDP port 137 to resolve the hostname.



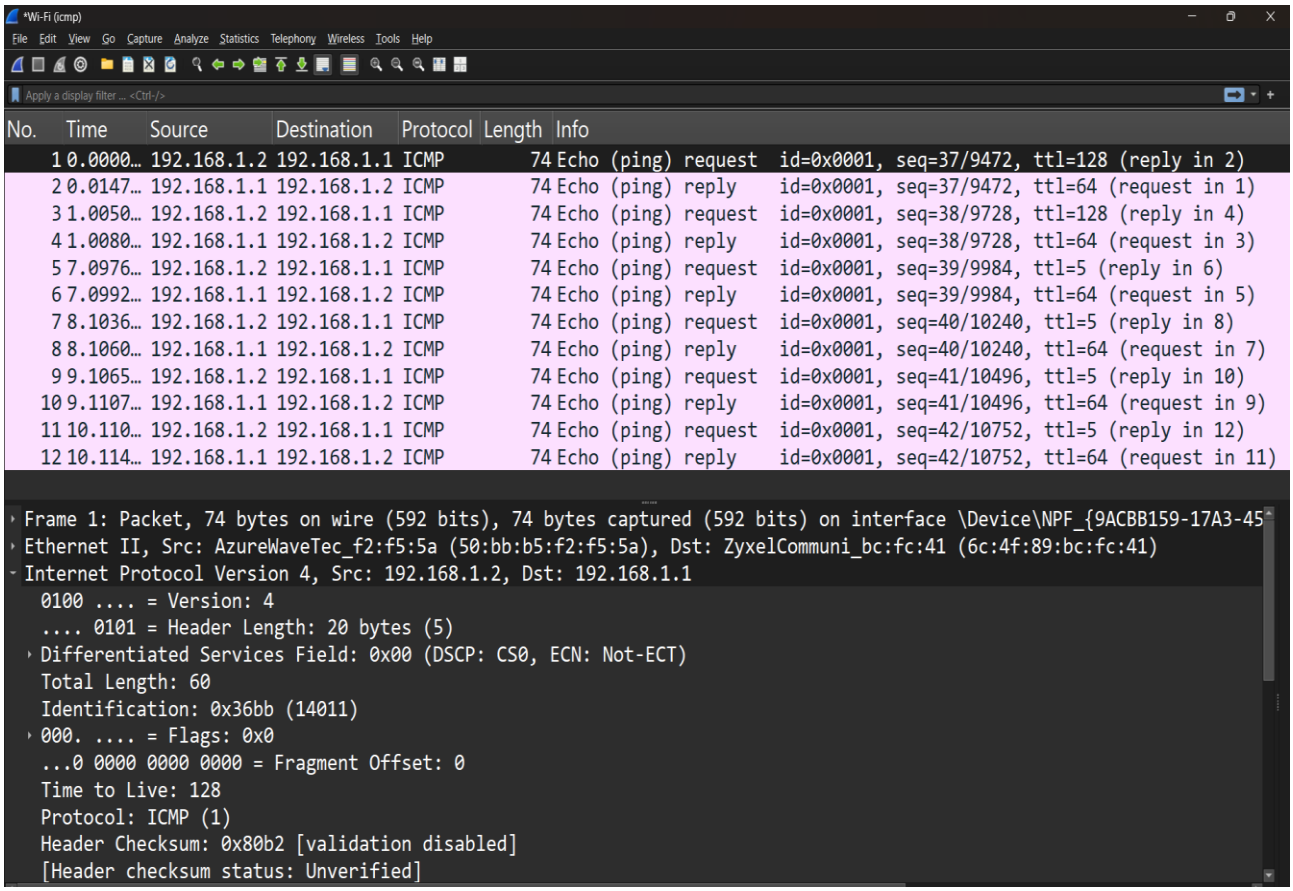
```
C:\Users\itzzi>ping -a 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=4ms TTL=64
Reply from 192.168.1.1: bytes=32 time=4ms TTL=64
Reply from 192.168.1.1: bytes=32 time=3ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

5. Customizing Packet Count and TTL

1. Execute a ping with a custom packet count: ping -n 2 192.168.1.1 to send exactly 2 packets instead of 4.
2. Execute a ping with a custom Time To Live: ping -i 5 192.168.1.1
3. In Wireshark, verify that the echo request packet has a TTL of 5, while the returning echo reply maintains the default TTL of 128.



```
C:\Users\itzzi>ping -n 2 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=15ms TTL=64
Reply from 192.168.1.1: bytes=32 time=4ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 15ms, Average = 9ms
```

6. ICMP Payload Inspection

1. Select an ICMP Echo Request packet in Wireshark.
2. Inspect the Data payload section at the bottom of the details pane.
3. Observe that the default 32-byte payload consists of a sequence of alphabets.

```
▸ Ethernet II, Src: AzureWaveTec_f2:f5:5a (50:bb:b5:f2:f5:5a), Dst: ZyxelCommuni_bc:fc:41 (6c:4f:89:bc:fc:41)
▸ Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.1
▾ Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x4d33 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 40 (0x0028)
    Sequence Number (LE): 10240 (0x2800)
    [Response frame: 8]
  ▸ Data (32 bytes)
```

7. Exercise Execution

1. Open Command Prompt and type ipconfig /all to retrieve local machine network parameters.

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
   Physical Address. . . . . . . . . : 50-BB-B5-F2-F5-5A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2401:4900:8826:c7c8:7449:3713:7b86:7d5a(Preferred)
   Temporary IPv6 Address. . . . . . : 2401:4900:8826:c7c8:6049:8cf3:72dd:a06c(Preferred)
   Link-local IPv6 Address . . . . . : fe80::ddee:d13a:f220:4899%16(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.2(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 21 September 2026 11:11:40 PM
   Lease Expires . . . . . . . . . . : 22 September 2026 11:11:37 PM
   Default Gateway . . . . . . . . . : fe80::6e4f:89ff:febc:fc41%16
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 122731445
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-CE-9D-7C-50-BB-B5-F2-F5-5A
   DNS Servers . . . . . . . . . . . : fe80::6e4f:89ff:febc:fc41%16
                                       192.168.1.1
                                       fe80::6e4f:89ff:febc:fc41%16
```

Parameter	Value
Your Machine IP Address	192.168.1.2
Your Machine MAC Address	50-BB-B5-F2-F5-5A
Default Gateway address	192.168.1.1
DNS Server IP Address	192.168.1.1

2. Ping any website (e.g., ping google.com), filter for icmp in Wireshark, and extract the ICMP Request Message field values.

```
C:\Users\itzzi>ping google.com

Pinging google.com [2404:6800:4007:802::200e] with 32 bytes of data:
Reply from 2404:6800:4007:802::200e: time=58ms
Reply from 2404:6800:4007:802::200e: time=67ms
Reply from 2404:6800:4007:802::200e: time=84ms
Reply from 2404:6800:4007:802::200e: time=101ms

Ping statistics for 2404:6800:4007:802::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 58ms, Maximum = 101ms, Average = 77ms
```

Field Name	Field Length (no of bits)	Field Value
Type	8	128
Code	8	0
Checksum	16	0x235c
Identifier	16	1 (0x0001)
Sequence Number	16	9 (0x0009)

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION: In this experiment, Wireshark was successfully utilized to capture and analyze ICMP traffic. By manipulating ping command parameters, we identified the structural differences between ICMP Query and Error messages, analyzed the network layer headers (such as TTL and Payload size), and successfully extracted packet metadata including Checksums, Identifiers, and Sequence Numbers.

RESULT: Thus, the capture and analysis of ICMP Request and Response packets using Wireshark and the command prompt was executed and verified successfully.

Ex.No:	9a
Name of the Experi ment	Network Protocol analysis: a) Capture and Analyze IP Packets. b) Capture and Analyze TCP Segment. c) Capture and Analyze UDP Datagram. d)Capture and Analyze HTTP Packets.
Date	

OBJECTIVE(S):

To capture and analyse TCP and IP packet using Wireshark.

INTRODUCTION:

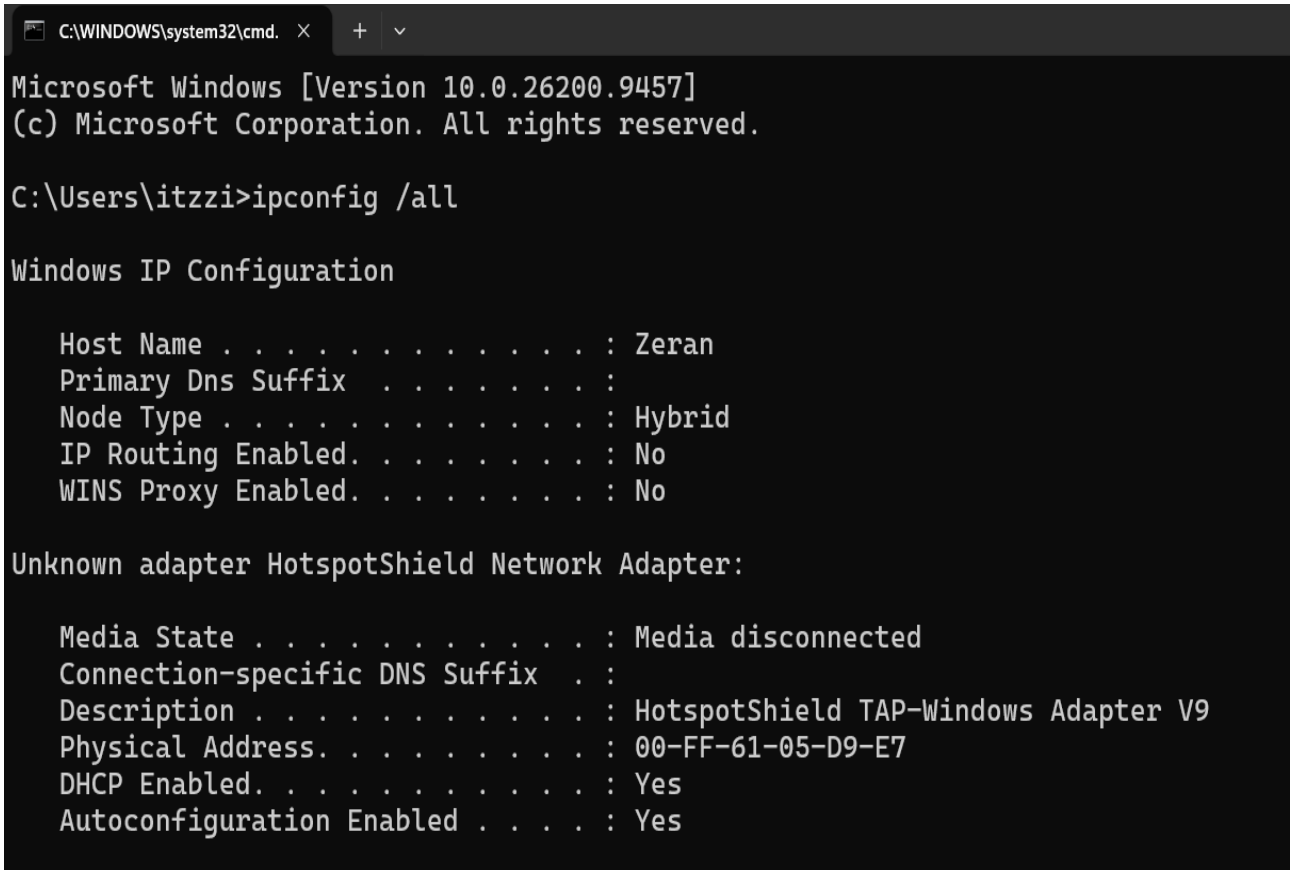
Packet Analysis is a technique used to intercept data in information security, where many of the tools that are used to secure the network can also be used by attackers to exploit and compromise the same network. The core objective of sniffing is to steal data, such as sensitive information, email text, etc., or sniff the traffic that is being transmitted between two parties.

Packet Analysis involves intercepting network traffic between two target network nodes and capturing network packets exchanged between nodes. A packet sniffer is referred to as a network monitor that is used legitimately by a network administrator to monitor the network for vulnerabilities by capturing the network traffic and should there be any issues, proceeds to troubleshoot the same. Similarly, sniffing tools can be used by attackers in promiscuous mode to capture and analyze all the network traffic. Once attackers have captured the network traffic they can analyze the packets and view the user name and password information in a given network as this information is transmitted in a cleartext format. An attacker can easily intrude into a network using this login information and compromise other systems on the network.

Hence, it is very crucial for an Information Security Auditor or a Penetration Tester to be familiar with network traffic analyzers and he or she should be able to maintain and monitor a network to detect rogue packet sniffers, MAC attacks, DHCP attacks, ARP poisoning, spoofing, or DNS poisoning, and know the types of information that can be detected from the captured data and use the information to keep the network running smoothly.

Exercise 1: Machine and Target Details

Visit any one website by opening a browser and fill your machine details.

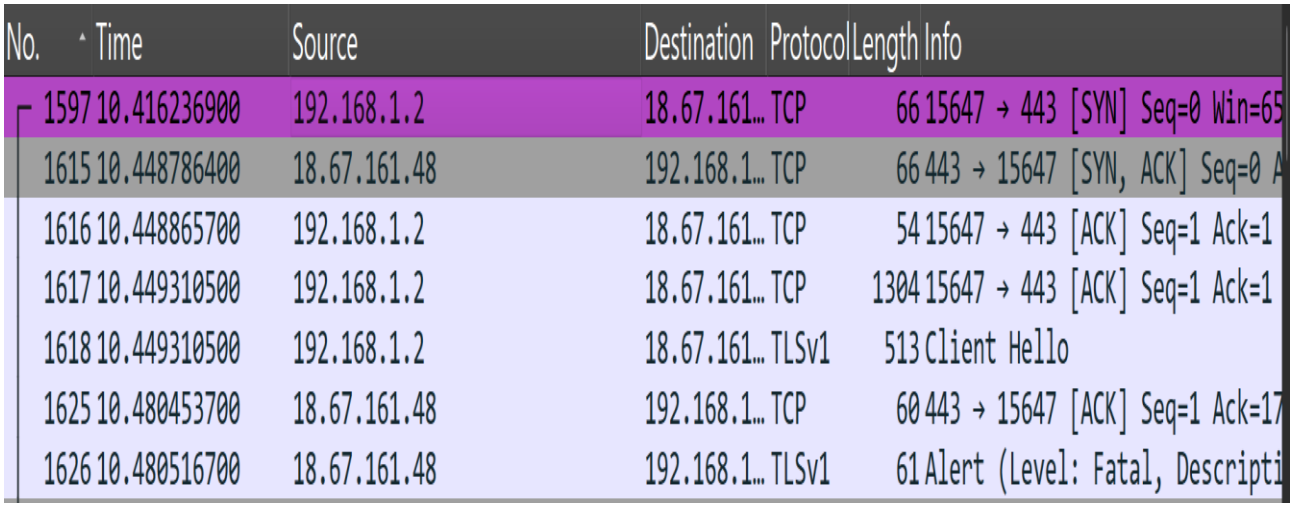


Parameter	**Value**
Your Machine IP Address	192.168.1.2
Your Machine MAC Address	50-BB-B5-F2-F5-5A
Default Gateway address	192.168.1.1
Website URL	https://www.kalasalingam.ac.in
Website IP Address	18.67.161.48

Exercise 2: IP Packet Details

IP Packet Details

Field Name	Field Length (no of bits)	Field value
Destination MAC address	48 bits	50:bb:b5:f2:f5:5a
Source MAC address	48 bits	6c:4f:89:bc:fc:41
Destination IP address	32 bits	192.168.1.2
Source IP Address	32 bits	18.67.161.48
Destination TCP port	16 bits	15647
Source TCP port	16 bits	443



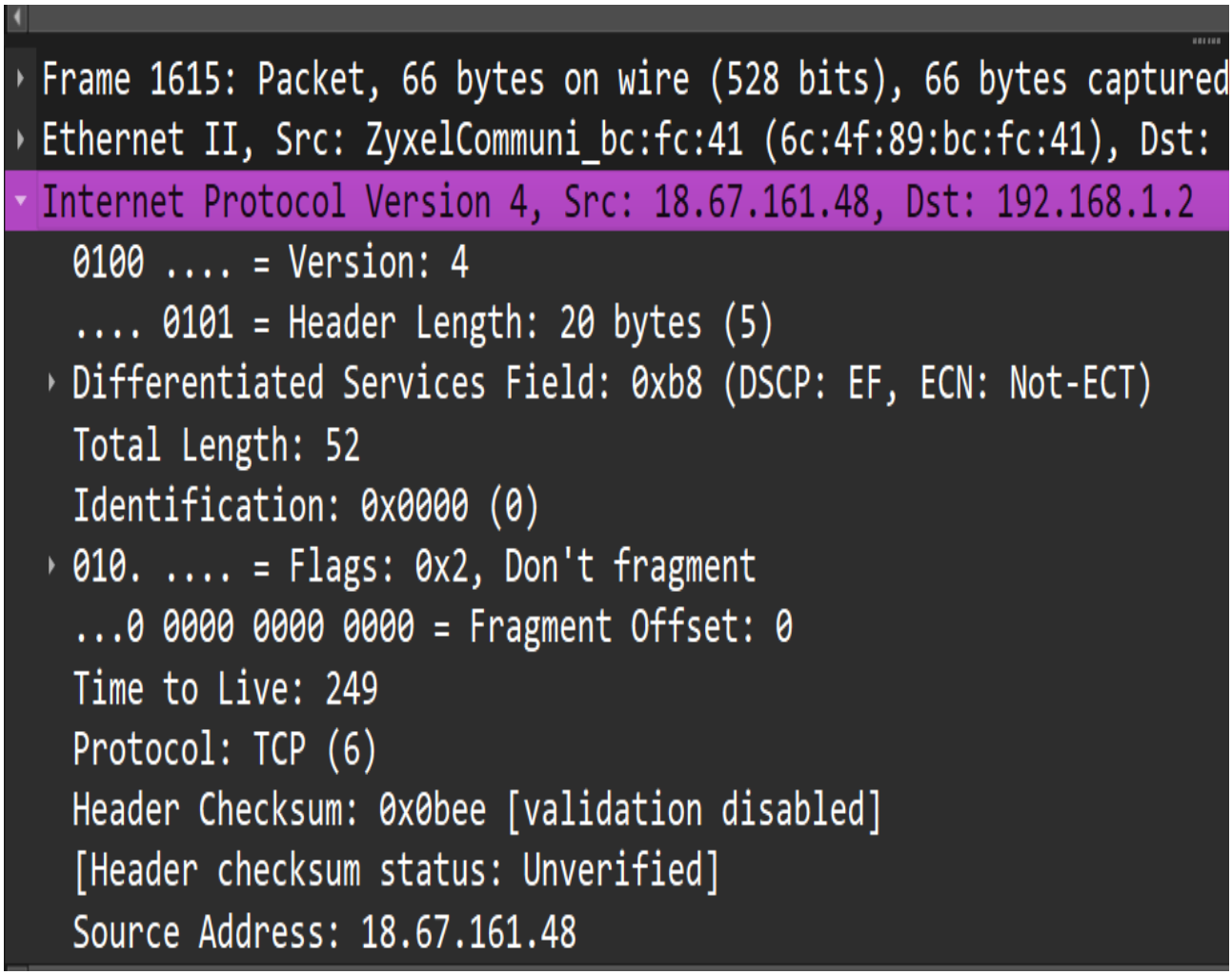
No.	Time	Source	Destination	Protocol	Length	Info
1597	10.416236900	192.168.1.2	18.67.161…	TCP	66	15647 → 443 [SYN] Seq=0 Win=65
1615	10.448786400	18.67.161.48	192.168.1…	TCP	66	443 → 15647 [SYN, ACK] Seq=0 A
1616	10.448865700	192.168.1.2	18.67.161…	TCP	54	15647 → 443 [ACK] Seq=1 Ack=1
1617	10.449310500	192.168.1.2	18.67.161…	TCP	1304	15647 → 443 [ACK] Seq=1 Ack=1
1618	10.449310500	192.168.1.2	18.67.161…	TLSv1	513	Client Hello
1625	10.480453700	18.67.161.48	192.168.1…	TCP	60	443 → 15647 [ACK] Seq=1 Ack=17
1626	10.480516700	18.67.161.48	192.168.1…	TLSv1	61	Alert (Level: Fatal, Descripti

Exercise 3: IP and TCP Header Analysis

IP Frame Format

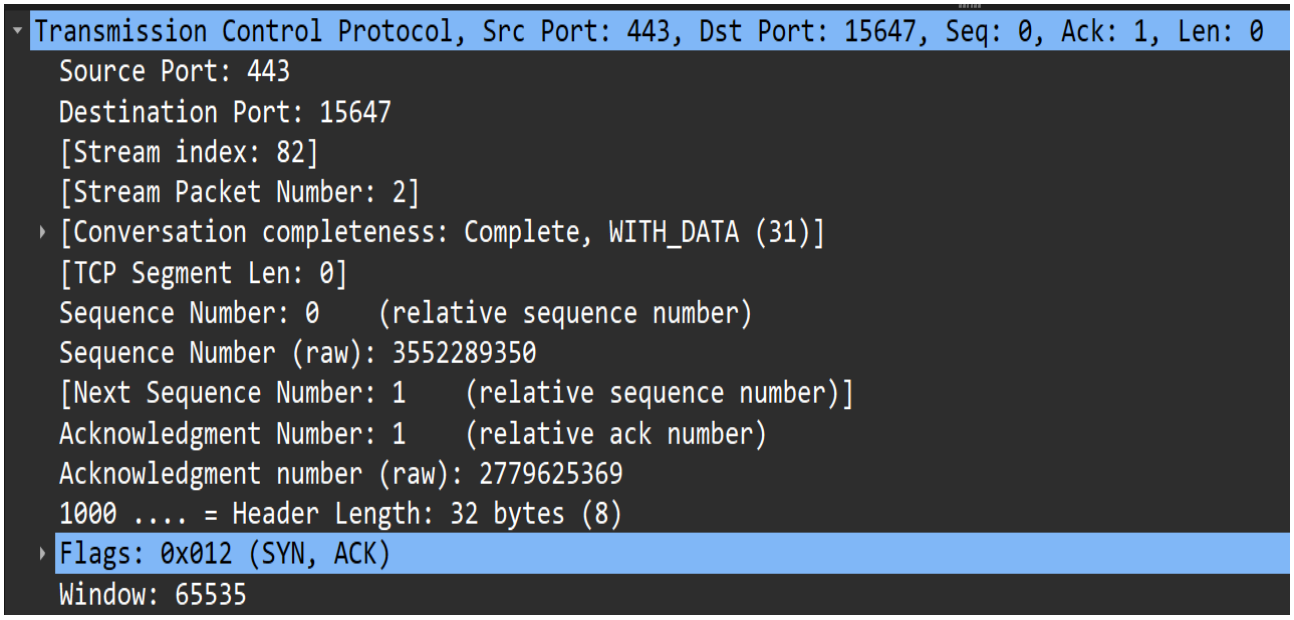
Field Name	Field Length (# of bits)	Field Value (Binary or Hex Value)
Version	4	0100 (4)
Header Length	4	0101 (5)
Type of service	8	0xb8

Datagram Length	16	52
16 bit Identifier	16	0x0000
Flags	3	010
13-bit Fragment offset	13	0 0000 0000 0000
Time-to-live	8	249
Upper layer protocol	8	TCP (6)
Header Checksum	16	0x0bee
32 bit Source Address	32	18.67.161.48
32 bit destination address	32	192.168.1.2



TCP Header Format

Field Name	Field Length (# of bits)	Field Value (Binary or Hex Value)
Source Port	16 bits	443
Destination Port	16 bits	15647
Sequence No.	32 bits	0
Acknowledgement No	32 bits	1
Header Length	4 bits	8 (32 bytes)
FLAGS (URG,PSH,ACK,RST,SYN,FIN)	6 bits	010010 (SYN, ACK)
Receive Window Size	16 bits	65535
Checksum	16 bits	0xe88f
Urgent Pointer	6 bits	0
Options	-	Maximum segment size, NOP, NOP, SACK permitted, NOP, Window scale
Data	-	0 bytes



OBJECTIVE(S):

To capture and analyze HTTP packets using Wireshark.

INTRODUCTION:

The full form of HTTP is HyperText Transfer Protocol. HTTP is an application layer protocol in the ISO or TCP/IP model. It is used by the World Wide Web (WWW) and defines how messages are formatted and transmitted by a browser. HTTP defines the rules for what action should be taken when a browser receives an HTTP command, as well as the rules for transmitting HTTP commands to request data from a server.

For example, when you enter a URL into a browser (Internet Explorer, Chrome, Firefox, Safari, etc.), it sends an HTTP command to the server. The server then replies with the appropriate command and data.

HTTP Methods:

There is a specific set of methods for HTTP/1.1 (the HTTP version): GET, HEAD, POST, PUT, DELETE, CONNECT, OPTION, and TRACE. The most frequently observed methods are:

- **GET:** Asks for data from the web server. This is the primary method used for document retrieval.
- **POST:** Used when it is required to send data (like form submissions) to the server.

HTTP in Wireshark (Theory & Example):

When downloading a file (e.g., alice.txt) from a server, HTTP uses port 80 (or 443 for HTTPS) and relies on TCP as its transport layer protocol.

1. **Connection Setup:** A TCP 3-way handshake (SYN, SYN+ACK, ACK) occurs first.
2. **HTTP Request:** The client sends an HTTP GET request. Key fields include:
    - Request Method: GET
    - Request URI: The specific file or path requested.
    - User-Agent: The client-side browser type (e.g., Mozilla).
    - Host: The web server name.
3. **Data Transfer:** The server sends the requested TCP Data.
4. **HTTP Response:** The server sends an HTTP OK message. Key fields include:
    - Status Code: 200
    - Response Phrase: OK (meaning the GET request succeeded).
    - Content-Type: The type of file being sent (e.g., text/plain).

Exercise Data Collection

1. Machine and Target Details

Visit any one website by opening a browser and fill your machine details.

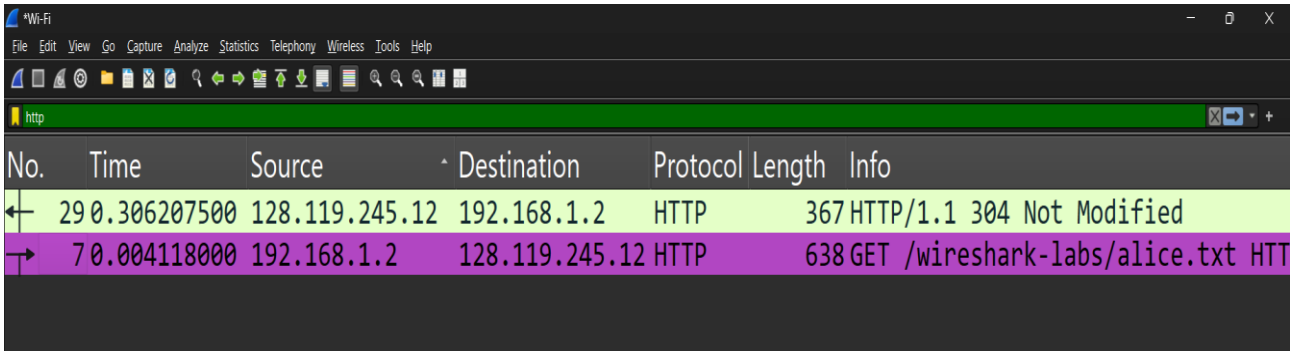
Parameter	**Value**
Your Machine IP Address	192.168.1.2
Your Machine MAC Address	50-BB-B5-F2-F5-5A
Default Gateway address	192.168.1.1
Website URL	gaia.cs.umass.edu
Website IP Address	128.119.245.12

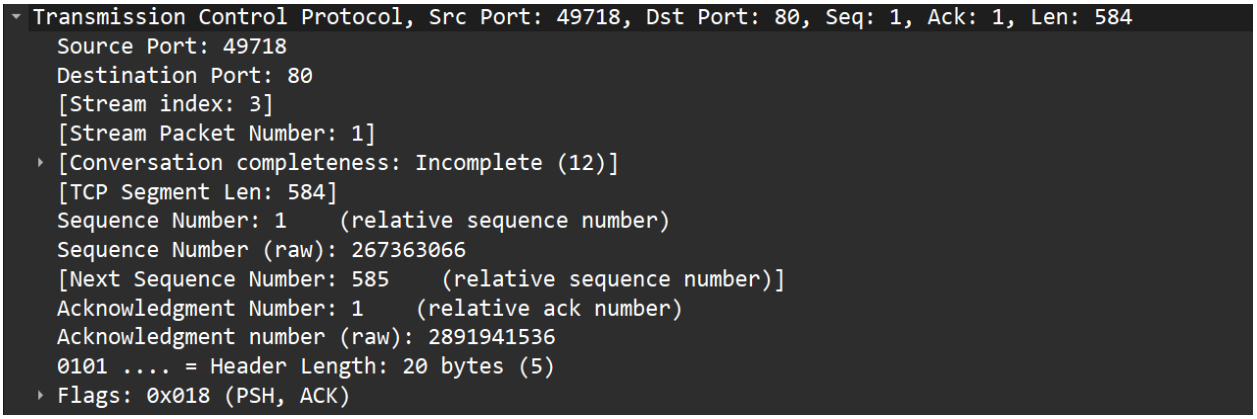
2. TCP Connection Segment Details

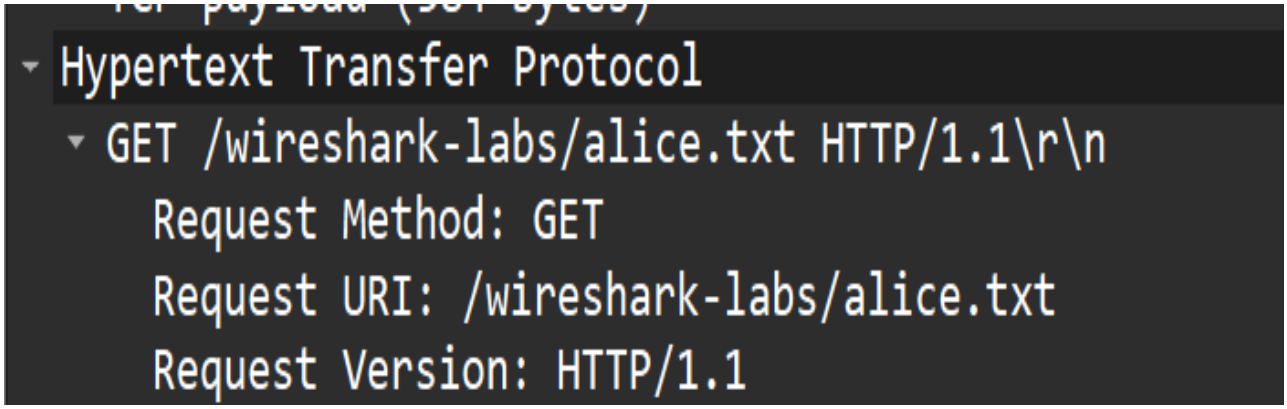
Field Name	**Field Length (no of bits)**	**Field value**
Destination MAC address	48	6c:4f:89:bc:fc:41
Source MAC address	48	50:bb:b5:f2:f5:5a
Destination IP address	32	128.119.245.12
Source IP Address	32	192.168.1.2
Destination TCP port	16	80
Source TCP port	16	49718

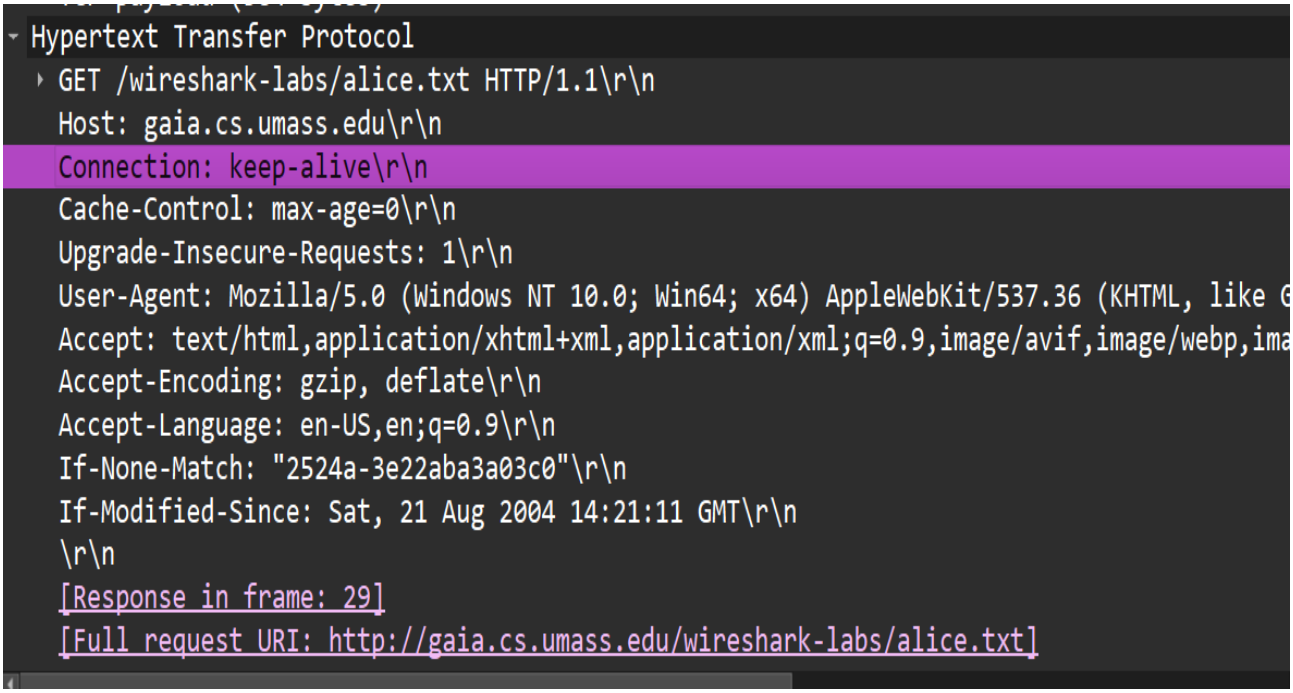
3. HTTP Request Message Details

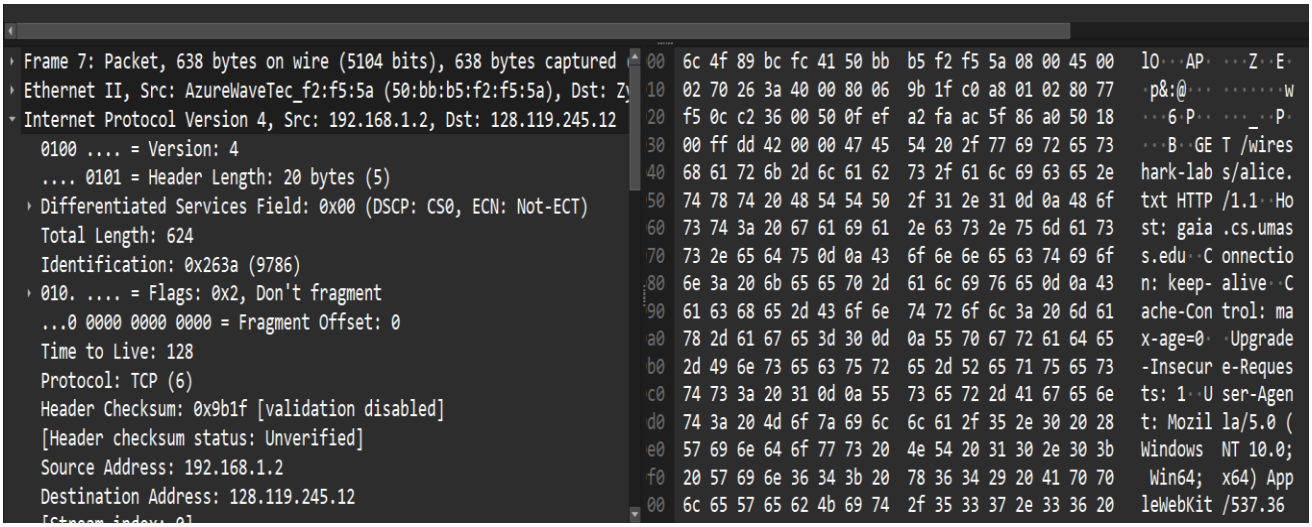
Field Name	Field Length (# of Bits)	Field Value (Binary or Hexa value)
Method	24	GET
Host	104	gaia.cs.umass.edu
Accept	304	text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,...
User-Agent	144	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36...
Accept-Language	40	en-US,en;q=0.9
Accept-Encoding	24	gzip, deflate
Connection	80	keep-alive






Hypertext Transfer Protocol
GET /wireshark-labs/alice.txt HTTP/1.1\r\n
Request Method: GET
Request URI: /wireshark-labs/alice.txt
Request Version: HTTP/1.1


Hypertext Transfer Protocol
GET /wireshark-labs/alice.txt HTTP/1.1\r\n
Host: gaia.cs.umass.edu\r\n
Connection: keep-alive\r\n
Cache-Control: max-age=0\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like G
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,ima
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
If-None-Match: "2524a-3e22aba3a03c0"\r\n
If-Modified-Since: Sat, 21 Aug 2004 14:21:11 GMT\r\n
\r\n
[Response in frame: 29]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/alice.txt]


Frame 7: Packet, 638 bytes on wire (5104 bits), 638 bytes captured
Ethernet II, Src: AzureWaveTec_f2:f5:5a (50:bb:b5:f2:f5:5a), Dst: Zy
Internet Protocol Version 4, Src: 192.168.1.2, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 624
Identification: 0x263a (9786)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x9b1f [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.2
Destination Address: 128.119.245.12
00 6c 4f 89 bc fc 41 50 bb b5 f2 f5 5a 08 00 45 00
10 02 70 26 3a 40 00 80 06 9b 1f c0 a8 01 02 80 77
20 f5 0c c2 36 00 50 0f ef a2 fa ac 5f 86 a0 50 18
30 00 ff dd 42 00 00 47 45 54 20 2f 77 69 72 65 73
40 68 61 72 6b 2d 6c 61 62 73 2f 61 6c 69 63 65 2e
50 74 78 74 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f
60 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73
70 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 74 69 6f
80 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 43
90 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 6d 61
a0 78 2d 61 67 65 3d 30 0d 0a 55 70 67 72 61 64 65
b0 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73
c0 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e
d0 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28
e0 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b
f0 20 57 69 6e 36 34 3b 20 78 36 34 29 20 41 70 70
00 6c 65 57 65 62 4b 69 74 2f 35 33 37 2e 33 36 20
lO AP Z E
p&:@ w
6 P _ P
B GE T /wires
hark-lab s/alice.
txt HTTP /1.1 Ho
st: gaia .cs.umas
s.edu C onnectio
n: keep- alive C
ache-Con trol: ma
x-age=0 Upgrade
-Insecur e-Reques
ts: 1 U ser-Agen
t: Mozil la/5.0 (
Windows NT 10.0;
Win64; x64) App
leWebKit /537.36

Rubrics for Experiment Assessment:

Rubrics	**Good**	**Normal**	**Poor**	**Marks**
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION: In this experiment, Wireshark was successfully used to capture and analyze IP, TCP, and HTTP packets. By inspecting network traffic from a live web session, key parameters such as MAC/IP addresses, TCP ports, and HTTP request headers were successfully extracted, demonstrating how packet analyzers are used to monitor data transmission across different network layers.

RESULT: Thus, the capture and detailed analysis of TCP and HTTP packets using Wireshark was executed and verified successfully.

Ex.No:	10
Name of the Experi ment	Configure DNS, FTP, Email Servers (DNS Packet Analysis)
Date	

OBJECTIVE(S):

To capture and analyze DNS Query and Response packets using Wireshark.

INTRODUCTION:

What is DNS?

The Domain Name System (DNS) is a hierarchical and decentralized naming system for computers, services, or other resources connected to the internet or a private network. It translates easily memorized domain names to the numerical IP addresses needed for locating and identifying computer services and devices with the underlying network protocols.

What is nslookup?

nslookup is a network administration command-line tool available in many computer operating systems for querying the Domain Name System (DNS) to obtain a domain name or IP address mapping, or other DNS records.

To perform DNS analysis in Wireshark, the nslookup command is typically used to generate controlled DNS traffic.

PROCEDURE (Wireshark Analysis Steps):

1. **Initiate nslookup:** Open the terminal (Command Prompt) and run the nslookup command for a specific domain (e.g., nslookup wireshark.org) to view the site's IP address and non-authoritative replies.

2. **Capture Traffic:** Run Wireshark on your active network interface and apply the display filter dns to isolate the traffic generated by your terminal command.
3. **Analyze DNS Query:** Select the first DNS Query packet. In the packet details pane, you will observe the encapsulation layers: Link layer (Ethernet), Network layer (IPv4), Transport layer (UDP), and Application layer (Domain Name System).
    - In the Application layer, expand the Flags section. The first flag bit (Response) will be 0, indicating this is a **Query**.
    - The "Answers RRs" (Resource Records) part is 0 because there is no answer yet.
4. **Analyze DNS Response:** Select the corresponding DNS Response packet.
    - In the Domain Name System section, verify the Transaction ID matches the query.
    - Expand the Flags section. The Response bit will be 1, indicating this is a **Response**.
    - The "Answer RRs" section will populate with the number of answers (e.g., 3), matching the output seen in the terminal.

EXERCISE DATA COLLECTION:

1. Network Configuration Details (ipconfig /all)

Use the command prompt and fill in the following details using the ipconfig /all command.

```
C:\Users\itzzi>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Zeran
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Unknown adapter HotspotShield Network Adapter:

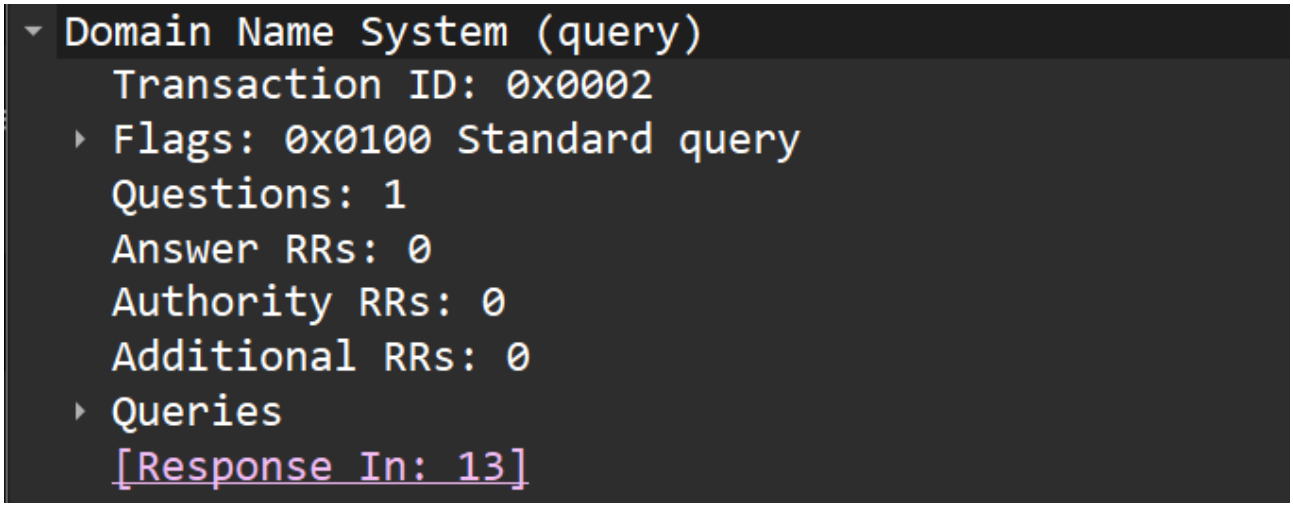
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : HotspotShield TAP-Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-61-05-D9-E7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

Parameter	Value
Your Machine IP Address	192.168.1.2
Your Machine MAC Address	50-BB-B5-F2-F5-5A
Default Gateway address	192.168.1.1
DNS Server IP Address	192.168.1.1

2. DNS Query Message Details

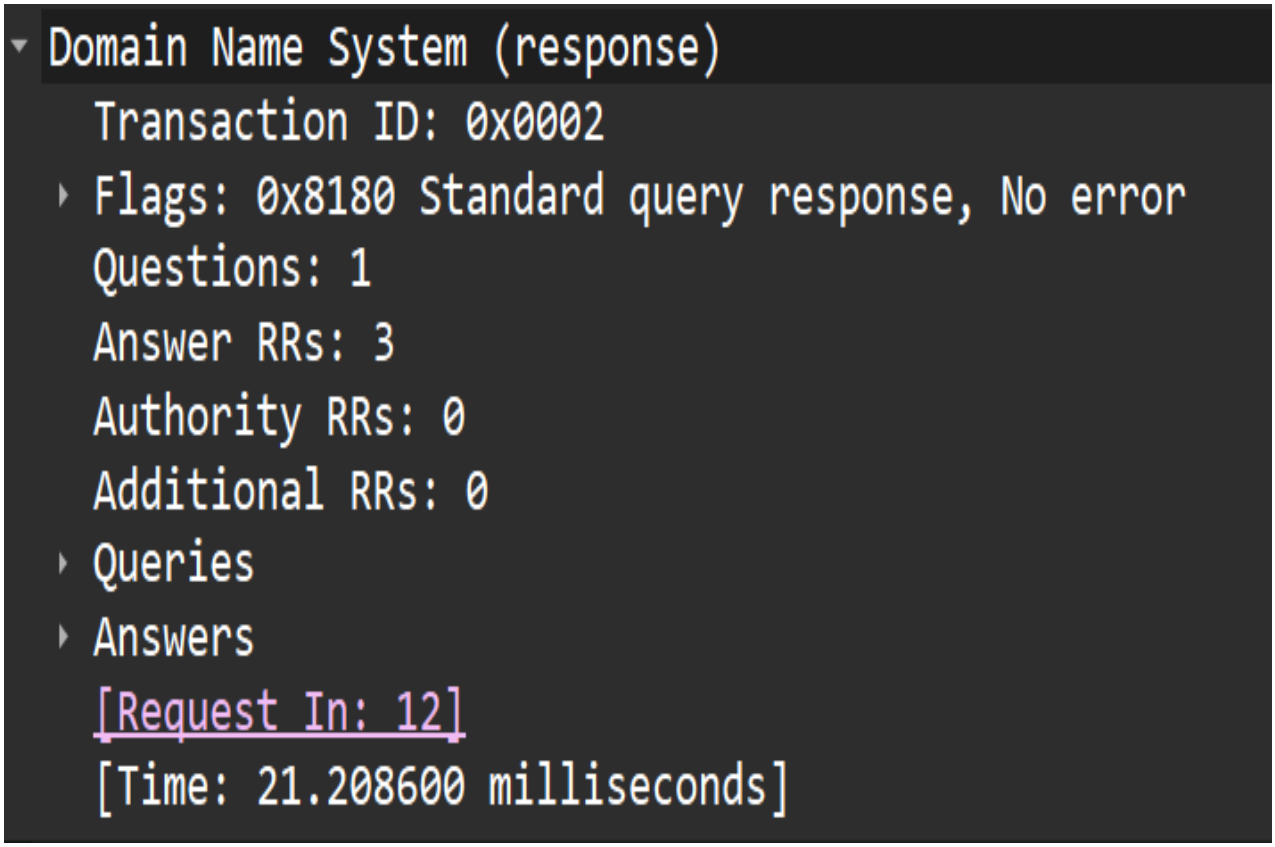
Ping any website or run nslookup through the command prompt and fill the following details by applying the filter as dns in Wireshark.

Field Name	Field Length (no of bits)	Field value
Destination MAC Address	48	6c:4f:89:bc:fc:41
Source MAC Address	48	50:bb:b5:f2:f5:5a
Destination IP Address	128	fe80::6e4f:89ff:febc:fc41
Source IP Address	128	fe80::ddee:d13a:f220:4899
Destination UDP port	16	53
Source UDP port	16	51889
DNS Tx id	16	0x0002
DNS Flags	16	0x0100
DNS Questions	16	1
DNS Queries	variable	events.data.microsoft.com



3. DNS Response Message Details

Field Name	Field Length (no of bits)	Field value
Destination MAC Address	48	50:bb:b5:f2:f5:5a
Source MAC Address	48	6c:4f:89:bc:fc:41
Destination IP Address	128	fe80::ddee:d13a:f220:4899
Source IP Address	128	fe80::6e4f:89ff:febc:fc41
Destination UDP port	16	51889
Source UDP port	16	53
DNS Tx id	16	0x0002
DNS Flags	16	0x8180
DNS Questions	16	1
DNS Queries	Variable	Office-g.microsoft.com




```
C:\Users\itzzi>nslookup events.data.microsoft.com
Server:  UnKnown
Address:  fe80::6e4f:89ff:febc:fc41

Non-authoritative answer:
Name:    onedscolprdcus54.centralus.cloudapp.azure.com
Address:  4.150.223.112
Aliases:  events.data.microsoft.com
          global.asimov.events.data.trafficmanager.net
```

Rubrics for Experiment Assessment:

Rubrics	Good	Normal	Poor	Marks
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
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Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION: In this experiment, Wireshark was successfully utilized to capture and analyze Domain Name System (DNS) traffic. By executing the nslookup command, we generated specific DNS requests and isolated them using display filters. Analyzing these packets allowed us to extract crucial MAC/IP parameters, observe UDP port 53 in action, and understand the structural differences between DNS Query and Response messages, particularly the Transaction IDs and Flag bits.

RESULT: Thus, the capture and detailed analysis of DNS Query and Response packets using Wireshark and the command prompt was executed and verified successfully.

OBJECTIVE(S):

To design and implement Email Server configuration using Cisco Packet Tracer.

INTRODUCTION:

An email server stores and sends email messages to email clients on request. Whenever you compose and send an email, the message first goes to a mail server. The mail server then delivers the email when it is requested by the email client (e.g., mail app) on the recipient's device. To configure an email server in Cisco Packet Tracer, both mail services (SMTP/POP3) and DNS services are required to resolve and route messages between local networks.

Here is the Hardware & Software Requirements section formatted for your lab manual:

HARDWARE & SOFTWARE REQUIREMENTS

- **Software:** Cisco Packet Tracer Simulation Tool
- **Devices:**
    - **End Devices:** PC0, PC1
    - **Networking Devices:** PT Switch (2 Units) and Router (3 Units)
    - **Connections:** Copper Straight-Through cables, Copper Cross-Over cables

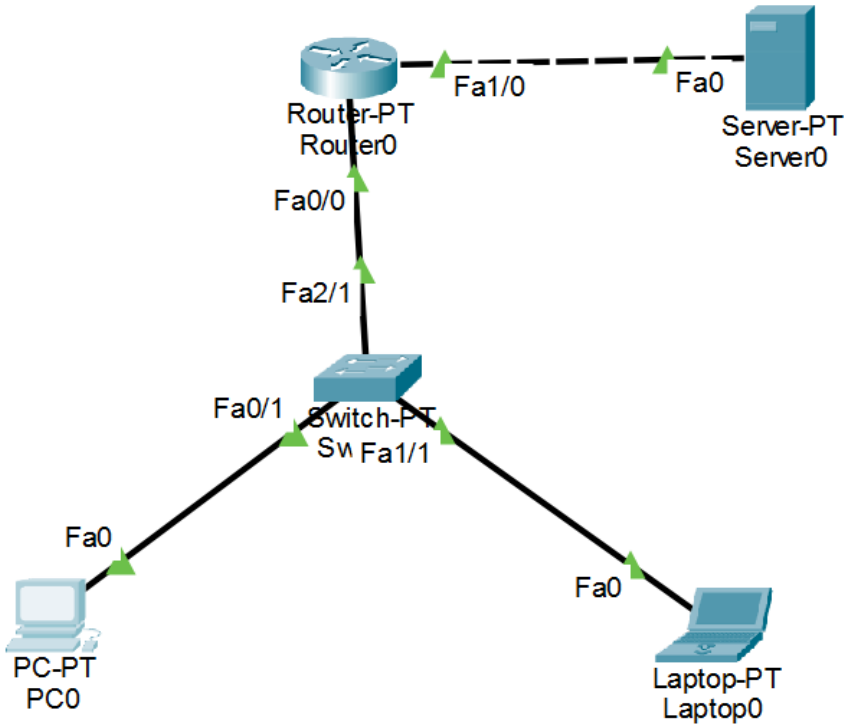
CONFIGURATION DETAILS

Device Name	Interface Name	IP Address	Subnet Mask	Default Gateway
Router	Fa0/0	192.168.1.1	255.255.255.0	N/A
Server	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
Laptop 0	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1

PROCEDURE WITH DESCRIPTION

Step 1: Build the Network Topology

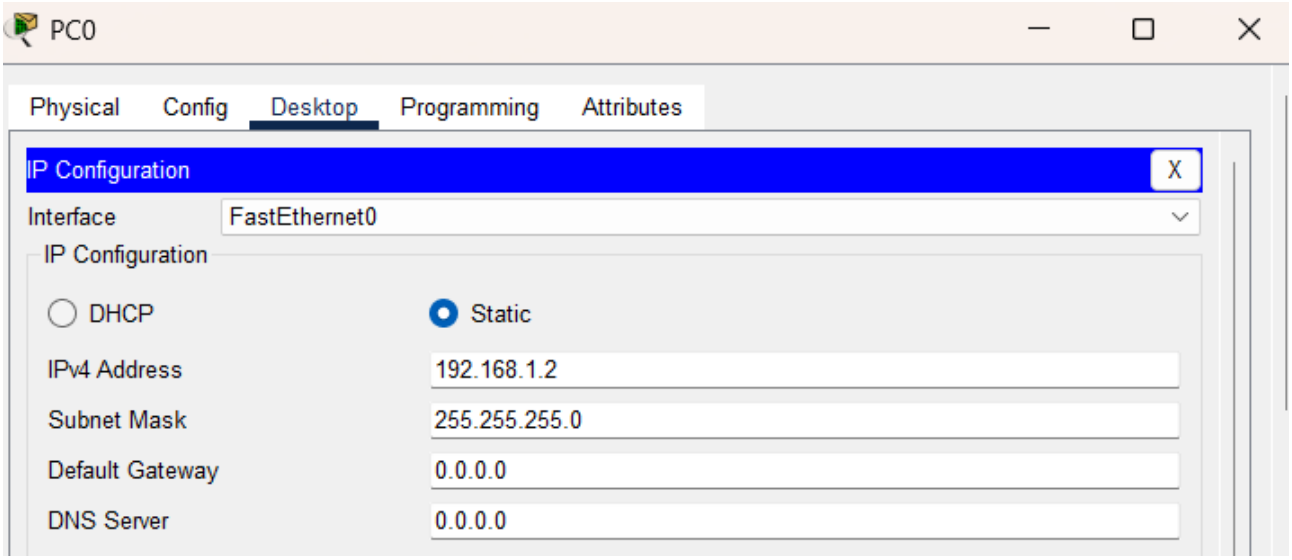
- Open Cisco Packet Tracer and place one Router, one Switch, one Server, PC0, and Laptop 0 onto the workspace.
- Connect PC0 and Laptop 0 to the Switch using Copper Straight-Through cables via their FastEthernet interfaces.
- Connect the Switch to the Server and the Router using Copper Straight-Through cables to complete the physical network layout.



Step 2: IP Configuration for End Devices and Server

- Click on **PC0**, navigate to **Desktop** > **IP Configuration**, and set the static IPv4 Address to 192.168.1.2, Subnet Mask to 255.255.255.0, and Default Gateway to 192.168.1.1.
- Click on **Laptop 0**, navigate to **Desktop** > **IP Configuration**, and set the static IPv4 Address to 192.168.1.3, Subnet Mask to 255.255.255.0, and Default Gateway to 192.168.1.1.

- Click on the **Server**, navigate to **Desktop** > **IP Configuration**, and set the static IPv4 Address to 192.168.2.2, Subnet Mask to 255.255.255.0, and Default Gateway to 192.168.2.1.



Step 3: Router Interface Configuration (CLI)

- Access the **Router**, open the **CLI** tab, and enter global configuration mode.
- Ensure all the connections are correct.
- Run the following commands to configure the router interface and activate it:

CLI

Router> enable

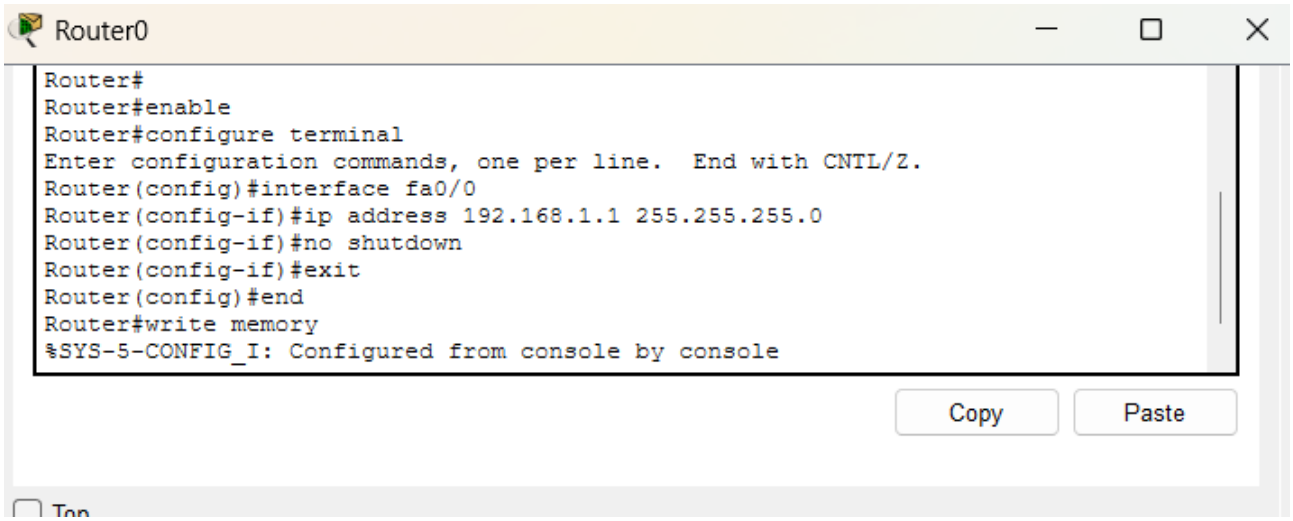
Router# configure terminal

Router(config)# interface fa0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

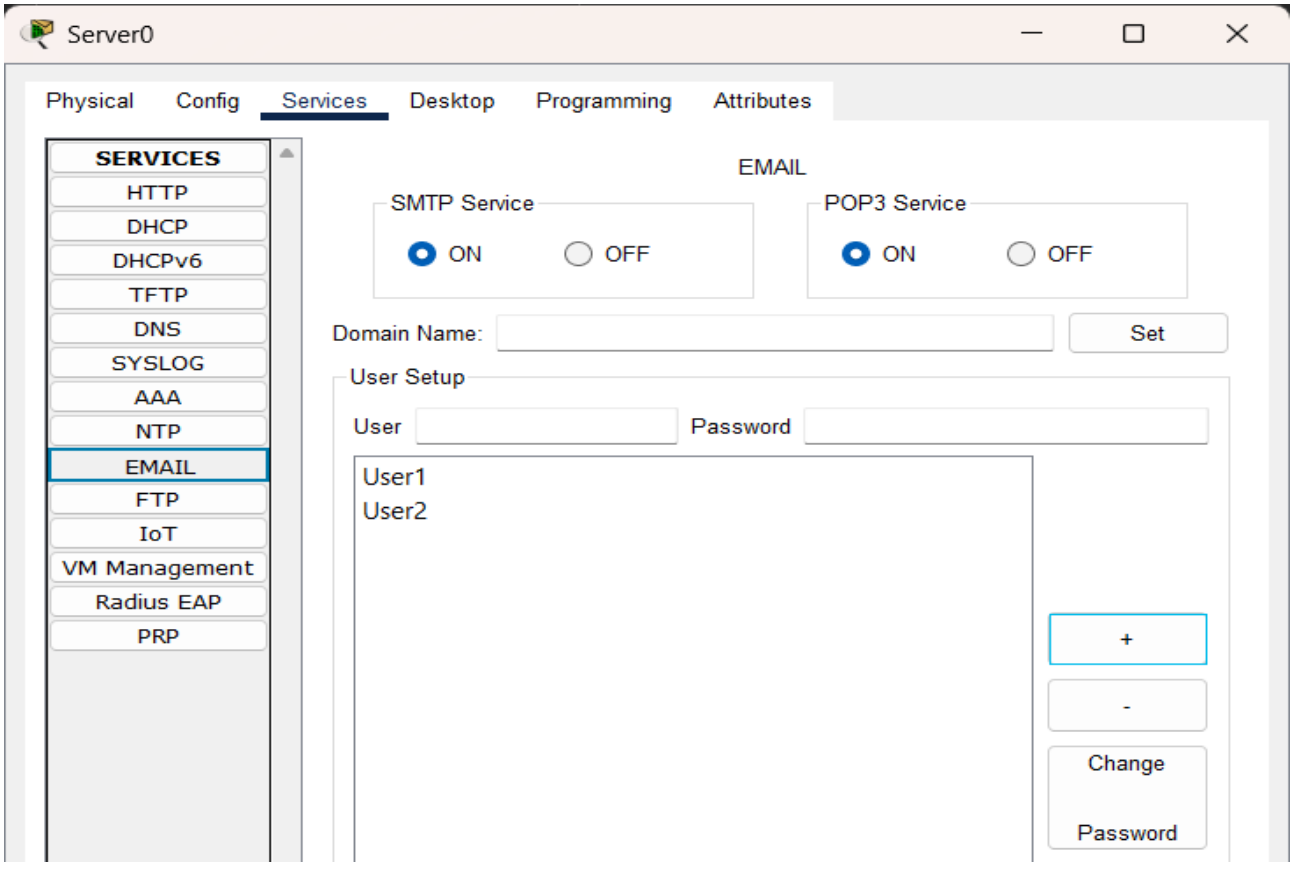
Router(config-if)# no shutdown

Router(config-if)# end

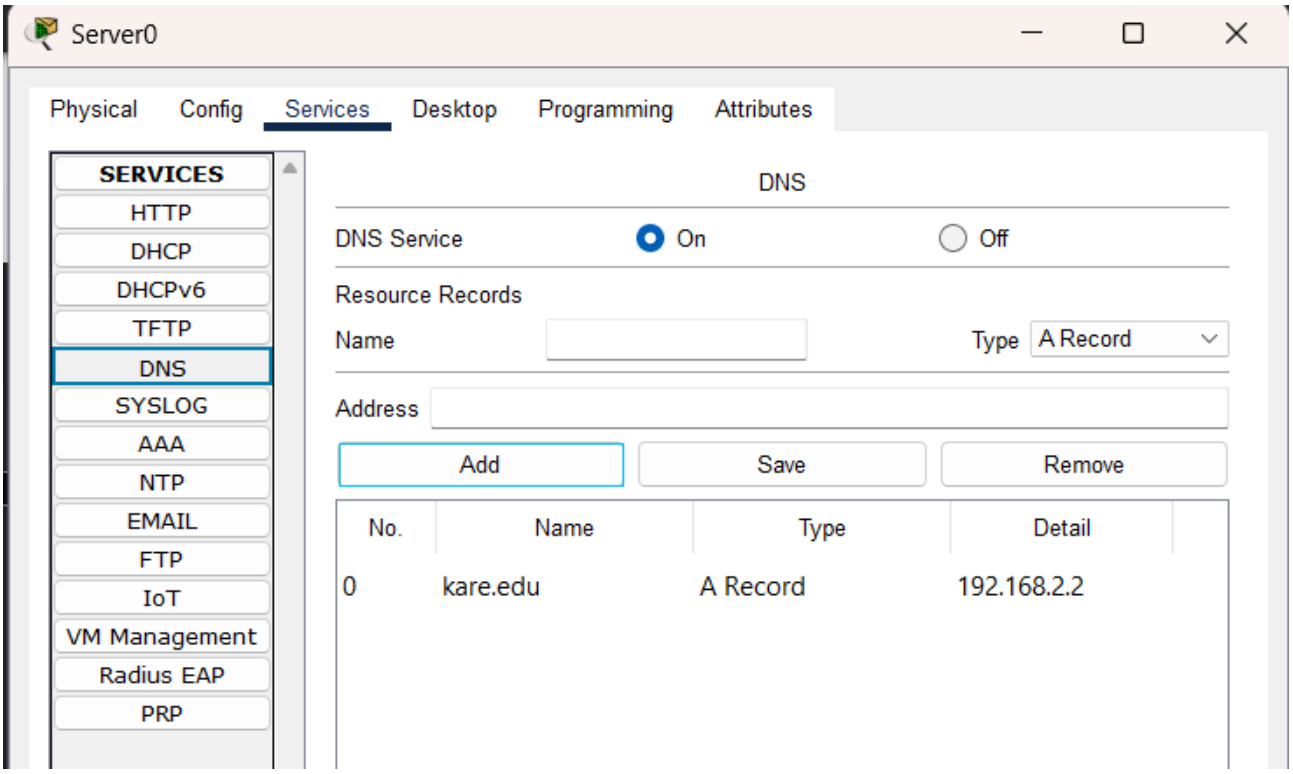


Step 4: Configure Mail and DNS Services on the Server

- Click on the **Server**, select the **Services** tab, and click on **EMAIL**.
- Turn both **SMTP Service** and **POP3 Service** to **ON**. Enter a Domain name (e.g., kare.edu).
- Create user accounts by adding usernames (user1, user2) and passwords, clicking the + button for each.

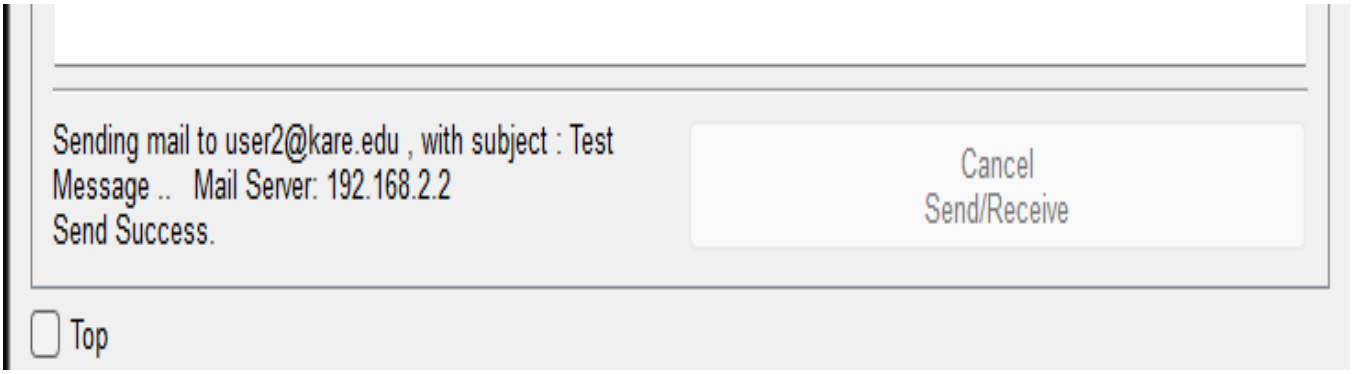


- Navigate to the **DNS** service tab on the Server, turn DNS **ON**, enter the domain name (kare.edu), map it to the server IP address (192.168.2.2), and click **Add**.

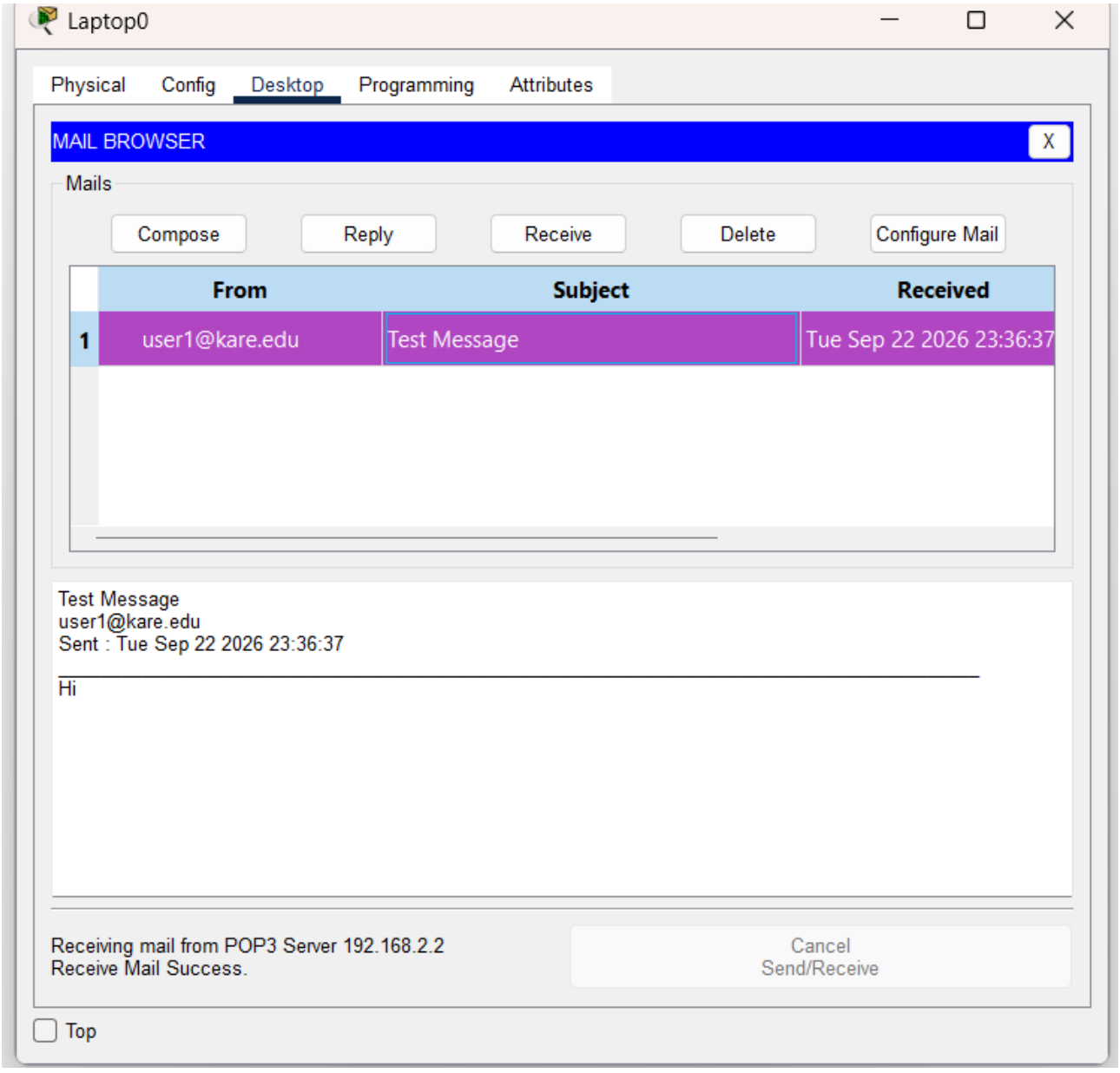


Step 5: Compose, Send, and Receive Emails

- Open the Email application on PC0, configure the user profile for user1@kare.edu by setting the incoming (POP3) and outgoing (SMTP) servers to 192.168.2.2 with username user1, compose a message addressed to user2@kare.edu, and click Send.



- Open the **Email** application on **Laptop 0**, configure the user profile for user2@kare.edu, click **Receive** to view the incoming email from PC0, and click **Reply** to send a response back.



Rubrics for Experiment Assessment:

Rubrics	**Good**	**Normal**	**Poor**	**Marks**
Creation of Topology(4)	Created the topology, Identify the proper devices and making the connections **(4)**	Created the topology, Identify the proper devices, making the connections But missing some features **(3)**	Created wrong topology,Failed to Identify the proper devices and making connections **(1)**	
Verify the connectivity (4)	Verified the connectivity in all the levels **(4)**	Verified the connectivity at some levels (only some nodes) **(2)**	Verified the connectivity is not done. **(1)**	
Timely Completion (2)	Completed the lab before the allotted time **(2)**	Completed the lab after the deadline **(1)**	Did not submitted before grading**(0)**	
Total				

CONCLUSION: In this experiment, The successful configuration of an email server using Cisco Packet Tracer demonstrates the practical application of network services. By enabling SMTP and POP3 protocols alongside DNS resolution, end devices like PC0 and Laptop 0 can reliably communicate, authenticate, and exchange messages across subnets.

RESULT: Thus, the design and implementation of the Email Server configuration using Cisco Packet Tracer were executed and verified successfully.